Overvalued Local Expertise in International Equity Funds

Kevin Zheyuan Cui* Yinan Qiu†

November 13, 2026

Abstract

Managers familiar with foreign countries may bring local expertise, but international equity funds invest across many countries beyond their managers' familiar ones. Using foreign education as a proxy, we ask how far this expertise reaches and whether funds recognize its limits. Greater foreign-educated representation is associated with lower factor-adjusted net returns, about 70 basis points a year between fully foreign- and U.S.-educated teams. The expertise is real but narrow. In managers' education countries, high-conviction stocks earn a 4% annual alpha but are a small share of portfolios. Elsewhere, foreign-educated teams select stocks worse in markets bordering or more socially connected to their education countries. Funds do not appear to recognize these limits. Following poor returns in a country, funds are more likely to appoint a manager educated there, but fund performance shows no clear improvement afterward. Additionally, foreign-educated recruits have weaker prior performance than U.S.-educated recruits, and their retention is less sensitive to poor fund returns. The findings suggest that funds overvalue local expertise as a signal of broader investment ability.

Keywords: Manager selection; local expertise; familiarity; manager education; international mutual funds; stock selection.
JEL Codes: G11, G15, G23, G41, J24.

Previously circulated as "Familiarity in International Equity Investment: Concentrated Value, Pervasive Allocation Bias."

*Department of Economics, Princeton University.
†Guanghua School of Management, Peking University.

1 Introduction

International equity funds appoint managers to invest across many countries, but the information those managers possess may be valuable only in particular markets. Prior research documents informational advantages in geographically proximate investments and education-linked holdings (Coval and Moskowitz, 2001; Cohen et al., 2008), as well as home-country stock-selection advantages among international fund managers, particularly when their backgrounds align with regional or country mandates (Jagannathan et al., 2022). Yet familiarity can also influence portfolio choices without superior investment performance. Investors favor familiar assets (Huberman, 2001), and mutual fund managers overweight home-state stocks that do not outperform (Pool et al., 2012). Even when familiarity reflects information, its value to a fund investing across many markets depends on how much of the portfolio it informs. Local advantages, even when genuine, may be too narrow to offset weaker stock selection elsewhere, while managers' educational backgrounds may still be valued as signals of broader investment ability. Because team-based fund returns make individual contributions difficult to observe (Holmström, 1982), such observable backgrounds may be particularly salient in hiring and evaluation. Using foreign education as a proxy for familiarity with particular markets, we examine whether funds distinguish the local investment value of expertise from broader investment ability.

Our evidence suggests that funds do not fully recognize the limits of local expertise. Greater foreign-educated representation is associated with lower factor-adjusted net fund returns, even though selected high-conviction holdings in managers' education countries earn substantial alphas. These positions account for little portfolio capital. Outside their education countries, foreign-educated teams select stocks worse in markets that border or are more socially connected to those countries than in other markets. Hiring and subsequent employment patterns reinforce this interpretation. Funds are more likely to appoint managers educated in countries where their holdings have underperformed, yet the subsequent local improvement is short-lived and comes almost entirely from stocks held before the hire. In

broader hiring comparisons, foreign-educated recruits have weaker prior performance than U.S.-educated recruits. Among colleagues on the same fund, foreign-educated managers' departures are less sensitive to deteriorating fund performance than those of U.S.-educated managers.

We study 1,663 actively managed U.S. international equity funds from 2000 to 2024, linking Morningstar manager education histories and dated fund assignments to portfolio holdings and returns. We use the countries where managers studied as a proxy for their familiarity with those markets and thus for potential local expertise. The funds' portfolios extend far beyond these markets. Among teams with foreign-educated managers, holdings span 18.4 foreign countries on average, 17.2 outside the team's education countries. This breadth lets us compare stock selection in education countries and elsewhere and assess whether localized gains translate into better overall fund performance. Manager assignment histories allow us to compare the same fund as its team changes, examine recruits' prior performance and study how subsequent departures and new assignments vary with fund performance.

We first examine returns across the whole portfolio. We use changes in the foreign-educated share within a fund, linking each quarter-end team to factor-adjusted returns over the next three months. Comparing the same fund over time removes persistent fund differences, while comparing funds with the same benchmark in the same month accounts for common shocks. We also control for prior performance and flows that may shape staffing. The estimated association implies annual net returns about 70 basis points lower for a fully foreign-educated team than for a fully U.S.-educated team at the same fund. The gap is nearly identical before expenses.

Education-country ties shape where funds allocate capital. Relative to their benchmarks, funds overweight a country by about 1.1 percentage points when a current manager studied there. This overweight increases when the first country-linked manager arrives and declines when the last departs. Yet this tilt is not accompanied by superior average stock selection.

When we compare holdings in the same country and quarter, adjusted for stock characteristics, foreign-educated teams show no statistically significant advantage over U.S.-educated teams in their education countries. The point estimate is 0.2 basis points a year, though it is imprecise.

The average can conceal a concentrated advantage, and fund-level underperformance can coexist with valuable local stock selection when the gains are confined to a small share of holdings. The strongest evidence of a local advantage comes from high-conviction positions, the stocks that funds overweight most heavily within an education country. They earn a 4.0% annual alpha, 3.6 percentage points more than other positions in the same countries and 3.9 points more than high-conviction positions outside education countries, with every portfolio reweighted to the same country mix. These positions make up only 3.6% of funds' observed equity holdings, however, so even a 4.0% alpha would add only about 14 basis points a year to portfolio returns. Familiarity thus has value in selected local investments, but that value contributes little to the overall portfolio.

We next examine whether ties to managers' education countries extend useful information or merely familiarity to other markets, measuring these ties by shared borders, common official languages and social connectedness (Bailey et al., 2018). Foreign-educated teams select stocks 184 basis points a year worse in markets bordering their education countries than in other outside markets, and their selection also deteriorates as social connectedness rises. The border gradient is 484 basis points more negative than for U.S.-educated teams, with similarly unfavorable relative gradients for language and social connectedness. This underperformance in more closely connected markets is consistent with familiarity bias (Huberman, 2001; Pool et al., 2012).

We then ask when funds seek local expertise and what performance follows these appointments. Comparing countries held by the same fund in a given quarter, we find that weak market returns and poor performance of the fund's own holdings in a country each predict the appointment of a manager educated there. Relative to matched non-hiring funds

investing in the same country, local stock selection improves in the first year, but the gain disappears by the second. A decomposition of the first-year improvement attributes 174 of 178 annualized basis points to stocks held before the appointment and only an estimated 4 basis points to changes in holdings. This short-lived recovery is consistent with a rebound following prior underperformance and provides no clear evidence that the incoming manager's stock choices generated the improvement.

We next turn from these appointments to all experienced recruits and ask what prior performance funds accept. For foreign- and U.S.-educated recruits hired by the same fund, we compare their previous teams' stock-selection performance in foreign markets over the four quarters before appointment. Foreign-educated recruits' prior teams underperform by 86 basis points a year, after accounting for previous funds' benchmarks, school rankings and other observed qualifications. Their previous funds also attracted weaker investor flows. Even in countries the hiring fund holds but in which none of its continuing managers studied, recruits educated there bring weaker prior performance in that market than U.S.-educated recruits, by 274 basis points a year. Funds thus accept weaker prior performance from foreign-educated recruits than from U.S.-educated recruits.

Beyond recruitment, we examine whether foreign education is associated with different performance sensitivity in managers' retention and new fund assignments. Comparing colleagues on the same fund, we find greater relative retention for foreign-educated managers after poor returns and smaller relative gains in new assignments after good returns. A ten-percentage-point deterioration in benchmark-relative quarterly returns raises the foreign-minus-U.S. retention difference by 1.45 percentage points. A ten-point improvement lowers their relative probability of receiving a new fund assignment by 1.07 points. These patterns are consistent with less performance-sensitive career evaluation of foreign-educated managers.

Finally, we consider whether employing foreign-educated managers offers funds a commercial payoff through greater asset gathering or higher fees. We find no statistically clear

association with greater subsequent flows or higher reported fee rates. The measured commercial outcomes do not establish a compensating benefit for funds.

Together, these results show that funds apply lower performance standards to foreign-educated managers at hiring and afterward, with no offsetting advantage in country-specific performance, fundraising or fees. Given the narrow investment value of local expertise, the pattern is most consistent with funds overvaluing it as a signal of broader ability.

The paper contributes to research on the scope of investment information. Geographic, educational and offshore connections can improve investment decisions (Coval and Moskowitz, 2001; Cohen et al., 2008; Baik et al., 2010; Bai et al., 2022), while familiarity and perceived competence can affect portfolios without a corresponding advantage (Heath and Tversky, 1991; Huberman, 2001; Grinblatt and Keloharju, 2001; Graham et al., 2009; Pool et al., 2012). Relatedness also shapes international allocations (Schumacher, 2018; Kuchler et al., 2022). We separate the concentrated value of local stock choices from the poorer selection that accompanies familiarity beyond education countries, and relate that distinction to how funds select and evaluate managers. The concentration of value complements evidence on industry specialization, active share and best ideas (Kacperczyk et al., 2005; Cremers and Petajisto, 2009; Antón et al., 2021).

The personnel evidence contributes to research on manager selection and evaluation. Education and background predict fund outcomes (Chevalier and Ellison, 1999a; Gottesman and Morey, 2006; Chuprinin and Sosyura, 2018), and performance matters for replacement and reassignment (Khorana, 1996; Chevalier and Ellison, 1999b; Fang et al., 2014). Qualifications provide signals of productivity (Spence, 1973; Phelps, 1972; Aigner and Cain, 1977), and subsequent outcomes allow employers to update those assessments (Farber and Gibbons, 1996; Altonji and Pierret, 2001), although fund returns are noisy (Fama and French, 2010). We show that funds accept weaker general and country-specific prior performance from foreign-educated recruits, and that these managers' careers respond less to realized fund performance than those of U.S.-educated colleagues. The results complement work

on professionals' beliefs (Linnainmaa et al., 2021) by examining how a single qualification relates to both portfolio outcomes and personnel decisions.

Our closest comparison is Jagannathan et al. (2022), who document superior stock selection in home-country holdings of home-linked managers, with stronger evidence among regional and country funds than in their full international-fund sample. Their findings highlight the potential value of local expertise, but its importance to fund investors depends on how much of the portfolio benefits from it. Broad global funds account for 70% of funds and 82% of assets in our sample. Among broad global funds that hold stocks in their managers' education countries, these markets average only 13% of equity portfolios, compared with 67% for country-focused funds. Our characteristic-adjusted tests find no statistically significant average stock-selection advantage in education countries, although high-conviction positions there earn substantial alpha.[1] We show that these concentrated local gains need not translate into stronger fund-wide returns, extend the analysis to markets beyond education countries and examine whether hiring and career outcomes reflect the limits of local expertise.

Section 2 describes the institutional setting, the data and measures, and the prevalence of foreign-educated managers. Section 3 asks how far local expertise reaches, proceeding from whole-fund returns to education-country allocation and stock selection, then to familiarity in outside countries. Section 4 asks whether funds recognize its limits, examining hiring after country losses and the source of the subsequent recovery, the prior performance of recruits, later career responses and commercial outcomes. Section 5 concludes.

[1] Jagannathan et al. (2022) report a monthly four-factor alpha of 31 basis points ($t = 1.69$) for a portfolio long home-country holdings of home-linked funds and short same-country holdings of unlinked funds in their full international-fund sample (their Table 4, Panel A). Our country-level tests differ in sample period, education-link definition and performance measurement. Appendix D provides a detailed reconstruction of their tests and discusses the differences.

2 Setting, Data and Measures

2.1 Institutional setting

International equity funds give U.S. investors access to equity markets abroad. Morningstar classifies a fund in this group when foreign holdings account for at least 40% of its equity portfolio on average over the preceding three years. In a typical sample quarter, about 760 active funds operate, with mean assets of $1.6 billion. Most have global or global-ex-U.S. benchmarks, and others follow regional or single-country mandates. Funds invest in 19.0 countries on average and hold a median of roughly 80 stocks, compared with more than 5,000 stocks in about 50 countries in the Morningstar Global ex-US index. This setting lets us distinguish the allocation across countries from the selection of stocks within each country, while observing the overall return investors receive.

We classify funds by Morningstar investment category into broad global, regional and country-focused funds. Broad global funds are those in Morningstar's foreign and world large-cap and small/mid-cap categories. Regional funds focus on emerging markets, Asia-Pacific, Europe or Latin America. Country-focused funds concentrate on markets such as Japan, India or Greater China. We hold these classifications fixed over time. Broad global funds account for 70% of sample funds and 82% of their assets.[2]

Investment decisions are delegated to small teams. The average fund has 3.1 managers with mean tenure of 5.6 years, and managers often serve on several funds simultaneously. Among funds with at least one foreign-educated manager, the average team has 4.1 managers, including 2.0 foreign-educated managers, and holds stocks in 18.4 foreign countries. Of these markets, 17.2 lie outside all recorded education countries of the current team. The team's mandate extends well beyond the countries linked to its members' education. We observe returns and holdings for the team, even when its members divide research respon-

[2]Education-country holdings are measured as a share of each fund's covered equity holdings, including U.S. equities. For the shares reported in the introduction, we give equal weight to fund-quarters with holdings in at least one of the team's education countries.

sibilities. The performance tests therefore use team education, and the appointment tests use the performance of a recruit's previous teams. Each manager's dated fund assignments identify later departures and new assignments, and the manager's education countries mark a potential specialty.

The investment management firm responsible for a fund selects its portfolio managers. Appointees may come from another firm, from elsewhere in the same fund family or from outside the funds we observe. For experienced recruits we observe the performance of their previous teams as well as the degrees listed in their biographies, so we can see what funds accept at appointment. Later assignments and departures show how the careers of colleagues on the same fund vary with its performance.

2.2 Funds, holdings and returns

Our sample consists of actively managed U.S. international equity funds in Morningstar over 2000–2024, including funds that later liquidated or merged.[3] For each fund we observe quarterly stock holdings, the holdings of its assigned benchmark, monthly net returns and assets in U.S. dollars, reported gross returns, and annual expenses and turnover. Security returns and market capitalizations come from Datastream. We assign each company to the country of its primary business activity rather than its listing, which aligns country weights with economic exposure (Demirci et al., 2022).[4]

2.3 Manager education

We use the countries where managers studied as a proxy for familiarity with particular markets, which can inform investment decisions. Studying in a country can help managers learn

[3]We select funds in Morningstar's U.S. International Equity category group, keep each fund's oldest share class and exclude index funds, without conditioning on survival. Each fund is identified by the ISIN of its oldest share class, whose returns we use. ISINs that share a Morningstar fund code form one economic fund, the unit of clustering.

[4]Where total-return series are unavailable, returns are constructed from prices and dividends. Missing returns are excluded rather than set to zero.

about local firms, business practices and institutions. Relationships with classmates and alumni can help them find and interpret information about local companies. We refer to the investment-relevant knowledge that familiarity may provide as local expertise. Morningstar biographies report managers' post-secondary institutions. We map them to countries by hand using reported degrees and attendance, excluding professional certificates and nondegree executive programs.[5]

Dated fund spells link these education records to the managers serving together at each quarter-end and identify appointments and departures. Education is known for 85.5% of manager-fund-quarters in the main fund-return sample. A manager is foreign-educated if they have recorded post-secondary study outside the United States and U.S.-educated if their recorded study is exclusively in the United States. Our main variable, ForeignShare_{jq}, is the fraction of fund j's managers at quarter-end q who are foreign-educated, among those with known education. The fraction with unknown education, UnknownShare_{jq}, enters separately.[6] Coefficients compare an entirely foreign-educated team with an entirely U.S.-educated team. The estimated difference for a ten-percentage-point change is one tenth as large. Appointment and career tests use the individual indicator $\mathbb{1}_i^{\text{Foreign}}$, which equals one if manager i is foreign-educated.

For a given team, education countries are the non-U.S. countries in which at least one current manager studied, and we refer to holdings there as local. All other foreign markets are outside countries. In the country panel, $\mathbb{1}_{jcq}^{\text{Link}} = 1$ if a current manager studied in country c. Because a team can include foreign-educated managers without a link to a particular country, $\mathbb{1}^{\text{Link}}$ identifies education in that market, whereas ForeignShare measures foreign education anywhere. We measure how closely a market is related to an education country by shared borders and common official languages from CEPII GeoDist (Mayer and Zignago, 2011) and by social connectedness, measured with the Facebook Social Connectedness Index

[5] A manager who studied in several countries is linked to each. Conflicting profiles are coded as unknown education.

[6] For a team with two foreign-educated, one U.S.-educated and one unknown-education manager, the foreign-educated share is 2/3 and the unknown-education share is 1/4.

(Bailey et al., 2018). For U.S.-educated managers, these ties are measured relative to the United States.

Except where noted, the regressions control for five observed team traits, covering gender, CFA charters, the availability of biographies and fund-management experience.[7]

2.4 Performance measures

Our first measure is the net return on the fund's net asset value (NAV), which captures the outcome for investors across the entire portfolio, including country allocation, stock selection, cash, trading and costs (Kacperczyk et al., 2008). To compare funds with different mandates and styles, we deduct the return explained by predetermined exposures to thirteen factors, namely the five factors and momentum for developed ex-U.S. markets and for emerging markets (Fama and French, 2015), plus the U.S. market. For fund j in month m,

$$r^a_{jm} = R^{\text{net}}_{jm} - R^f_m - \hat{\beta}'_{j,m}\mathbf{f}_m. \tag{1}$$

Here R^{net}_{jm} is the net return, R^f_m the one-month Treasury-bill return, $\mathbf{f}_m$ the factor returns and $\hat{\beta}_{j,m}$ the fund's loadings estimated over the preceding 60 months. Using only prior returns allows exposures to change over time without using the evaluated return to estimate them.[8] We express this factor-adjusted return in annualized basis points, $R^a_{jm} = 12 \times 10^4 r^a_{jm}$.

Our second measure evaluates stock selection within a country. A fund can benefit from a strong national market without picking better companies there, so comparing its holdings with similar local stocks separates the choice of stocks from the choice of countries (Brinson

[7]The traits are the female share among managers with known gender, the CFA share among managers with biographies, the fractions with unknown gender and no biography, and the team average of log one plus years since each manager's first recorded fund spell. Appointment tests also include indicators for degrees from top-100 and 101–500 institutions in the 2003 Academic Ranking of World Universities, so ranking-controlled appointment comparisons begin in 2004.

[8]Loadings require at least 36 observed returns. Without that history, we use loadings estimated on other funds with the same benchmark over the same window, excluding the focal fund, with broader peer groups where needed. Peer loadings are used in 32,584 of the 206,847 main-sample months. Only the factor component of the loading regression is deducted, not its intercept.

and Fachler, 1985; Coval and Moskowitz, 2001). For each country c in which fund j holds stocks at quarter-end q, we compare the return on the fund's holdings in that country over the next three months with the return on a matched portfolio of similar local stocks, following characteristic-adjusted performance evaluation (Daniel et al., 1997; Wermers, 2000). Each stock k the fund holds receives a benchmark return. Within the country and quarter, stocks are sorted into terciles by market capitalization at the quarter-end and, within each size tercile, into terciles by their cumulative return over the eleven months before the quarter-end month. The benchmark return $\tilde{r}_{km}$ in month m is the capitalization-weighted return of the other stocks in the same size and past-return group, excluding stock k itself.[9] In each of the three months m that follow quarter-end q, the return on the fund's holdings in country c in excess of these benchmarks is

$$r^{\text{adj}}_{jcm} = \frac{\sum_k \omega_{jkm} \left(r_{km} - \tilde{r}_{km}\right)}{\sum_k \omega_{jkm}}, \tag{2}$$

where r_{km} is stock k's return in month m, and ω_{jkm} is stock k's weight in the fund's portfolio at quarter-end q, grown with the stock's returns from the quarter-end through month $m-1$. The sums run over the fund's stocks in country c that have a return in month m. We refer to the fund's holdings in a country as its country basket. The fund's adjusted country return, R_{jcq}, is the average of r^{adj}_{jcm} over the three months, annualized and expressed in basis points per dollar of the basket, which we call basket basis points. A positive value means that the fund's holdings in country c beat similar local stocks. Holdings and teams observed at quarter-end q are thus paired with returns over the following quarter, the payoff quarter, and both are updated each quarter. The same measure underlies the prior-performance tests in Section 4.

[9] The comparison stocks are all stocks held by sample funds in that country and quarter, and their capitalization weights drift with returns within the quarter. When fewer than five other stocks in the group have a return that month, the benchmark uses the size tercile alone and, failing that, the country's market return. About three quarters of holdings capital receives the full size and past-return match.

NAV coefficients describe performance per dollar of fund capital, whereas country-return coefficients describe performance per dollar invested in a particular market. The first measure shows how foreign-educated teams perform across the whole mandate, and the second shows in which markets local expertise helps. Because the regressions use different comparisons and weights, their coefficients are not components of the NAV coefficient. Appendix Table A19 summarizes the outcomes, units and timing.

2.5 Sample and summary statistics

All analyses in the main text start from the same 1,663 funds with manager records, an assigned benchmark and valid country weights, and each test uses the observations with the returns, holdings and manager information it requires. The main NAV sample has 206,847 fund-months at 1,418 funds with at least 24 months of eligible return history. Including all 1,536 funds with usable returns gives a similar coefficient. The joint country-selection panel has 974,924 fund-country-quarters at 1,428 funds, including 38,806 in education countries, and the outside-country panel used in Section 3.5 has 936,076 at 1,423 funds.

Table 1 reports summary statistics, including the fund, team and country counts described above. Annual expenses average 1.22%. The sample funds' combined net assets averaged $1.3 trillion a year, rising from $497 billion in 2000 to $2.2 trillion in 2024 and peaking at $2.5 trillion in 2021.[10] The mean foreign-educated share is 0.35, and about half of fund-quarters have no foreign-educated manager. Panel E reports manager and country counts for fund-quarters with at least one foreign-educated manager. Among the 2,992 managers with recorded education, 26.9% studied only outside the United States, and the United Kingdom is the most common foreign education country. Figure 1 shows that the

[10] Combined assets sum the fund-year net assets of Panel B across funds, including all of each fund's share classes. Assets are reported for 567 funds in 2000 and 869 in 2024.

foreign-educated share of managers with recorded education rose from 31.9% in 2000–2004 to 40.7% in 2020–2024.[11]

Panel standard errors are clustered by fund, allowing for dependence over time and across a fund's countries (Petersen, 2009; Cameron and Miller, 2015). Unless stated otherwise, fund-months and fund-quarters receive equal weight, and precision-weighted estimates are supplementary checks (Solon et al., 2015). Portfolio alphas use monthly time series.[12]

3 Investment Performance and Local Expertise

This section asks how far local expertise reaches. We begin with NAV returns to assess performance across the whole fund, including any local gains. We then examine allocation and returns in managers' education countries, where direct experience should be most useful. Finally, we ask whether familiarity also helps stock selection in markets outside those countries.

3.1 Fund returns and expenses

We first ask whether funds with more foreign-educated managers perform better across the entire portfolio, after accounting for market and style exposures (Carhart, 1997; Fama and French, 2010). The outcome R^a_{jm} is the factor-adjusted net return of Equation (1), in annualized basis points. We follow the same fund as its team changes and compare it with funds sharing its benchmark in the same month:

$$R^a_{jm} = \alpha_j + \lambda_{bm} + \rho \text{ForeignShare}_{j,q-1} + \Gamma' X_{j,q-1} + u_{jm}. \tag{3}$$

[11] Each manager counts once per quarter, including managers who run several funds. Managers without recorded education are excluded rather than classified as U.S.-educated. Because the figure covers all active sample funds, its shares differ from the NAV-sample statistics in Table 1.

[12] Portfolio regressions use a monthly risk-free rate converted from the Treasury-bill yield, conventional standard errors for individual alphas and robust Wald tests for differences between alphas.

Here q is the quarter containing month m, so the team at the preceding quarter-end determines ForeignShare for the three months of quarter q and team composition is measured before the evaluated returns. Fund effects α_j account for persistent differences across funds, and benchmark-month effects λ_{bm}, where b is the fund's assigned benchmark, compare funds with the same benchmark in the same month. The coefficient ρ is the annualized return difference associated with a zero-to-one change in the foreign-educated team share.

Staffing can respond to the fund's recent performance and business conditions. We control for those conditions using prior-year benchmark-relative returns and volatility, preceding-quarter flows and lagged assets. The remaining controls capture team scale and observed qualifications, expenses, turnover and top-ten holdings concentration.[13] Because expenses, turnover and concentration may themselves reflect earlier staffing decisions, we also report the model without these operating controls. Fund-months receive equal weight, with standard errors clustered by economic fund.

Funds earn lower factor-adjusted net returns when a larger share of their managers is foreign-educated. Table 2, column 1, estimates −69.5 annual basis points for the full zero-to-one difference in the team share, with a standard error of 35.4. At the sample mean share of 0.35, the implied gap relative to a team with no foreign-educated managers, holding other characteristics fixed, is approximately 24 basis points a year. For scale, the full-share coefficient exceeds half the sample's average annual expense ratio of 1.22%. The alternative samples and controls in Table 2 and Appendix Table A6 give estimates between −66 and −70 basis points, and weighting funds by the precision of their returns gives −70 with $p = 0.002$. We treat the equally weighted model as the primary estimate. Because NAV includes education-country investments, the estimate already incorporates any local gains.

[13] X contains unknown education, log team size and log lagged assets with spline terms, prior-year relative performance and volatility, preceding-quarter flows, prior-year expenses and turnover, prior-quarter top-ten concentration, and the five team traits defined in Section 2. Prior-year returns require nine observed months and use benchmark peers excluding every share class of the focal fund. Volatility is log one plus annualized relative volatility. Flows enter directly in percentage points of assets at the beginning of the preceding quarter, with a missing-flow indicator. Expenses are annual basis points, turnover is log one plus the reported percentage divided by 100, and concentration is the observed top-ten equity weight. Missing-value indicators retain unavailable controls.

To ask whether the shortfall arises before or after expenses, we replace net returns with reported gross returns in Equation (3), keeping exposures, observations, controls, effects and weights fixed. On 205,773 common fund-months, both coefficients are approximately -71 annual basis points (Table 3). Their paired difference, $\rho_{\text{gross}} - \rho_{\text{net}}$, is -0.47 basis points, with a 95% interval of $[-2.45, 1.51]$. This interval is narrow relative to the return gap, and the comparison without operating controls gives the same conclusion. The shortfall arises in gross returns, not in the expenses deducted from them.

The appendix examines three other explanations for this gap. First, a poor match between education countries and mandates does not explain it. Returns are weaker, not stronger, where managers' education countries carried more weight in the fund's benchmark before appointment, holding the foreign-educated share and the mix of education countries fixed (Appendix B.1). Second, reverse timing does not explain it. Country losses do predict the hiring of managers educated in those countries (Section 4.1), but adding the team's composition four quarters ahead leaves the current-share coefficient essentially unchanged (Appendix B.3). Third, disruption from new appointments does not explain it. The gap is also present among teams with foreign-educated managers who have served at the fund for more than two years (Appendix B.4).

3.2 Allocation to education countries

We turn next to holdings, asking first whether funds put more capital in a country when one of their managers studied there. Because a large national market receives a large weight mechanically, we measure each country's weight relative to the fund's assigned benchmark. For fund j, country c and month m, we estimate

$$\text{Overweight}_{jcm} = \alpha_j + \lambda_{cm} + \psi \mathbb{1}^{\text{Link}}_{jcm} + \Gamma' X_{jm} + e_{jcm}. \tag{4}$$

Here c ranges over non-U.S. countries, Overweight_{jcm} is the percentage of the fund's portfolio invested in country c minus that country's weight in the fund's assigned benchmark, in percentage points, and $\mathbb{1}^{\text{Link}}_{jcm}$ equals one if a current manager studied in c. The coefficient ψ measures how much more a fund overweights a country when one of its current managers studied there. A second specification replaces the indicator with LinkedShare_{jcm}, the fraction of the fund's current managers, including those with unknown education, who studied there, so that ψ compares an entirely linked team with an unlinked one. Country-month effects remove what all funds do in a country in a given month. We first include fund effects α_j and then replace them with fund-month effects α_{jm}. The latter compare countries within the same portfolio in the same month, so ψ then measures how a fund distributes its capital between linked and unlinked countries. Controls X contain the five team traits and are absorbed by fund-month effects.[14] The allocation coefficient describes where funds place capital, and the following return tests assess the stock choices made there.

Both measures show an active education-country tilt (Table 4). The indicator coefficients are 1.11 percentage points with fund effects and 1.09 with fund-month effects. The linked-share coefficients are 2.63 and 2.59, and all four estimates are significant at 1%. The linked-share coefficients compare a team with no linked manager with an entirely linked team, so a ten-percentage-point increase in the linked share corresponds to about 0.26 percentage points more country overweight. The similar within-fund-month results show that the education-country tilt persists when comparing countries within the same portfolio and month.

Figure 2 traces the overweight around changes in the team. An event is the arrival of the first manager educated in country c, or the departure of the last such manager. With $\text{Overweight}_{jcq} = w_{jcq} - w^b_{jcq}$, where w_{jcq} and w^b_{jcq} are the fund's and its benchmark's

[14] The equally weighted panel starts with all 1,663 funds and contains approximately 5.6 million fund-country-months from January 1999 to December 2024. Because these tests use only holdings and benchmark weights, not subsequent returns, they also include the 1999 holdings that precede the return sample. The fund-month models retain 1,660 economic funds after singleton fixed-effect groups are dropped. A missing-trait indicator retains teams without biographies, and the team record assigned to each month defines $\mathbb{1}^{\text{Link}}$ and X. Errors are clustered by economic fund.

percentage weights in country c at quarter-end q, we estimate separately for arrivals and departures

$$\text{Overweight}_{jcq} = \alpha_{jc} + \lambda_{cq} + \mu_{jq} + \sum_{\ell \neq -1} \psi_\ell \, \mathbb{1}\{q - q^*_{jc} = \ell\} + e_{jcq}. \tag{5}$$

Here q^*_{jc} is the quarter of the event for fund j and country c, so $\mathbb{1}\{q - q^*_{jc} = \ell\}$ equals one ℓ quarters after the event, for ℓ from -6 to 12, and each ψ_ℓ measures the change in the country's overweight relative to the quarter before the event, $\ell = -1$, which is omitted. Fund-country effects remove the fund's usual weight in that country, country-quarter effects remove changes common to all funds investing there, and fund-quarter effects remove changes in the fund's overall portfolio.[15] Weights rise about 0.5–0.7 percentage points after arrivals and fall 0.2–0.55 after departures, without a clear movement beforehand. Capital follows the team's education countries as its membership changes.

3.3 Stock selection in education countries

Funds allocate more capital to managers' education countries. We now test whether foreign-educated teams also select stocks better there, on average, than U.S.-educated teams investing in the same markets. This test covers the entire country basket, and the next subsection examines high-conviction positions separately. We use the adjusted country return, R_{jcq}, defined in Section 2.4, in a joint panel of education and outside countries:

$$R_{jcq} = \alpha_{jc} + \lambda_{cq} + \beta \mathbb{1}^{\text{Link}}_{jcq} + \gamma \text{ForeignShare}_{jq} + \Gamma' X_{jcq} + \varepsilon_{jcq}. \tag{6}$$

Here $\mathbb{1}^{\text{Link}}_{jcq}$ equals one if a current manager studied in country c, and ForeignShare_{jq} is the team's foreign-educated share. Two coefficients describe distinct comparisons. The coefficient γ is the difference between an entirely foreign-educated team and an entirely

[15] The 977 arrivals and 793 departures use up to twelve-quarter windows, a six-quarter clean pre-period, no overlapping events and truncation if the link reverses.

U.S.-educated team in markets where no current manager studied. The coefficient β is the additional difference when a current manager studied in the market. The comparison of interest, a fully foreign-educated team investing in its education country versus a U.S.-educated team investing in the same country, is therefore $\beta+\gamma$. Fund-country effects compare the same fund in the same market over time, and country-quarter effects compare funds investing in the same market and quarter, so β is identified when a link to a market begins or ends within a fund. Controls X contain log country stock count, log team size, unknown-education share and the five team traits. Countries have equal weight within each fund-quarter, and standard errors are clustered by fund.

The joint panel contains 974,924 fund-country-quarters at 1,428 funds, including 38,806 in education countries, and links begin or end in 1,037 fund-country pairs at 525 funds.[16]

In column 1 of Table 5, $\beta + \gamma$ is +0.21 annualized basket basis points (standard error 56.61), so in their education countries foreign-educated teams show no clear average advantage over U.S.-educated teams investing in the same markets. Neither coefficient is statistically significant on its own, and their sum measures the education-country comparison of interest. The precision-weighted comparison in column 2 is similar (+5.57 basis points, standard error 51.39). This is an average across all holdings in those countries, distinct from whether a fund improves after appointing a manager educated in the country, which we examine in Section 4.2.

3.4 The concentrated value of local expertise

An average across all holdings in a country can conceal an advantage concentrated in a few stocks. We therefore compare the positions managers overweight most, which we call high-conviction positions, with their ordinary holdings under a common country mix, which isolates the choice of stocks, and then ask how much these positions add to the funds' returns.

[16]The identifiers represent 1,429 oldest share classes. Formation quarters run from 1999Q4 through 2024Q3, with payoffs from January 2000 through December 2024. Country weights are renormalized over all foreign countries with usable returns within each fund-quarter.

We ask whether a fund's high-conviction positions within an education country outperform its other holdings there. We identify those positions before observing their returns using a country-normalized version of the I-measure of Antón et al. (2021):

$$I_{jkq} = \frac{\omega_{jkq}}{\sum_{k' \in \Omega_{jcq}} \omega_{jk'q}} - \frac{\mathrm{MC}_{kq}}{\sum_{k' \in \Omega_{jcq}} \mathrm{MC}_{k'q}}. \tag{7}$$

Here k is a stock that fund j holds in country c at quarter-end q, Ω_{jcq} is the set of such stocks and k' runs over it, ω_{jkq} is the stock's weight in the fund's portfolio, as in Equation (2), and MC_{kq} is its market capitalization at quarter-end q. The measure subtracts the stock's capitalization share among those holdings from its weight in the fund's country basket. A positive value identifies an active overweight among the fund's holdings in that country. Normalizing within countries prevents a large country allocation from mechanically making every stock there a high-conviction position. High conviction here means a large active weight relative to the fund's other holdings in the country.

Within each fund-country-quarter, the top I-measure quartile defines high conviction and the other three quartiles define ordinary holdings. We form calendar-time portfolios for each group, separately in education and outside countries. Each portfolio pools the group's positions across all funds. Within a country, positions are weighted by their dollar value, and countries are weighted by their shares in the Morningstar Global ex-US index. Reweighting to these benchmark country shares removes differences in aggregate national-market weights from the group comparisons. Each portfolio's alpha comes from the calendar-time regression

$$R_{pm} - R^f_m = \alpha_p + \beta_p' \mathbf{f}^{(10)}_m + \beta^G_p (R^G_m - R^f_m) + \varepsilon_{pm}. \tag{8}$$

Each regression uses the 300 months from January 2000 to December 2024. The portfolio index p identifies the country group and weighting rule, and all local and outside conviction portfolios use this same model. Here R_{pm} is the constructed portfolio's return, $\mathbf{f}^{(10)}_m$ contains the developed- and emerging-market five factors, and R^G_m is the return on the Morningstar

Global ex-US index, with loading β_p^G. All returns are monthly decimals. This common portfolio benchmark differs from the fund-specific assigned benchmarks in the NAV model. The intercept α_p is monthly excess performance beyond these exposures, which we report annualized. Direct alpha differences use stacked portfolio series and robust Wald tests on intercept equality. The intercept measures the performance of a strategy that holds the disclosed positions, not the funds' realized returns.

High-conviction local positions have a substantial alpha advantage. The top education-country quartile earns a 4.0% annual alpha (standard error 1.3), compared with 0.4% for ordinary local holdings (Table 6). Their direct 3.6-percentage-point difference is significant at 1%.

Comparing high-conviction positions across country groups shows whether they are especially valuable in education countries. High-conviction positions in education countries outperform those outside them by 3.9 percentage points a year ($p = 0.002$). Together with the 3.6-point difference from ordinary local holdings, this places the advantage in high-conviction positions in education countries. Figure 3 plots the four alphas, their confidence intervals and the direct tests. Appendix B.5 reports eight alternative constructions, including two alternative conviction measures, the exclusion of U.K. equities, undergraduate-only education links, other risk models and returns net of trading costs, and the top-versus-ordinary difference remains significant at 5% in six of them (Appendix Table A9).

Figure 4 shows the cumulative gross returns of the same strategies over 2000–2024, alongside the outside-country comparison.

How much do these positions add to the funds' returns? They account for 3.6% of the sample funds' observed equity holdings. Even if every dollar in them earned the 4.0% annual alpha, they would add only about 14 basis points a year to portfolio returns ($0.036 \times 4.0\%$). The estimated contribution is smaller still and close to zero. The 4.0% alpha weights each country by its benchmark share. Weighted instead by the capital that funds actually hold in each country, the same positions earn an imprecisely estimated alpha of about 0.2% a year

(Appendix B.6). The local advantage thus lies in a few high-conviction positions and adds little to the portfolio as a whole.

3.5 Stock selection and familiarity outside education countries

Does local expertise reach beyond education countries? We examine markets closely connected to them, again using characteristic-adjusted country returns.[17] If information acquired in one country extends to nearby or closely connected markets, foreign-educated teams should select stocks better in those markets. If familiarity instead creates confidence beyond the reach of useful information, selection can be worse. We compare these patterns for foreign- and U.S.-educated teams, using only countries where no current manager studied. The key comparison is how stock selection changes with familiarity for each education group.

We measure how closely a market is tied to where a manager studied in three ways, each in a separate regression, using a shared border, a common official language and the log of Facebook social connectedness from the October 2021 snapshot. For foreign-educated managers, the tie is measured between the market and the manager's non-U.S. education countries, using the closest tie when there are several. For U.S.-educated managers, it is measured between the market and the United States. Both groups therefore face the same question of whether stock selection is better in markets close to where the manager studied.

Ties are defined for managers, but returns are observed for the team, so we average the ties over the team. Let z_{ic} measure how closely market c is tied to where manager i studied, using a zero-one indicator for border and language ties or log social connectedness. For fund j at quarter-end q, with N_{jq} managers of known education, we form two team averages:

$$\text{Tie}_{jcq} = \frac{1}{N_{jq}} \sum_i z_{ic}, \qquad \text{ForeignTie}_{jcq} = \frac{1}{N_{jq}} \sum_i \mathbb{1}_i^{\text{Foreign}} z_{ic},$$

[17] Appendix B.7 shows that the outside-country gap between foreign- and U.S.-educated teams widens over the sample period.

where both sums run over the team's managers with known education and $\mathbb{1}_i^{\text{Foreign}}$ equals one if manager i is foreign-educated. Tie averages every manager's tie, whereas ForeignTie averages the same ties but counts U.S.-educated managers as zero. For example, consider a team with one manager educated in Germany and one educated only in the United States. For its holdings in Austria, which borders Germany but not the United States, Tie = ForeignTie = 1/2. For its holdings in Canada, which borders the United States but not Germany, Tie = 1/2 and ForeignTie = 0. ForeignTie is therefore an average of manager-level interactions, not the product of the team's foreign share and its average tie. For each tie measure, we regress the fund's adjusted country return in the market, R_{jcq}, on these team averages:

$$\begin{aligned} R_{jcq} = \alpha_{jc} + \lambda_{cq} + \gamma \text{ForeignShare}_{jq} + \delta \text{Tie}_{jcq} \\ + \theta \text{ForeignTie}_{jcq} + \Gamma' X_{jcq} + \varepsilon_{jcq}. \end{aligned} \tag{9}$$

For an entirely U.S.-educated team, only Tie varies with familiarity, so δ describes how stock selection changes with closeness to the United States. For an entirely foreign-educated team, Tie and ForeignTie vary together, so $\delta + \theta$ describes how stock selection changes with closeness to the team's education countries. The focal coefficient θ is the difference between the two slopes. It is negative if closeness is associated with less favorable selection for foreign-educated managers than for U.S.-educated managers. Finally, γ is the foreign-versus-U.S. difference in markets with no tie to either group's education.[18] For the border and language measures, a unit change compares an entirely tied team with an entirely untied team, and for social connectedness, slopes are per log point.

Fund-country effects α_{jc} compare the same fund in the same market over time, and country-quarter effects λ_{cq} remove returns common to all funds investing in that market in that quarter. Because each manager's tie to a market is fixed, the slopes are identified from

[18] For social connectedness, γ refers to zero log connectedness, outside the observed range, so we interpret only the slopes in that model.

changes in a team's composition within a fund and market, measured relative to other funds investing in the same market that quarter. The sample contains only markets outside the team's current education countries. Controls X contain $\text{MissingShare}_{jcq}$, log country stock count, log team size, unknown-education share and the five team traits.[19] Countries receive equal weight within each fund-quarter, and standard errors are clustered by fund.

Table 7, column 1, reports the border results. A shared border is worth far less to foreign-educated teams than to U.S.-educated teams. The difference between their slopes, $\theta = -484$ basis points (standard error 158), is significant at 1%, so a border with the team's education country is associated with selection 4.84 percentage points a year less favorable for an entirely foreign-educated team than for an entirely U.S.-educated team. Foreign-educated teams select stocks worse in markets bordering an education country than in other outside markets, with a slope $\delta + \theta$ of -184 basis points (standard error 67, significant at 1%).[20] U.S.-educated teams select stocks better in markets bordering the United States, with a slope δ of +300 basis points (standard error 135, significant at 5%).

The other two measures give the same relative result. With social connectedness (column 3), the difference from U.S.-educated managers is $\theta = -110$ basis points per log point (standard error 42, significant at 1%), or 117 basis points a year for a one-standard-deviation increase in log connectedness (about 1.06 log points). Foreign-educated teams' own slope is -43 basis points per log point (standard error 17, significant at 5%). With a common official language (column 2), the difference is $\theta = -231$ basis points (standard error 67, significant at 1%). Here the gap reflects mainly what U.S.-educated teams gain. They select stocks 155 basis points better in markets that share an official language with the United States (standard error 75, significant at 5%), while the foreign-educated slope of -75 basis points is negative but not significant.

[19] Unavailable pair measures contribute zero to the tie averages, and $\text{MissingShare}_{jcq}$ records the share of managers whose tie cannot be classified. These are U.S.-educated managers with unclassified geographic ties in the border and language models and foreign-educated managers with a missing social-connectedness pair in the social model.

[20] Standard errors of $\delta + \theta$ use the joint covariance of the two coefficients from the same regression.

A similar pattern appears when we drop the size and past-return adjustment and compare raw country returns among funds holding the same country in the same quarter (Appendix B.8).

Across all three measures, familiarity is worth less to foreign-educated teams than to U.S.-educated teams, by 484 basis points a year for a shared border, 231 for a common official language and 117 for a one-standard-deviation increase in social connectedness. For borders and social connectedness, foreign-educated teams' own selection also worsens with familiarity, consistent with familiarity bias. Together with the high-conviction local gains and the whole-fund shortfall, these results show that the value of local expertise is real but narrow.

4 Recruitment and Career Evaluation

We now ask whether funds recognize the limits of local expertise when they select and evaluate managers, a question that matters because foreign-educated managers have become more common (Figure 1). A concentrated local advantage can justify seeking country expertise. The question is whether funds hold foreign-educated managers to the same performance standards as U.S.-educated managers when they hire and evaluate them. Sections 4.1 and 4.2 examine when funds seek local expertise and whether the recovery that follows reflects the new manager's choices. Section 4.3 compares the prior investment performance, fundraising and country-specific performance that recruits bring. Section 4.4 relates later fund performance to retention and new assignments, and Section 4.5 asks whether a commercial benefit could explain these choices.

4.1 Country losses and hiring

A fund may seek a manager educated in a country when investing there becomes difficult. We test whether disappointing country returns predict these appointments, using countries where the fund held stocks at the preceding quarter-end.

An observation is a fund j, country c and calendar quarter q, including quarters with no appointment. The outcome equals one when a manager who studied in c joins fund j during q, which we call a linked appointment. We express appointment probabilities in basis points by multiplying the zero-one indicator by 10,000, so one probability basis point equals 0.01 percentage point. We distinguish two prior-year signals. The market signal L^M_{cq} is minus the compounded country-market return over the preceding twelve months, and the fund signal L^P_{jcq} is minus the fund's average country-market-adjusted holdings return over the preceding four quarters. Both are expressed in percentage points, so a larger value means worse performance.[21] We estimate

$$Y^g_{jcq} = \alpha_{jc} + \lambda_{jq} + \beta^g_M L^M_{cq} + \beta^g_P L^P_{jcq} + \Gamma'_g X_{jcq} + \varepsilon^g_{jcq}. \tag{10}$$

Here g denotes the direct education link or, in a separate comparison, education in a bordering country without education in c itself. Fund-country effects remove persistent country preferences, while fund-quarter effects compare countries within the same fund and calendar quarter. Controls X include the preceding country weight and stock count in logs and the incumbent team's direct and bordering-country education shares. Fund-quarter effects also absorb team size and other common fund conditions. We weight countries equally within each fund-quarter and cluster by fund. Country-quarter effects would absorb the country-market signal and are not included.

[21] The market proxy weights the covered stocks by lagged market capitalization, with monthly stock returns winsorized at the 0.1st and 99.9th percentiles. The fund-specific signal averages annualized country-market-adjusted basket returns, requiring at least three of the preceding four payoff quarters. It differs from the size-and-momentum-adjusted country return used in the stock-picking tests. All predictors precede the appointment quarter, so no appointment-quarter return enters.

Both signals predict direct-country hiring (Table 8, column 1). A ten-percentage-point lower prior market return raises the quarterly probability of a linked appointment by 0.43 probability basis points, about 3.7% of its average of 11.80 (0.118%). A ten-point deterioration in the fund's own country-relative performance raises this probability by 0.48 basis points, about 4.1%, holding the market signal constant. Both coefficients are significant at 5%. The comparison uses 978,521 fund-country-quarters at 1,540 funds, with 1,027 linked country appointments. Demand for an education link rises after disappointing performance in the relevant market. Column 2 examines whether the same losses predict hiring someone educated in a bordering country instead. Neither return signal is statistically significant there. The market-loss coefficient turns slightly negative, and the own-underperformance coefficient is similar in size to column 1 but imprecise. The clearest evidence of demand is for expertise in the country itself, and the next subsection asks whether these appointments improve performance there.

4.2 Where the recovery comes from

After a linked appointment, does country performance improve relative to comparable non-hiring funds, and does improvement extend to the rest of the portfolio? We compare each linked appointment after disappointing performance with up to three funds holding the same country, under the same benchmark and in the same quarter, with similar prior country performance, allocation, stock count and existing managers' education links, but no appointment that quarter. We follow both groups for four quarters before and four quarters after appointment, omitting the appointment quarter. We choose the matches before looking at subsequent returns.

To make the local, outside-country and whole-fund comparisons consistent, we use the same observations, controls and fixed effects for all three outcomes. Each matched case

consists of a country-linked appointment and its non-hiring comparison funds. Let a index the case, j a fund in it and t the calendar quarter in which returns are earned. We estimate

$$Y_{jat} = \alpha_{ja} + \lambda_{at} + \beta_H \mathbb{1}_{ja}^{\text{Hire}} \times \mathbb{1}_{at}^{\text{Post}} + \Gamma' X_{jt} + \varepsilon_{jat}. \tag{11}$$

Here Y_{jat} is the adjusted return on the focal country's holdings, the adjusted return on holdings outside the hiring team's education countries, or factor-adjusted NAV, each annualized in basis points. The first two outcomes are returns per dollar invested in the respective basket, whereas NAV is a whole-fund return. The education-country set is fixed at appointment and applied to both hiring and comparison funds throughout the window. Fund-case effects α_{ja} remove persistent differences between the funds, and case-quarter effects λ_{at} remove common changes within the comparison. The indicators identify the hiring fund and the post-appointment quarters. Thus β_H measures the hiring fund's change relative to the change at its comparison funds. Controls X_{jt} are those of the NAV model in Equation (3), including team size, and errors are clustered by economic fund. Appendix B.9 describes the matching procedure, weights and sample requirements.

Local performance improves after appointment relative to the comparison funds. On 511 country appointments at 267 hiring funds, adjusted returns in the focal country improve by 228 annualized basket basis points relative to non-hiring funds (standard error 80), significant at 1% (Table 9). For the same hiring and comparison funds, the estimates are −15 basis points for outside-country holdings (standard error 34) and +37 for whole-fund NAV (standard error 37). Neither is precisely estimated. The improvement is local, with no clear gain in whole-fund NAV, and its importance for the fund depends on the capital invested in that country.

Does this local improvement persist? We extend the comparison through the second year after appointment, retaining the same funds for the pre-appointment year and both subsequent years. In Equation (11), we replace the single hiring-by-post interaction with

separate interactions for the first and second years, with the pre-appointment year as the baseline. Fixed effects, controls and fund clustering are unchanged. The common cohort contains 430 country appointments at 231 hiring funds, with 514 fund clusters. The first-year local improvement is 195 annualized basket basis points (standard error 78), compared with −11 in the second year (standard error 84). The direct second-minus-first-year difference is −205 basis points (standard error 97), significant at 5% (Table 10). The 228-basis-point estimate above uses the larger one-year cohort. The corresponding NAV estimates show no clear change between years. The local recovery appears temporary, and the next test asks what produces it.

A recovery after losses can occur without better stock choices by the incoming manager. To separate recovery in existing holdings from the contribution of portfolio changes, we construct a passive portfolio from the fund's country holdings at the quarter-end before appointment, retain those stocks and let their weights drift with returns. We compare its characteristic-adjusted return with the return on the fund's observed holdings, which are updated each quarter. Both portfolios use the same stock benchmarks. The difference captures the contribution of changes in holdings through purchases, sales and reweighting of retained stocks. We construct the same decomposition before appointment and for the non-hiring funds, and estimate the same two-year specification for all three outcomes. The question is whether the hiring funds' improvement exceeds the recovery already present in their inherited portfolios.

Table 11 decomposes the relative local recovery on the 430 cases of Table 10, excluding the 258 quarters in which the two portfolio returns are not both observed. The first-year coefficient is 178.45 basis points for observed holdings, 174.37 for the passive pre-hire portfolio and 4.08 for changes in holdings. The coefficients add exactly. At the point estimates, inherited-portfolio recovery accounts for 97.7% of the improvement and portfolio changes for 2.3%. The passive-portfolio coefficient is significant at 5%, while the portfolio-change

estimate has a standard error of 64.29 basis points. Nearly all of the estimated first-year improvement is present without replacing the pre-hire stocks.[22]

In the second year, the observed-holdings coefficient falls to −16.03 basis points, and Table 11 reports both components of this decline. The decomposition thus places the first-year recovery in the inherited holdings rather than in portfolio changes after the hire. Hiring after local losses reveals demand for local expertise, but the recovery itself does not demonstrate that the recruit supplied new stock-picking value.

4.3 Recruits' prior performance

The performance that funds observe before an appointment offers a more direct view of what recruits bring. What investment performance do funds accept from foreign-educated recruits? We compare foreign- and U.S.-educated recruits appointed by the same fund, first on their previous teams' investment performance and then on their previous funds' flows. We refer to the funds a recruit previously managed as origin funds. The comparison is not limited to the linked appointments above. It covers all experienced appointees, with one observation for each manager's appointment to a fund. We measure the previous teams' average adjusted country return over the four full quarters before hiring, including both education countries and other non-U.S. countries. As in Section 2.4, these returns compare country holdings with similar local stocks. Including education countries allows any demonstrated local advantage to enter the recruit's overall international performance, and the last test in this subsection examines education-country performance separately.[23]

[22] The 95% interval for the incremental contribution of portfolio changes is $[-122, 130]$ annualized basket basis points. Appendix B.10 gives the construction, joint-covariance accounting and support checks.

[23] Countries have equal weight within each origin fund-quarter. We average quarters, combine share classes of each economic fund and then average across distinct origin funds. This characteristic-adjusted measure covers non-U.S. stocks and is not a whole-fund NAV return.

Let h identify manager i's appointment to fund j in year y. The prior average return, $\bar{R}_h^{\text{pre}}$, is expressed in annualized basis points per dollar of country holdings. We first set the outcome Y_h equal to this measure of prior investment performance:

$$Y_h = \alpha_j + \lambda_y + \delta \mathbb{1}_i^{\text{Foreign}} + \eta' \mathbf{B}_h + \Gamma' X_h + v_h. \tag{12}$$

The coefficient of interest is δ, on the indicator for a foreign-educated recruit. A negative δ means that foreign-educated recruits had weaker prior performance than U.S.-educated recruits accepted by the same fund, conditional on the controls. Hiring-fund effects α_j account for each fund's persistent recruitment standards, and appointment-year effects λ_y remove common recruitment conditions. The comparison uses hires across years within a fund. Because prior performance partly reflects the previous funds' mandates, $\mathbf{B}_h$ controls for the benchmarks of the recruit's previous funds.[24] Controls X_h include school-ranking categories, gender, CFA, experience, the size and unknown-education share of the origin and destination (hiring) teams, and the origin teams' country stock count. Appointments receive equal weight, and standard errors are clustered by hiring fund.

The comparison covers 1,321 appointments of experienced recruits at 300 hiring funds, all with observable previous-team performance.[25]

Foreign-educated recruits arrive with weaker prior performance. Table 12, column 1, estimates a difference of −86 annualized basket basis points (standard error 38), significant at 5%. Their previous teams' adjusted country returns are about 0.86 percentage points lower

[24] These are benchmark shares, weighted as in the outcome, and 551 appointments have more than one origin benchmark. Global ex-U.S. is the omitted category among 16. Prior NAV does not enter this equation, so the outcome is not conditioned on another realized performance measure.

[25] Of the appointments at the 1,663 funds, 1,600 have observable prior performance, and requiring the histories and the fund and year comparisons used here leaves 1,322 at 300 hiring funds. We use the 1,321 that also have a same-benchmark peer, so that this comparison and the peer-adjusted check in Appendix B.13 cover the same recruits. The outcome here uses all non-U.S. country baskets with usable returns for these appointments, including education countries, and does not require peer returns. The same-round and education-country tests use their own samples. Managers without a recorded prior fund role cannot enter this comparison.

per year than those of U.S.-educated recruits accepted by the same hiring fund, accounting for origin benchmarks and observed qualifications.

Two finer comparisons in Appendix B.13, which use performance outside education countries, give similar gaps. Measuring the previous teams' returns relative to other funds with the same benchmark, country and quarter gives −72 basis points. Comparing only recruits hired by the same fund in the same quarter gives −81. Both are significant at 10%.

Weaker prior investment performance could be acceptable if the recruit brings a stronger fundraising record. We test that possibility using flows to the previous funds before appointment. In column 2, we replace prior investment performance with the prior-flow record, $Y_h = \text{Flow}_h^{\text{pre}}$, in Equation (12). This outcome is the quarterly net flow into the recruits' previous funds, in percentage points of beginning-of-quarter assets, minus the mean flow of other funds with the same benchmark, averaged over the four quarters before the appointment quarter and across previous funds. Because flows follow returns, this regression controls for the previous funds' benchmark-relative NAV return, $R_h^{\text{NAV,pre}}$, in place of $\mathbf{B}_h$. It uses the 1,322 experienced appointments and otherwise follows Equation (12).[26]

The foreign-education coefficient is −1.8 percentage points per quarter, significant at 10%, so relative to U.S.-educated recruits, foreign-educated recruits come from funds with weaker fundraising records even after accounting for prior returns. Weaker prior investment performance is not offset by stronger fundraising records. An average over all foreign countries could still conceal an advantage in the recruit's education country. The last test examines prior performance in that potential specialty, where local expertise might compensate for weaker performance overall.

For this test, we identify countries where the hiring fund already holds stocks but none of its continuing managers studied. This set is defined before considering the recruit's edu-

[26]This signal is the mean quarterly relative NAV return in the payoff quarters from five to two quarters before the appointment quarter, in percentage points. It is observed before hiring and does not enter either the main or the peer-adjusted investment-performance equation.

cation. A recruit educated in one of these countries adds an education link the team lacks.[27] An observation is an appointment h paired with one such country c. The outcome $\bar{R}^{\text{pre}}_{hc}$ is the adjusted return the recruit's previous teams earned on holdings in c over the four full quarters before appointment:[28]

$$\bar{R}^{\text{pre}}_{hc} = \alpha_{jc} + \lambda_{cy} + \beta_G \mathbb{1}_i^{\text{Foreign}} + \beta_L \mathbb{1}_{ic}^{\text{OwnLink}} + \Gamma' X_h + u_{hc}. \tag{13}$$

Here $\mathbb{1}_{ic}^{\text{OwnLink}}$ equals one if the recruit studied in c, so $\beta_G + \beta_L$ compares recruits educated in the country with U.S.-educated recruits appointed by the same fund, on their previous teams' performance in the same market. Fund-country and country-year effects and the manager and team controls are included, and the previous funds' benchmark-relative NAV return replaces the benchmark shares, so that the comparison holds the previous funds' overall performance fixed. Each appointment's countries have total weight one, and standard errors are clustered by hiring fund.

Recruits educated in the country bring weaker prior performance there too. The gap $\beta_G + \beta_L$ is -274 annualized basket basis points (standard error 101), significant at 1% (Table 13), or about 2.7 percentage points a year on holdings in the market where the education link is relevant.[29] The gap does not depend on holding prior NAV fixed (Appendix B.13). Because the specialty gap measures performance in a single market, on a different sample and with different controls, it does not add to the -86-basis-point general gap. Stronger demonstrated performance in the potential specialty therefore does not explain the weaker general performance accepted at recruitment.

[27] Continuing managers are present before and at the end of the appointment quarter. Departures and simultaneous recruits do not supply incumbent coverage. Missing prior holdings remain unclassified.

[28] We average across share classes of the same previous fund in each quarter, then across quarters and distinct previous funds. Every origin fund must be observed in all four quarters.

[29] The sum combines $\beta_G = -55$ and $\beta_L = -219$, using their joint covariance. The comparison uses 9,446 country-appointment observations at 1,083 appointments and 257 funds, and linked observations cover 145 appointments and 85 funds. The 152 linked observations span 22 countries, including 72 in the United Kingdom. Starting with 1,131 appointments with usable pre-hire holdings and hiring-fund/year histories, prior-country availability supplies 11,117 observations at 1,105 appointments and 269 funds before singleton fixed-effect groups are dropped. Appointments span 2004Q1–2024Q4 and prior payoffs 2003Q1–2024Q3. No single fund accounts for more than about 5% of the identifying variation.

4.4 Careers and performance evaluation

The hiring comparisons show that foreign-educated recruits enter with weaker prior performance, including in their potential specialty. We next ask whether performance after appointment also matters less for their careers. If funds give foreign education more weight and realized performance less, foreign-educated managers should be less likely than their U.S.-educated colleagues on the same team to leave as fund returns worsen, and should gain less in new assignments as returns improve. We test departures and new assignments using the same return signal among all current managers. This comparison extends the question from recruitment to ongoing evaluation and is not restricted to the experienced recruits in Section 4.3.

Each observation is a manager i on fund j at quarter-end q. We study two outcomes over the following quarter. The indicator $\mathbb{1}_{ijq}^{\text{Exit}}$ equals one if manager i departs from fund j between quarter-ends q and $q+1$, and $\mathbb{1}_{ijq}^{\text{NewFund}}$ equals one if the manager receives a new assignment at another fund. We multiply both indicators by 100, so coefficients are in percentage points. The signal R_{jq}^{rel} is the fund's compounded net return over quarter q minus the mean return of other funds sharing its benchmark, also in percentage points. It uses reported whole-fund NAV, covering both U.S. and foreign investments. We estimate one common specification separately for departures and new assignments:

$$Y_{ijq} = \mu_{jq} + \nu_{ij} + \kappa \mathbb{1}_{i}^{\text{Foreign}} \times R_{jq}^{\text{rel}} + \Gamma' X_{ijq} + e_{ijq}. \tag{14}$$

Here Y_{ijq} is either $100 \times \mathbb{1}_{ijq}^{\text{Exit}}$ or $100 \times \mathbb{1}_{ijq}^{\text{NewFund}}$. The coefficient κ measures how a one-percentage-point improvement in fund returns changes the foreign-minus-U.S. difference in the outcome probability among colleagues on that fund. We denote it by κ_{Exit} for departures and κ_{NewFund} for new assignments. Fund-quarter effects μ_{jq} absorb the common return and fund characteristics shared by colleagues, including the benchmark, mandate and team size. Fund-manager effects ν_{ij} absorb each manager's education and persistent differences in their

career prospects at the fund. The coefficient is therefore identified from how colleagues' outcomes diverge as their shared fund return varies.

Controls X_{ijq} include log spell length, log one plus management experience, and interactions of R^{rel} with the female, unknown-gender, CFA and unknown-biography indicators and log experience. Each fund-quarter receives equal total weight, divided equally among its managers, and errors are clustered by economic fund.

Departures indicate whether an existing assignment continues, and new assignments indicate additional opportunities. A new assignment is a first appearance on another fund's team in the next quarter, including a new fund launch, and can occur without leaving the original fund.

The departure interaction is $\kappa_{\text{Exit}} = +0.145$ percentage points per return percentage point, significant at 1%, on 154,677 manager-quarters at 1,334 funds (Table 14, columns 1–2). A ten-point deterioration lowers the foreign-minus-U.S. departure gap by 1.45 percentage points, so foreign-educated managers become less likely to leave relative to their U.S.-educated teammates. For new assignments, $\kappa_{\text{NewFund}} = -0.107$, significant at 5%, so a ten-point improvement lowers the foreign-minus-U.S. assignment-probability gap by 1.07 percentage points. The paired signs describe relative protection as returns worsen and a smaller relative gain as returns improve. Both signs are consistent with career evaluations that give the qualification more weight and shared fund returns less.

Weaker sensitivity to whole-fund performance could reflect evaluation on the manager's country specialty. We therefore add the observed performance of education-country holdings and ask whether the foreign-minus-U.S. response to common returns becomes smaller.[30]

Columns 3–4 add this local performance to Equation (14). The local return R^{loc}_{ijq} is the return on the fund's holdings in a foreign-educated manager's education countries over the same quarter, less those countries' index returns, in percentage points. Unlike R_{jcq}, it adjusts only for the national market. We add it as $\chi \mathbb{1}^{\text{Observed}}_{ijq} \times R^{\text{loc}}_{ijq}$, where $\mathbb{1}^{\text{Observed}}_{ijq}$ equals one

[30] Appendix B.14 shows that the relative retention is not confined to managers whose country knowledge no colleague shares.

for foreign-educated managers whose local return is observed and zero otherwise, including for all U.S.-educated managers, together with the indicator itself, $\phi \mathbb{1}_{ijq}^{\text{Observed}}$. The coefficient χ measures the association between observed local performance and career outcomes, conditional on shared fund performance. If differences in observed local returns account for the groups' different career responses, conditioning on those returns should move κ_{Exit} and κ_{NewFund} toward zero.[31]

Adding local performance leaves the career pattern largely unchanged. The fund-return interactions are +0.138 for departures and −0.108 for assignments, close to the baseline +0.145 and −0.107, and remain statistically significant. The local-return slopes themselves are small and imprecise, at 0.017 percentage points for departures and 0.002 for new assignments per return percentage point. Observed local performance therefore does not account for the different career responses to shared fund returns. Together with the weaker prior education-country performance, this result makes demonstrated local expertise a less convincing explanation for both the recruitment and career patterns.

Funds may also base new assignments on a longer performance history than the latest quarter. We therefore replace the latest-quarter return with the mean of the four quarterly benchmark-relative returns ending in q in all performance interactions. Columns 5–6 use a common sample of 116,550 manager-quarters at 1,016 funds in which each manager has been on the team for the preceding four quarters, retaining the fixed effects, controls and fund clustering of Equation (14). On this sample the foreign-education interaction is −0.163 (standard error 0.054) using the latest quarter, compared with −0.107 in the full sample, and −0.278 (0.101) using the four-quarter mean. Both are significant at 1%. The latter implies that a one-percentage-point increase in average quarterly performance lowers the foreign-minus-U.S. difference in next-quarter assignment probabilities by 0.278 percentage points. The weaker assignment response is not confined to the most recent quarter, as stronger

[31] All 154,677 manager-quarters remain. Of these, 48,073 at 832 funds have observed local returns, and the added return term is zero elsewhere. The local return weights holdings by initial country capital. The coefficients χ and ϕ are estimated separately for each outcome, and controls, effects, weights and clustering are unchanged.

performance over the preceding year also brings foreign-educated managers smaller relative gains in new assignments. The corresponding departure interaction is 0.056 (standard error 0.108), positive but imprecise.

4.5 Flows and fees

A commercial payoff could also explain why funds accept weaker prior performance or place less weight on subsequent returns. Foreign education could attract investors or support higher fees. Beyond the prior-flow records in Section 4.3, we ask whether employing more foreign-educated managers is associated with subsequent asset gathering or higher fee rates. The monthly flow regression compares the same fund over time with other funds sharing its benchmark in the same month, accounting for prior performance and fund and team characteristics.

In the monthly comparison, a change from an entirely U.S.-educated to an entirely foreign-educated team is associated with monthly net flows 0.21 percentage points higher, with a standard error of 0.15, which is not statistically significant (Appendix Table A15). Quarterly flows and expense ratios show no clear difference either (Appendix Table A16). These comparisons provide no statistically clear commercial payoff to explain why funds value the background despite the weaker prior performance.

Funds thus do not appear to recognize the limits of local expertise. Country losses predict linked hiring, but the ensuing local recovery comes almost entirely from stocks held before the hire. In the broader recruitment sample, foreign-educated recruits bring weaker prior performance, including in their potential specialty. Among colleagues, retention and new-assignment responses are consistent with less weight on shared fund performance, even after accounting for observed local returns. The commercial comparisons show no clear compensating payoff. Combined with weaker stock picking in familiar outside markets, these findings suggest that funds overvalue local expertise as a signal of broader ability.

5 Conclusion

Using foreign education as a proxy for familiarity, we ask how far local expertise reaches and whether funds recognize its limits. Local expertise is real but narrow. Greater foreign-educated representation accompanies lower factor-adjusted net and gross returns. Funds overweight education countries without a clear average stock-selection advantage. High-conviction stocks in managers' education countries earn high alphas but add little to portfolio alpha. Outside their education countries, foreign-educated teams select stocks worse in markets that border or are more socially connected to their education countries, precisely where familiarity is expected to help. Familiarity is also worth far less to them than to U.S.-educated teams.

Funds do not appear to recognize these limits. They are more likely to hire managers educated in a country after their holdings there underperform, but the short-lived recovery that follows comes almost entirely from stocks held before the hire, with no clear gain in whole-fund returns. Recovery alone therefore does not establish the recruit's ability. In the broader set of experienced recruits, foreign-educated recruits bring weaker prior performance and weaker prior-fund flows than U.S.-educated recruits. Their prior performance is weaker even in their potential country specialty. After appointment, their retention is less sensitive to deteriorating fund performance than that of U.S.-educated colleagues, and their assignment opportunities benefit less as returns improve. Controlling for observed local returns leaves these career differences largely unchanged.

Together, the investment and personnel evidence shows that funds apply lower performance standards to foreign-educated managers, both in the prior performance they accept at hiring and in how their careers respond to performance, and we find no offsetting advantage in their country-specific performance, fundraising or fees. Given how narrow the investment value of local expertise is, this pattern is most consistent with funds overvaluing it as a signal of broader ability.

References

Aigner, D. J. and Cain, G. G. (1977). Statistical theories of discrimination in labor markets. *Industrial and Labor Relations Review*, 30(2):175–187.

Altonji, J. G. and Pierret, C. R. (2001). Employer learning and statistical discrimination. *The Quarterly Journal of Economics*, 116(1):313–350.

Antón, M., Cohen, R. B., and Polk, C. (2021). Best ideas. Working paper, available at SSRN.

Bai, J. J., Tang, Y., Wan, C., and Yüksel, H. Z. (2022). Fund manager skill in an era of globalization: Offshore concentration and fund performance. *Journal of Financial Economics*, 145(2):18–40.

Baik, B., Kang, J.-K., and Kim, J.-M. (2010). Local institutional investors, information asymmetries, and equity returns. *Journal of Financial Economics*, 97(1):81–106.

Bailey, M., Cao, R., Kuchler, T., Stroebel, J., and Wong, A. (2018). Social connectedness: Measurement, determinants, and effects. *Journal of Economic Perspectives*, 32(3):259–280.

Brinson, G. P. and Fachler, N. (1985). Measuring non-U.S. equity portfolio performance. *Journal of Portfolio Management*, 11(3):73–76.

Cameron, A. C. and Miller, D. L. (2015). A practitioner's guide to cluster-robust inference. *Journal of Human Resources*, 50(2):317–372.

Carhart, M. M. (1997). On persistence in mutual fund performance. *The Journal of Finance*, 52(1):57–82.

Chevalier, J. and Ellison, G. (1999a). Are some mutual fund managers better than others? Cross-sectional patterns in behavior and performance. *The Journal of Finance*, 54(3):875–899.

Chevalier, J. and Ellison, G. (1999b). Career concerns of mutual fund managers. *The Quarterly Journal of Economics*, 114(2):389–432.

Chuprinin, O. and Sosyura, D. (2018). Family descent as a signal of managerial quality: Evidence from mutual funds. *The Review of Financial Studies*, 31(10):3756–3820.

Cohen, L., Frazzini, A., and Malloy, C. (2008). The small world of investing: Board connections and mutual fund returns. *Journal of Political Economy*, 116(5):951–979.

Coval, J. D. and Moskowitz, T. J. (2001). The geography of investment: Informed trading and asset prices. *Journal of Political Economy*, 109(4):811–841.

Cremers, K. J. M. and Petajisto, A. (2009). How active is your fund manager? A new measure that predicts performance. *The Review of Financial Studies*, 22(9):3329–3365.

Daniel, K., Grinblatt, M., Titman, S., and Wermers, R. (1997). Measuring mutual fund performance with characteristic-based benchmarks. *The Journal of Finance*, 52(3):1035–1058.

Demirci, I., Ferreira, M. A., Matos, P., and Sialm, C. (2022). How global is your mutual fund? International diversification from multinationals. *The Review of Financial Studies*, 35(7):3337–3372.

Fama, E. F. and French, K. R. (2010). Luck versus skill in the cross-section of mutual fund returns. *The Journal of Finance*, 65(5):1915–1947.

Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.

Fang, J., Kempf, A., and Trapp, M. (2014). Fund manager allocation. *Journal of Financial Economics*, 111(3):661–674.

Farber, H. S. and Gibbons, R. (1996). Learning and wage dynamics. *The Quarterly Journal of Economics*, 111(4):1007–1047.

Gantchev, N., Giannetti, M., and Li, R. (2024). Sustainability or performance? Ratings and fund managers' incentives. *Journal of Financial Economics*, 155:103831.

Gottesman, A. A. and Morey, M. R. (2006). Manager education and mutual fund performance. *Journal of Empirical Finance*, 13(2):145–182.

Graham, J. R., Harvey, C. R., and Huang, H. (2009). Investor competence, trading frequency, and home bias. *Management Science*, 55(7):1094–1106.

Grinblatt, M. and Keloharju, M. (2001). How distance, language, and culture influence stockholdings and trades. *The Journal of Finance*, 56(3):1053–1073.

Heath, C. and Tversky, A. (1991). Preference and belief: Ambiguity and competence in choice under uncertainty. *Journal of Risk and Uncertainty*, 4(1):5–28.

Holmström, B. (1982). Moral hazard in teams. *The Bell Journal of Economics*, 13(2):324–340.

Huberman, G. (2001). Familiarity breeds investment. *The Review of Financial Studies*, 14(3):659–680.

Jagannathan, M., Jiao, W., and Karolyi, G. A. (2022). Is there a home field advantage in global markets? *Journal of Financial Economics*, 143(2):742–770.

Kacperczyk, M., Sialm, C., and Zheng, L. (2005). On the industry concentration of actively managed equity mutual funds. *The Journal of Finance*, 60(4):1983–2011.

Kacperczyk, M., Sialm, C., and Zheng, L. (2008). Unobserved actions of mutual funds. *The Review of Financial Studies*, 21(6):2379–2416.

Khorana, A. (1996). Top management turnover: An empirical investigation of mutual fund managers. *Journal of Financial Economics*, 40(3):403–427.

Kuchler, T., Li, Y., Peng, L., Stroebel, J., and Zhou, D. (2022). Social proximity to capital: Implications for investors and firms. *The Review of Financial Studies*, 35(6):2743–2789.

Linnainmaa, J. T., Melzer, B. T., and Previtero, A. (2021). The misguided beliefs of financial advisors. *The Journal of Finance*, 76(2):587–621.

Mayer, T. and Zignago, S. (2011). Notes on CEPII's distances measures: The GeoDist database. CEPII Working Paper 2011-25.

Oster, E. (2019). Unobservable selection and coefficient stability: Theory and evidence. *Journal of Business & Economic Statistics*, 37(2):187–204.

Petersen, M. A. (2009). Estimating standard errors in finance panel data sets: Comparing approaches. *The Review of Financial Studies*, 22(1):435–480.

Phelps, E. S. (1972). The statistical theory of racism and sexism. *The American Economic Review*, 62(4):659–661.

Pool, V. K., Stoffman, N., and Yonker, S. E. (2012). No place like home: Familiarity in mutual fund manager portfolio choice. *The Review of Financial Studies*, 25(8):2563–2599.

Schumacher, D. (2018). Home bias abroad: Domestic industries and foreign portfolio choice. *The Review of Financial Studies*, 31(5):1654–1706.

Solon, G., Haider, S. J., and Wooldridge, J. M. (2015). What are we weighting for? *Journal of Human Resources*, 50(2):301–316.

Spence, M. (1973). Job market signaling. *The Quarterly Journal of Economics*, 87(3):355–374.

Wermers, R. (2000). Mutual fund performance: An empirical decomposition into stock-picking talent, style, transactions costs, and expenses. *The Journal of Finance*, 55(4):1655–1695.

Table 1: Summary statistics

	Mean	Median	Std. dev.	N
A. Portfolios and team education (fund-quarter)				
Number of countries held	18.99	19.00	8.44	67,380
Number of manager education countries (including U.S.)	1.57	1.00	0.92	63,510
Number of non-U.S. manager education countries	0.72	0.00	0.95	63,510
B. Funds (fund-year)				
Total net assets ($ billions)	1.59	0.23	6.66	21,084
Net expense ratio (%)	1.22	1.17	0.50	19,695
Turnover ratio (%)	64.31	48.00	60.96	18,999
Annual return (%)	8.68	10.86	22.59	18,773
Annualized volatility (%)	16.46	14.81	7.09	18,773
C. Managers (one observation per fund)				
Average manager tenure (years)	5.57	4.93	3.27	1,663
Number of managers	3.12	2.44	2.44	1,663
D. Foreign education in the net-return panel				
Foreign-educated manager share (fund-month)	0.35			206,847
Fund-quarters with ForeignShare = 0 (%)	48.6			68,949
Fund-quarters with partly unknown education (%)	28.3			68,949
E. Funds with foreign-educated managers (fund-quarter)				
Number of managers	4.13	3.00	4.15	33,761
Number of foreign-educated managers	2.00	1.00	1.72	33,761
Number of foreign countries held	18.44	18.00	8.41	33,761
Foreign countries outside recorded education countries	17.23	17.00	8.06	33,761

Notes: This table reports summary statistics for U.S. actively managed international equity funds (Morningstar), 2000–2024. Panel B summarizes fund characteristics, with net assets summed across each fund's share classes. The NAV regressions use returns and assets of the oldest share class. Economic-fund clustering combines ISINs that share a Morningstar fund code. Of the 2,992 managers with recorded post-secondary education, 26.9% studied only outside the United States. Panel A counts the distinct education countries of a fund's current managers. Panel E gives equal weight to 33,761 economic fund-quarters at 1,005 funds with positive covered non-U.S. holdings and at least one current manager with recorded post-secondary study outside the United States. The country and manager counts use the same observations. Outside countries have no education link to any current manager in the recorded education histories. ForeignShare is the share of current managers with any recorded non-U.S. education among managers whose education is known.

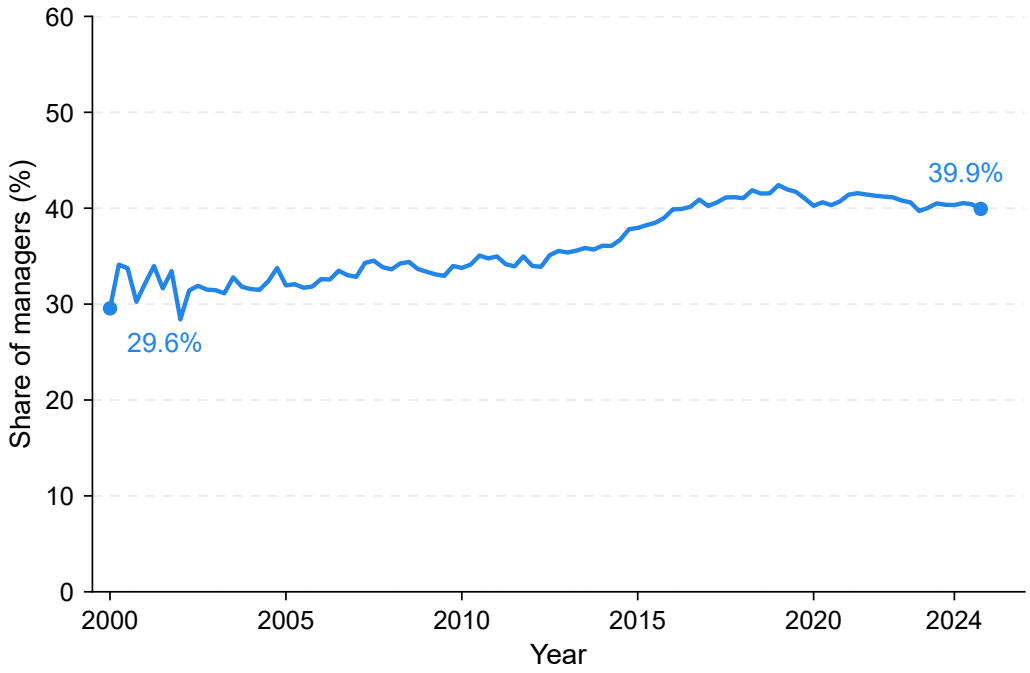


Figure 1: Foreign-educated share of fund managers

Notes: Quarterly percentage of foreign-educated managers among the current managers of active sample funds who have recorded education, 2000Q1–2024Q4. Active funds have positive reported non-U.S. equity holdings in that quarter. No return-availability or stock-classification requirement is imposed. Manager status is measured at quarter-end, and a manager who runs several funds counts once. Foreign-educated managers have recorded post-secondary study outside the United States, including managers educated both in the United States and abroad. Managers without recorded education are excluded. The endpoints are 73 of 247 managers (29.6%) and 414 of 1,037 managers (39.9%).

Table 2: Foreign education and net fund returns

Dependent variable	Factor-adjusted net NAV return			
Units	Annualized bps			
	(1)	(2)	(3)	(4)
Specification	Full controls	Broader sample	No operating controls	Full controls (WLS)
Foreign-educated share	-69.51^{**}	-69.30^{*}	-68.31^{*}	-69.66^{***}
	(35.42)	(35.43)	(35.61)	(22.66)
Log team size	−88.89	−87.85	−90.36	-72.93^{*}
	(59.10)	(59.06)	(59.22)	(37.80)
Unknown-education share	−108.22	−108.32	−100.94	-134.06^{***}
	(68.59)	(68.54)	(68.22)	(44.16)
Log lagged assets (USD millions)	-21.83^{**}	-22.38^{**}	-21.14^{**}	-17.30^{**}
	(9.90)	(9.84)	(9.91)	(7.38)
Assets spline basis 1	-100.49^{***}	-99.74^{***}	-99.81^{***}	-62.79^{***}
	(26.04)	(25.98)	(26.16)	(18.26)
Assets spline basis 2	559.63^{***}	556.76^{***}	553.00^{***}	327.21^{***}
	(158.52)	(158.37)	(158.96)	(106.46)
Team-size spline basis 1	−78.04	−79.95	−77.85	−24.61
	(137.48)	(137.40)	(137.52)	(88.00)
Team-size spline basis 2	695.54	701.71	686.45	394.77
	(590.07)	(589.75)	(589.82)	(374.04)
Female share (known gender)	−50.70	−50.08	−50.77	−37.88
	(38.54)	(38.52)	(38.58)	(27.60)
Unknown-gender share	−89.18	−90.07	−93.77	−0.34
	(93.07)	(93.05)	(92.41)	(53.01)
CFA share (observed biographies)	6.58	6.77	7.99	33.02
	(31.47)	(31.47)	(31.48)	(21.24)
Unknown-biography share	227.18^{**}	228.44^{**}	227.79^{**}	80.74
	(107.77)	(107.71)	(107.52)	(66.83)
Mean log(1 + experience)	-34.70^{**}	-34.15^{**}	-35.16^{**}	-39.13^{***}
	(16.75)	(16.73)	(16.88)	(10.63)
Prior-year relative NAV return (pp)	-4.60^{***}	-4.77^{***}	-4.63^{***}	-2.92^{***}
	(1.46)	(1.46)	(1.47)	(0.95)
Log(1 + prior relative volatility)	62.22^{**}	63.78^{**}	63.77^{**}	36.75^{**}
	(31.00)	(30.88)	(31.67)	(18.59)
Prior-quarter flow (pp)	1.57×10^{-6}	1.54×10^{-6}	1.55×10^{-6}	1.21×10^{-6}
	(2.25×10^{-6})	(2.24×10^{-6})	(2.24×10^{-6})	(1.51×10^{-6})
Missing prior return history	99.32	103.71^{*}	75.70	2.29
	(62.86)	(62.40)	(57.18)	(38.49)
Missing prior-quarter flow	−41.44	−38.72	−43.26	-66.19^{*}
	(55.66)	(53.41)	(55.30)	(36.44)
Lagged expense ratio (bps)	−0.40	−0.39	—	-0.46^{**}
	(0.34)	(0.34)		(0.23)
Lagged log turnover	84.01^{**}	82.51^{**}	—	30.18
	(35.50)	(35.41)		(25.07)
Top-ten equity concentration	75.23	68.69	—	79.59
	(138.23)	(137.84)		(90.84)
Missing lagged expense ratio	-111.04^{**}	-111.35^{**}	—	−23.98
	(54.10)	(53.80)		(34.88)
Missing lagged turnover	72.29^{**}	71.47^{*}	—	10.76
	(36.75)	(36.70)		(23.92)
Missing equity concentration	26.85	25.74	—	18.74
	(42.14)	(42.13)		(27.61)
Fund FE	Yes	Yes	Yes	Yes
Benchmark × month FE	Yes	Yes	Yes	Yes
Observations	206,847	208,425	206,847	206,847
Fund clusters	1,418	1,536	1,418	1,418

Notes: Monthly net NAV uses thirteen predetermined factor exposures, January 2000–December 2024. Coefficients are annualized bps. The foreign-educated share coefficient compares entirely foreign-educated with entirely U.S.-educated teams, among managers with known education. Column 2 adds shorter-history funds with benchmark-peer loadings. Column 3 omits expenses, turnover and concentration. Column 4 retains column 1's sample and controls with inverse fund residual-variance weights. Columns 1–3 give fund-months equal weight. Prior flow is uncapped and in raw percentage points, with a missingness indicator. A few extreme values dominate its variation, so its coefficient is very small. Outcomes are not winsorized or zero-filled. Standard errors, clustered by economic fund, appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table 3: Fees and the net-return shortfall

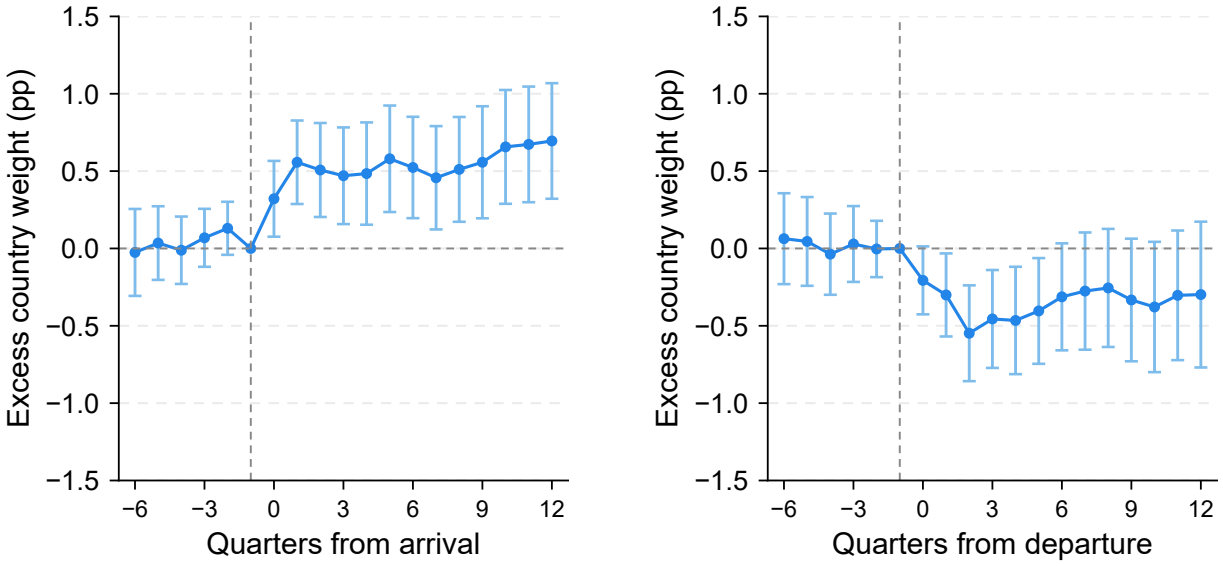
Dependent variable	Net NAV	Gross NAV	Net NAV	Gross NAV
Units		Factor-adjusted returns, annualized bps		
	(1)	(2)	(3)	(4)
Specification	Full controls	Full controls	No operating controls	No operating controls
Foreign-educated share	-70.56^{**}	-71.03^{**}	-69.29^{*}	-70.13^{**}
	(35.42)	(35.40)	(35.62)	(35.57)
Log team size	-98.60^{*}	-102.57^{*}	-99.66^{*}	-104.30^{*}
	(59.39)	(59.30)	(59.51)	(59.39)
Unknown-education share	−110.61	−112.41	−102.99	−105.70
	(68.97)	(69.14)	(68.57)	(68.76)
Log lagged assets (USD millions)	-21.41^{**}	-21.92^{**}	-20.73^{**}	-21.13^{**}
	(10.01)	(10.03)	(10.03)	(10.04)
Assets spline basis 1	-99.72^{***}	-102.85^{***}	-99.14^{***}	-103.73^{***}
	(26.23)	(26.23)	(26.36)	(26.34)
Assets spline basis 2	551.43^{***}	567.58^{***}	544.25^{***}	564.86^{***}
	(159.79)	(159.72)	(160.21)	(160.07)
Team-size spline basis 1	−53.74	−48.77	−55.10	−49.90
	(137.51)	(137.24)	(137.46)	(137.12)
Team-size spline basis 2	598.98	590.68	597.53	592.21
	(590.13)	(589.18)	(589.31)	(588.13)
Female share (known gender)	−54.16	−54.52	−53.78	−54.23
	(38.51)	(38.51)	(38.57)	(38.56)
Unknown-gender share	−89.22	−89.81	−94.35	−95.36
	(93.32)	(93.38)	(92.67)	(92.86)
CFA share (observed biographies)	6.40	5.06	8.11	6.54
	(31.83)	(31.88)	(31.86)	(31.91)
Unknown-biography share	243.98^{**}	248.18^{**}	245.10^{**}	250.14^{**}
	(108.10)	(108.14)	(107.86)	(107.95)
Mean log(1 + experience)	-34.30^{**}	-35.04^{**}	-34.56^{**}	-35.64^{**}
	(16.81)	(16.80)	(16.93)	(16.92)
Prior-year relative NAV return (pp)	-4.46^{***}	-4.50^{***}	-4.48^{***}	-4.50^{***}
	(1.48)	(1.48)	(1.48)	(1.48)
Log(1 + prior relative volatility)	54.63^{*}	55.14^{*}	56.26^{*}	57.68^{*}
	(31.13)	(31.17)	(31.82)	(31.85)
Prior-quarter flow (pp)	1.43×10^{-6}	1.47×10^{-6}	1.40×10^{-6}	1.47×10^{-6}
	(2.25×10^{-6})	(2.26×10^{-6})	(2.24×10^{-6})	(2.25×10^{-6})
Missing prior return history	107.10^{*}	104.68	78.87	78.88
	(63.81)	(63.90)	(57.63)	(57.65)
Missing prior-quarter flow	−67.65	−68.84	−69.12	−70.44
	(55.95)	(56.00)	(55.76)	(55.81)
Lagged expense ratio (bps)	−0.36	−0.07	—	—
	(0.34)	(0.35)		
Lagged log turnover	83.38^{**}	82.62^{**}	—	—
	(35.68)	(35.65)		
Top-ten equity concentration	71.59	67.49	—	—
	(139.26)	(139.32)		
Missing lagged expense ratio	-120.73^{**}	−84.63	—	—
	(53.46)	(54.45)		
Missing lagged turnover	80.45^{**}	80.34^{**}	—	—
	(35.65)	(35.75)		
Missing equity concentration	32.49	30.96	—	—
	(42.37)	(42.49)		
Fund FE	Yes	Yes	Yes	Yes
Benchmark × month FE	Yes	Yes	Yes	Yes
Observations	205,773	205,773	205,773	205,773
Fund clusters	1,418	1,418	1,418	1,418

Notes: Net and reported gross NAV use identical observations and thirteen predetermined factor exposures, 2000–2024. Columns 1–2 use full controls, and columns 3–4 omit expense, turnover and concentration controls. Fund-months have equal weight, prior flow enters raw with a missing indicator, and standard errors are clustered by economic fund. Standard errors appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%. Gross-minus-net differences in the foreign-educated share coefficient are −0.47 (SE 1.01) for columns 2 minus 1 and −0.84 (SE 1.33) for columns 4 minus 3. Standard errors use the joint covariance of each pair.

Table 4: Education links and country overweights

Dependent variable	Country overweight			
Units	Percentage points			
	(1)	(2)	(3)	(4)
Specification	Any link	Link share	Any link	Link share
Any manager educated in country	1.108***	—	1.091***	—
	(0.220)		(0.219)	
Share of managers educated in country	—	2.626***	—	2.590***
		(0.459)		(0.454)
Female share (known gender)	0.019	0.025	—	—
	(0.043)	(0.042)		
Unknown-gender share	0.087	0.085	—	—
	(0.060)	(0.059)		
CFA share (observed biographies)	−0.030	−0.032	—	—
	(0.032)	(0.032)		
Unknown-biography share	−0.116	−0.122	—	—
	(0.080)	(0.080)		
Mean log(1 + experience)	0.014	0.013	—	—
	(0.014)	(0.014)		
Missing team-trait record	−0.060	−0.067	—	—
	(0.103)	(0.103)		
Fund FE	Yes	Yes	No	No
Country × month FE	Yes	Yes	Yes	Yes
Fund × month FE	No	No	Yes	Yes
Observations	5,613,030	5,613,030	5,611,449	5,611,449
Fund clusters	1,662	1,662	1,660	1,660

Notes: The outcome is country weight minus benchmark weight, in pp, January 1999–December 2024. Any link is an indicator. Link share is the fraction of current managers, including those with unknown education, educated in the country. Fund-month effects in columns 3–4 absorb team traits, shown as dashes rather than zeros. Observations have equal weight. Standard errors, clustered by economic fund, appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%.



(a) Arrival of the first education-linked manager

(b) Departure of the last education-linked manager

Figure 2: Country overweight around manager arrivals and departures

Notes: This figure plots event-study coefficients for the fund's country weight minus the benchmark's country weight (percentage points) around the quarter in which the fund gains its first manager educated in the country (left) or loses the last such manager (right). Windows run up to twelve quarters and end if the status reverses. Events without a six-quarter clean pre-period or with overlapping windows are removed, leaving 977 arrivals and 793 departures. Regressions include fund × country, country × quarter and fund × quarter fixed effects, with quarter −1 as the reference. Bars are 95% confidence intervals with standard errors clustered by economic fund.

Table 5: Stock selection in education countries relative to U.S.-educated teams

Dependent variable	Adjusted country return	
Units	Annualized basket bps	
	(1)	(2)
Specification	OLS	WLS
Foreign-educated share (γ, outside gap)	−46.31	−31.20
	(31.35)	(28.87)
Education-country link (β, local increment)	46.52	36.77
	(49.90)	(47.44)
Log country stock count	−57.23***	−52.91***
	(11.42)	(11.36)
Log team size	−67.97***	−38.56*
	(21.46)	(19.84)
Unknown-education share	25.76	2.85
	(84.74)	(78.94)
Female share (known gender)	−96.56**	−117.46***
	(44.95)	(37.52)
Unknown-gender share	94.11	132.75
	(86.85)	(81.54)
CFA share (observed biographies)	51.48*	47.34
	(30.20)	(29.64)
Unknown-biography share	−39.73	−45.65
	(123.43)	(115.07)
Mean log(1 + experience)	−52.48***	−43.23***
	(15.43)	(15.14)
Fund × country FE	Yes	Yes
Country × quarter FE	Yes	Yes
Observations	974,924	974,924
Fund clusters	1,428	1,428

Notes: Equation (6) uses all held foreign countries and the adjusted country return of Equation (2) over the next three months. $\mathbb{1}^{\text{Link}} = 1$ if a current manager studied in the country, and ForeignShare is the fraction of known-education managers with foreign education. Countries have equal weight within each fund-quarter. WLS also weights by inverse fund residual variance. Outcomes cover January 2000–December 2024, and 1,429 identifiers map to 1,428 economic-fund clusters. Standard errors, clustered by economic fund, appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%. The focal comparison, $\beta + \gamma$, compares an entirely foreign-educated team investing in an education country with an entirely U.S.-educated team in the same market. The estimates are 0.21 (SE 56.61) and 5.57 (SE 51.39) bps in columns 1 and 2, using joint covariance.

Table 6: Local expertise and high-conviction stock selection

Dependent variable	Portfolio excess return			
Units	Annual alpha in bps, factor loadings dimensionless			
	(1)	(2)	(3)	(4)
Specification	Local top	Local ordinary	Outside top	Outside ordinary
Alpha (annualized bps)	400.2^{***}	40.1	13.4	38.1
	(133.3)	(95.9)	(54.8)	(39.9)
Global ex-U.S. benchmark excess return	0.148	0.135^{*}	0.208^{***}	0.183^{***}
	(0.106)	(0.076)	(0.043)	(0.032)
Developed ex-U.S. market	0.778^{***}	0.723^{***}	0.638^{***}	0.647^{***}
	(0.097)	(0.069)	(0.040)	(0.029)
Developed ex-U.S. size	−0.012	-0.146^{***}	0.059^{**}	-0.048^{***}
	(0.059)	(0.043)	(0.024)	(0.018)
Developed ex-U.S. value	−0.011	0.021	0.063^{**}	−0.013
	(0.068)	(0.049)	(0.028)	(0.020)
Developed ex-U.S. investment	0.030	−0.059	-0.080^{**}	−0.029
	(0.093)	(0.067)	(0.038)	(0.028)
Developed ex-U.S. profitability	0.015	−0.057	0.055	−0.008
	(0.093)	(0.067)	(0.038)	(0.028)
Emerging market	0.103^{**}	0.175^{***}	0.192^{***}	0.183^{***}
	(0.044)	(0.031)	(0.018)	(0.013)
Emerging size	0.000	−0.029	0.023	-0.040^{**}
	(0.064)	(0.046)	(0.026)	(0.019)
Emerging value	−0.011	−0.019	-0.065^{**}	-0.049^{**}
	(0.066)	(0.047)	(0.027)	(0.020)
Emerging investment	−0.041	0.001	0.129^{***}	−0.020
	(0.077)	(0.055)	(0.032)	(0.023)
Emerging profitability	−0.064	0.041	0.073^{**}	0.072^{***}
	(0.085)	(0.061)	(0.035)	(0.025)
Observations	300	300	300	300

Notes: Quarter-end I-measures (Equation (7)), computed within each fund-country-quarter, identify top-quartile and ordinary holdings. Portfolios are value-weighted and reweighted to benchmark country shares, January 2000–December 2024. Annual alpha is $12 \times 10^4 \alpha_p$ basis points. Factor loadings are dimensionless. Individual regressions use conventional OLS standard errors. Direct alpha differences and Wald tests use the robust stacked-system covariance without calendar-month clustering. Benchmark annual excess return is 4.30%. Standard errors appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%. The local top-minus-ordinary alpha is 360.1 bps (SE 137.1, $p = 0.009$). The outside counterpart is −24.7 (SE 60.3, $p = 0.682$). Local-minus-outside equality tests give $p = 0.002$ for top and $p = 0.981$ for ordinary portfolios. Figure 3 displays these four tests.

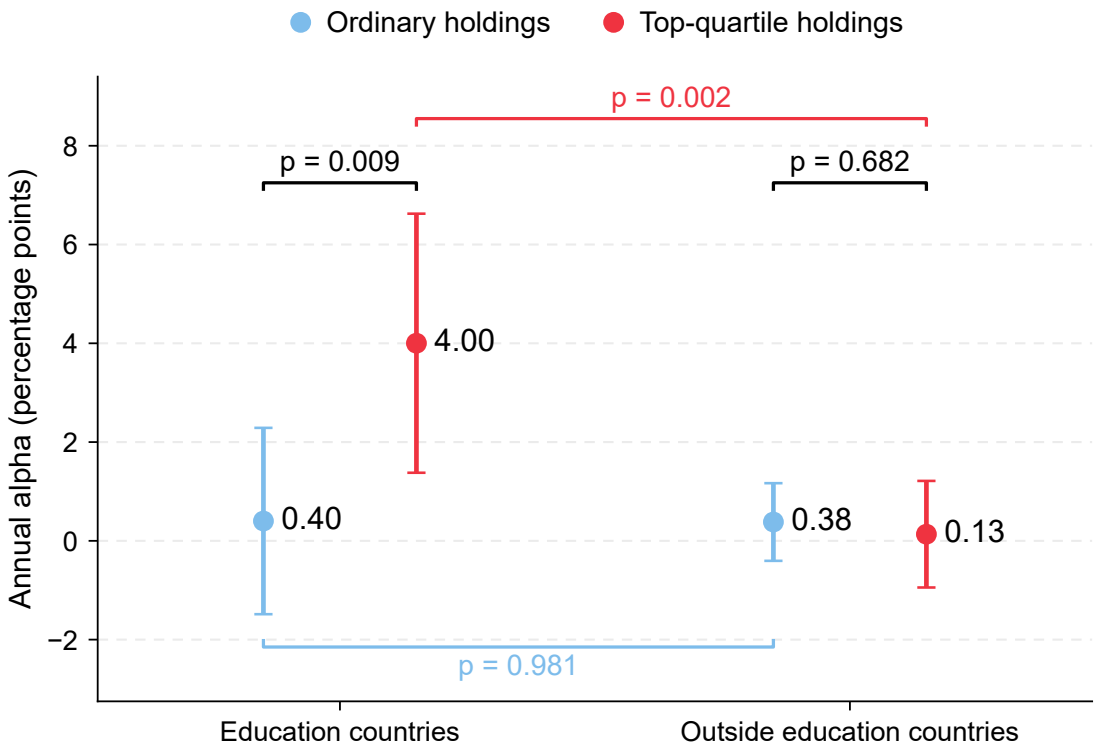
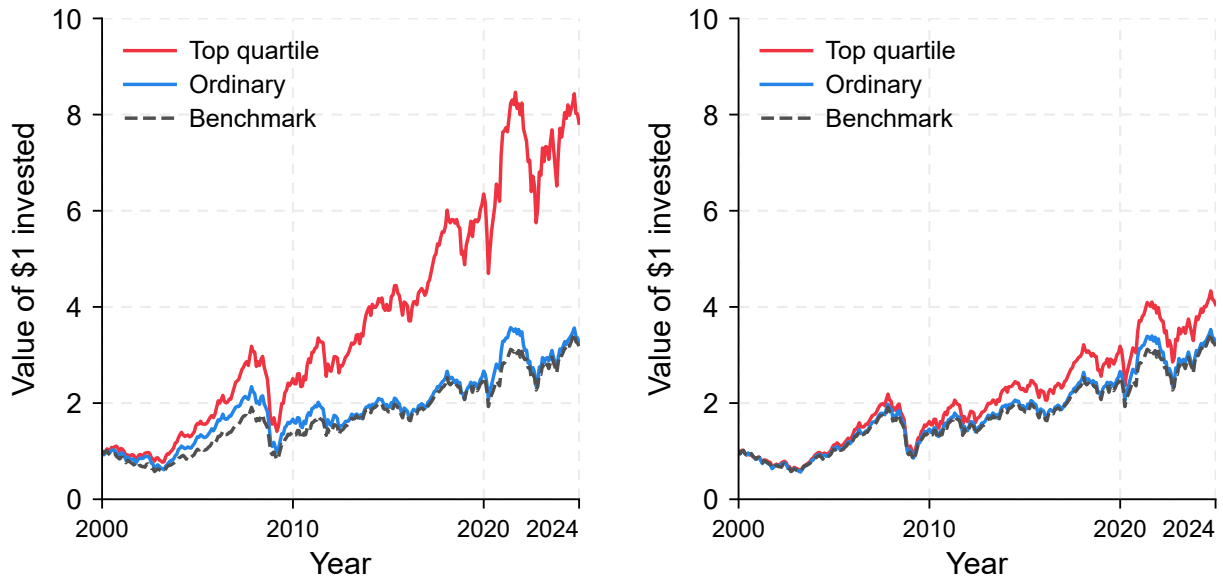


Figure 3: Conviction sorting: portfolio alphas and direct comparisons

Notes: The figure displays the four regressions in Table 6, using January 2000–December 2024. Alpha and its 95% confidence interval are annualized percentage points. Top denotes the highest within-country I-measure quartile. Ordinary denotes the other three quartiles. The brackets report all four Wald tests, comparing top with ordinary positions within each country group and education with outside countries within each conviction group. Individual intervals use conventional OLS inference. Differences use a robust stacked covariance. Country portfolios are value-weighted and reweighted to the common benchmark country mix.



(a) Education countries

(b) Outside education countries

Figure 4: Cumulative returns of conviction-sorted portfolios

Notes: Each panel compounds the monthly gross returns of the same top and ordinary strategies used in Table 6. One dollar is invested immediately before January 2000. All 300 monthly returns through December 2024 are included. The dashed line is the common Morningstar Global ex-US benchmark. These are returns on country-reweighted disclosed-holdings strategies, not actual fund NAV, capital contributions or cumulative risk-adjusted alpha. The two panels use the same vertical scale.

Table 7: Relatedness and stock selection outside education countries

Dependent variable	Adjusted country return		
Units	Annualized basket bps		
	(1)	(2)	(3)
Specification	Shared border	Official language	Social connectedness
Foreign education × tie (θ)	-484.20^{***}	-230.69^{***}	-109.98^{***}
	(158.36)	(67.31)	(42.05)
Tie (δ)	299.90^{**}	155.35^{**}	67.17
	(134.65)	(75.45)	(43.53)
Foreign-educated share (γ)	−28.66	1.14	826.14^{**}
	(32.33)	(33.18)	(330.63)
Missing-tie share (MissingShare)	8.29	38.62	-333.29^{*}
	(45.48)	(45.98)	(174.15)
Log country stock count	-62.09^{***}	-62.27^{***}	-65.81^{***}
	(11.73)	(11.73)	(11.60)
Log team size	-66.44^{***}	-67.15^{***}	-60.96^{***}
	(21.32)	(21.32)	(22.16)
Unknown-education share	59.19	60.97	43.81
	(84.35)	(85.02)	(83.59)
Female share (known gender)	-95.46^{**}	-96.27^{**}	-103.92^{**}
	(45.61)	(45.61)	(47.46)
Unknown-gender share	54.04	47.36	61.07
	(69.76)	(69.79)	(71.85)
CFA share (observed biographies)	43.11	42.28	44.36
	(30.85)	(30.91)	(31.05)
Unknown-biography share	22.70	30.46	7.04
	(119.49)	(119.55)	(114.17)
Mean log(1 + experience)	-44.05^{***}	-44.47^{***}	-38.68^{**}
	(15.57)	(15.60)	(16.04)
Fund × country FE	Yes	Yes	Yes
Country × quarter FE	Yes	Yes	Yes
Observations	936,076	936,076	876,532
Fund clusters	1,423	1,423	1,421

Notes: The specification is Equation (9), and the tie averages are defined in Section 3.5. Each column uses one familiarity measure. Column 1 is a shared border. Column 2 is a common official language. Column 3 is log social connectedness from the October 2021 snapshot, without standardization or centering. Its foreign-educated-share and missing-tie-share coefficients refer to log connectedness zero. At mean log connectedness 8.107, the comparable foreign-educated-share level is −65.38 bps. The text translates the raw gradient into a standard-deviation comparison. Foreign-educated managers' ties use their non-U.S. education countries. U.S.-educated managers' ties use the United States. The first row is θ, the coefficient on ForeignTie, which is the team average of each manager's foreign-education indicator interacted with that manager's tie, not foreign share times the team-average tie. It directly tests the foreign-minus-U.S. difference in slopes. The tie coefficient δ gives the U.S.-educated slope. The foreign-educated slope is their sum. All direct education countries are excluded. Countries receive equal weight within each fund-quarter. Standard errors are clustered by economic fund. Samples contain 1,424 ISINs and 1,423 economic-fund clusters for the border and language measures and 1,422 and 1,421 for social connectedness. Standard errors appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table 8: Country performance and linked appointments

Dependent variable	Direct-country appointment	Bordering-country appointment
Units	Quarterly appointment probability, bps	
	(1)	(2)
Prior country-market loss (pp)	0.043**	−0.014
	(0.021)	(0.028)
Prior own country underperformance (pp)	0.048**	0.036
	(0.024)	(0.023)
Log prior country weight	1.226	−1.227
	(0.818)	(0.920)
Log prior country stock count	0.834	2.061
	(1.520)	(1.527)
Incumbent direct-education share	−344.225***	5.472
	(40.964)	(14.110)
Incumbent bordering-country education share	5.629	−399.461***
	(13.202)	(43.656)
Fund × country FE	Yes	Yes
Fund × quarter FE	Yes	Yes
Observations	978,521	978,521
Fund clusters	1,540	1,540

Notes: Equation (10). Appointments use recorded education at all degree levels and dated spell starts. Column 1 indicates an entrant educated in the held country. Column 2 indicates a foreign-educated entrant who studied in a bordering country but not in the held country. Both outcomes are indicators multiplied by 10,000, so coefficients are probability basis points. One basis point is 0.01 percentage point. Return predictors are in percentage points, and the loss measures are minus prior returns, so positive values mean weaker performance. Country-market returns compound twelve preceding months. Own underperformance averages annualized country-market-adjusted returns over the preceding four quarters, requiring at least three. The remaining controls are measured before appointment. Incumbent shares divide by all team members. Team size and overall missing education are absorbed by fund-quarter effects. Each fund-quarter receives equal total weight, divided equally across its countries. Appointments span 2001Q1–2024Q4. Standard errors appear in parentheses below coefficients and are clustered by economic fund. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table 9: Local, outside-country and NAV performance following linked appointments

Dependent variable	Adjusted local return	Adjusted outside return	Factor-adjusted NAV
Units	Annualized basket bps	Annualized basket bps	Annualized bps
	(1)	(2)	(3)
Linked appointment × Post	228.18***	−14.94	36.82
	(80.48)	(33.93)	(37.37)
Log team size	−945.39**	−583.91***	−200.67
	(460.95)	(176.73)	(242.97)
Unknown-education share	1427.32*	6.06	496.76**
	(763.87)	(225.07)	(247.31)
Log lagged assets (USD millions)	−20.11	−113.83	−114.38*
	(115.84)	(70.79)	(69.29)
Assets spline basis 1	−103.60	60.01	−96.08
	(288.40)	(136.72)	(167.19)
Assets spline basis 2	−591.91	−896.10	−33.99
	(1823.30)	(756.91)	(937.12)
Team-size spline basis 1	928.86	938.84**	403.82
	(1005.21)	(383.95)	(523.96)
Team-size spline basis 2	−2699.45	−3006.56*	−1414.81
	(4093.10)	(1584.55)	(2156.91)
Female share (known gender)	−69.39	−16.20	8.81
	(320.13)	(146.05)	(165.84)
Unknown-gender share	−1155.27*	−368.35	−377.52
	(614.03)	(318.51)	(245.47)
CFA share (observed biographies)	346.36	160.48*	190.81
	(270.98)	(96.26)	(135.54)
Unknown-biography share	250.16	264.22	−49.08
	(894.74)	(371.03)	(325.80)
Mean log(1 + experience)	−290.48**	4.85	−56.31
	(142.55)	(50.04)	(64.65)
Prior-year relative NAV return (pp)	−25.32***	−27.24***	−35.58***
	(8.25)	(3.41)	(3.86)
Log(1 + prior relative volatility)	307.59	−55.50	−81.62
	(210.11)	(72.74)	(82.49)
Prior-quarter flow (pp)	-8.79×10^{-5}	$-3.56 \times 10^{-4***}$	$-2.00 \times 10^{-4***}$
	(1.84×10^{-4})	(7.67×10^{-5})	(6.24×10^{-5})
Missing prior return history	302.45	16.42	131.43
	(501.24)	(181.38)	(205.46)
Missing prior-quarter flow	169.39	−225.40	−451.38*
	(504.41)	(243.61)	(242.37)
Lagged expense ratio (bps)	3.17	−1.36	−0.19
	(3.93)	(1.13)	(1.65)
Lagged log turnover	94.77	161.03	358.33***
	(251.56)	(103.32)	(108.65)
Top-ten equity concentration	−904.39	−371.90	290.82
	(1160.77)	(553.58)	(565.07)
Missing lagged expense ratio	307.28	−121.37	−322.34
	(487.00)	(193.21)	(250.14)
Missing lagged turnover	15.94	33.22	145.02
	(283.24)	(112.15)	(119.25)
Fund × matched-case FE	Yes	Yes	Yes
Matched-case × quarter FE	Yes	Yes	Yes
Observations	13,237	13,237	13,237
Country-appointment cases	511	511	511
Hiring funds	267	267	267
Fund clusters	610	610	610

Notes: Equation (11). All columns use identical observations and controls. Linked appointments follow disappointing country-market or own country-relative performance. Comparison funds hold the same country under the same benchmark in the same appointment quarter, matched on prior conditions, with no appointment that quarter. Pre- and post-periods each cover four quarters, excluding quarter zero and requiring at least three common quarters per side. Education countries are fixed at appointment for both groups. Equal hiring-fund-quarter weights are divided across cases, hiring/comparison sides, phases and observed quarters. Concentration is never missing in this sample. Basket returns are characteristic-adjusted. NAV uses the factor adjustment of Table 2. Basket coefficients have separate capital denominators and do not add to NAV. Returns end in 2024Q4. Fund-clustered standard errors appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table 10: Persistence of local recovery following linked appointments

Dependent variable Units	Adjusted local return Annualized basket bps (1)	Factor-adjusted NAV Annualized bps (2)
Linked appointment × Year 1	194.80**	−13.90
	(78.39)	(38.08)
Linked appointment × Year 2	−10.54	12.12
	(83.83)	(40.61)
Log team size	−551.22	−91.36
	(347.78)	(225.61)
Unknown-education share	1274.38**	390.96
	(499.16)	(237.49)
Log lagged assets (USD millions)	−56.86	−40.56
	(99.45)	(52.69)
Assets spline basis 1	99.46	−218.64*
	(231.40)	(124.96)
Assets spline basis 2	−1878.87	757.42
	(1395.73)	(725.05)
Team-size spline basis 1	281.64	−141.41
	(879.99)	(435.46)
Team-size spline basis 2	−889.26	921.12
	(3750.55)	(1707.10)
Female share (known gender)	−110.56	−26.70
	(271.66)	(134.32)
Unknown-gender share	−517.29	−195.35
	(486.74)	(179.67)
CFA share (observed biographies)	329.78	105.77
	(217.80)	(97.87)
Unknown-biography share	−153.07	−255.32
	(624.31)	(279.48)
Mean log(1 + experience)	−302.30***	−118.01**
	(110.66)	(55.58)
Prior-year relative NAV return (pp)	−22.98***	−29.75***
	(7.56)	(3.66)
Log(1 + prior relative volatility)	173.83	−23.87
	(152.43)	(67.05)
Prior-quarter flow (pp)	-1.12×10^{-4}	-1.40×10^{-4}**
	(1.68×10^{-4})	(5.45×10^{-5})
Missing prior return history	79.59	136.21
	(518.35)	(189.33)
Missing prior-quarter flow	−284.37	−887.02***
	(597.20)	(292.45)
Lagged expense ratio (bps)	0.91	0.11
	(3.07)	(1.45)
Lagged log turnover	−54.73	304.88***
	(213.12)	(109.01)
Top-ten equity concentration	−1821.10*	171.01
	(1013.13)	(487.20)
Missing lagged expense ratio	479.59	−219.53
	(449.77)	(225.04)
Missing lagged turnover	−165.75	89.68
	(198.16)	(127.18)
Fund × matched-case FE	Yes	Yes
Matched-case × quarter FE	Yes	Yes
Observations	15,862	15,862
Country-appointment cases	430	430
Hiring funds	231	231
Fund clusters	514	514

Notes: Equation (11) with separate hiring interactions for quarters +1 to +4 and +5 to +8. Quarters −4 to −1 are the baseline and quarter zero is omitted. Both columns use the same cases, observations and controls, with the matched non-hiring funds of Table 9, and require at least three observed quarters in each phase. Appointments end by 2022Q4 and returns by 2024Q4. Equal hiring-fund-quarter weights are divided across cases, hiring/comparison sides, phases and observed quarters. Concentration is never missing in this sample. Year 2 minus Year 1 is −205.34 (SE 96.72, significant at 5%) for the local return and 26.03 (SE 34.02) for NAV, using the joint covariance. Standard errors, clustered by economic fund, appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table 11: Sources of local recovery following linked appointments

Dependent variable	Observed holdings	Passive pre-hire holdings	Portfolio changes
Units		Annualized basket bps	
	(1)	(2)	(3)
Linked appointment × Year 1	178.45**	174.37**	4.08
	(81.02)	(86.19)	(64.29)
Linked appointment × Year 2	−16.03	63.82	−79.86
	(87.91)	(90.96)	(74.92)
Log team size	−327.09	−372.40	45.32
	(360.58)	(383.77)	(288.39)
Unknown-education share	1200.86**	1205.90**	−5.04
	(486.66)	(490.13)	(431.72)
Log lagged assets (USD millions)	−77.34	32.48	−109.82
	(109.88)	(88.55)	(93.08)
Assets spline basis 1	129.40	−229.80	359.20*
	(246.27)	(209.78)	(189.30)
Assets spline basis 2	−1970.97	593.97	−2564.94**
	(1483.35)	(1381.18)	(1119.47)
Team-size spline basis 1	−224.44	−458.39	233.96
	(898.70)	(986.00)	(779.08)
Team-size spline basis 2	1088.04	3316.85	−2228.81
	(3818.17)	(3997.64)	(3171.99)
Female share (known gender)	−101.72	413.57	−515.29***
	(281.83)	(256.52)	(194.50)
Unknown-gender share	−425.37	−549.47	124.10
	(513.65)	(573.87)	(507.46)
CFA share (observed biographies)	283.35	537.87**	−254.52
	(233.04)	(229.03)	(188.24)
Unknown-biography share	−203.10	128.36	−331.46
	(638.17)	(660.68)	(632.80)
Mean log(1 + experience)	−350.85***	−64.79	−286.06***
	(114.32)	(109.12)	(102.22)
Prior-year relative NAV return (pp)	−24.88***	−12.34	−12.54**
	(7.74)	(8.74)	(5.72)
Log(1 + prior relative volatility)	154.84	133.37	21.47
	(158.39)	(161.36)	(132.84)
Prior-quarter flow (pp)	-1.90×10^{-4}	-1.66×10^{-4}	-2.44×10^{-5}
	(1.78×10^{-4})	(2.14×10^{-4})	(1.31×10^{-4})
Missing prior return history	476.65	869.89	−393.24
	(703.99)	(707.38)	(313.40)
Missing prior-quarter flow	525.98	2148.66	−1622.68
	(1656.61)	(2456.82)	(1071.74)
Lagged expense ratio (bps)	−0.40	−0.27	−0.13
	(3.10)	(2.74)	(2.81)
Lagged log turnover	12.32	95.82	−83.50
	(218.38)	(249.56)	(226.64)
Top-ten equity concentration	−1782.25*	−906.59	−875.65
	(1028.03)	(1048.49)	(905.44)
Missing lagged expense ratio	278.98	139.03	139.95
	(497.80)	(430.77)	(428.19)
Missing lagged turnover	−69.38	15.78	−85.15
	(207.12)	(227.35)	(215.73)
Fund × matched-case FE	Yes	Yes	Yes
Matched-case × quarter FE	Yes	Yes	Yes
Observations	15,604	15,604	15,604
Country-appointment cases	430	430	430
Hiring funds	231	231	231
Fund clusters	514	514	514

Notes: The two-year specification of Equation (11), on common observations. Column 2 carries forward pre-hire country holdings. Column 3 is observed minus passive returns. Columns 2 and 3 add to column 1. Year 2 minus Year 1 is −194.48 (SE 97.25, significant at 5%), −110.55 (SE 97.35) and −83.94 (SE 71.76) in columns 1–3, using the joint covariance. All 430 cases remain, and 258 rows without paired returns are omitted, with each unit's phase weight divided over remaining quarters. Weights and controls follow Table 10. See Appendix B.10. Concentration is never missing in this sample. Standard errors, clustered by economic fund, appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table 12: Prior performance and flows at appointment

Dependent variable	Prior adjusted return	Prior fund flow
Units	Annualized basket bps	Quarterly pp
	(1)	(2)
Foreign-educated appointee	−86.41**	−1.83*
	(38.18)	(1.06)
Log pre-hire destination team size	6.17	0.30
	(65.17)	(1.02)
Pre-hire destination unknown education	246.69	1.10
	(172.46)	(2.91)
Log origin team size	−13.50	−2.83**
	(49.99)	(1.18)
Origin-team unknown education	66.20	−7.84
	(287.65)	(7.12)
Log origin country stock count	−80.15	−1.47
	(63.12)	(1.14)
Appointee: top-100 school	54.42	−0.89
	(35.41)	(0.73)
Appointee: school ranked 101–500	106.35**	0.10
	(50.63)	(1.10)
Appointee: female	−21.00	0.83
	(33.75)	(0.85)
Appointee: unknown gender	58.50	1.30
	(85.40)	(2.53)
Appointee: CFA	−56.18*	1.91**
	(31.32)	(0.78)
Log(1 + manager experience)	50.49*	−1.89***
	(29.18)	(0.60)
Prior origin relative NAV (pp)	—	1.73***
		(0.47)
Prior benchmark: China	238.93	—
	(246.40)	
Prior benchmark: Developed Asia-Pacific	−1851.57*	—
	(969.62)	
Prior benchmark: Developed Asia-Pacific ex Japan	238.91*	—
	(139.68)	
Prior benchmark: Developed Europe	569.03***	—
	(193.48)	
Prior benchmark: Emerging Americas	−358.95**	—
	(176.43)	
Prior benchmark: Emerging markets	34.22	—
	(104.25)	
Prior benchmark: Global growth	336.90*	—
	(192.76)	
Prior benchmark: Global large cap	−55.15	—
	(242.52)	
Prior benchmark: Global	−82.44	—
	(127.64)	
Prior benchmark: Global small/mid cap	14.81	—
	(187.87)	
Prior benchmark: Global value	−107.66	—
	(228.04)	
Prior benchmark: Global ex-U.S. growth	7.68	—
	(120.20)	
Prior benchmark: Global ex-U.S. small/mid cap	180.53	—
	(135.85)	
Prior benchmark: Global ex-U.S. value	−170.82	—
	(112.42)	
Prior benchmark: Japan	3214.77	—
	(2386.77)	
Hiring-fund FE	Yes	Yes
Appointment-year FE	Yes	Yes
Observations	1,321	1,322
Hiring-fund clusters	300	300

Notes: Equation (12). The reference group is U.S.-educated appointees. Returns cover all non-U.S. countries with usable returns, using size/past-return adjustment and equal country weights. Flows are benchmark-relative. The omitted benchmark is Global ex-U.S. Appointments have equal weight, and hiring-fund-clustered standard errors appear in parentheses. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table 13: Prior performance in potential country specialties

Dependent variable	Prior adjusted country return
Units	Annualized basket bps
	(1)
Appointee's education-country link	−218.86**
	(103.19)
Foreign-educated appointee	−55.14
	(35.89)
Log pre-hire destination team size	9.54
	(74.69)
Pre-hire destination unknown education	1035.08***
	(257.23)
Log origin team size	1.43
	(55.61)
Origin-team unknown education	113.97
	(254.52)
Log origin country stock count	−193.92**
	(76.70)
Appointee: top-100 school	−46.52
	(34.20)
Appointee: school ranked 101–500	6.88
	(47.37)
Appointee: female	55.12
	(44.47)
Appointee: unknown gender	−23.44
	(126.33)
Appointee: CFA	−21.10
	(32.48)
Log(1 + manager experience)	57.99
	(36.41)
Prior origin relative NAV (pp)	245.86***
	(22.87)
Continuing-team unknown education	−582.72**
	(251.81)
Hiring-fund × country FE	Yes
Country × appointment-year FE	Yes
Country-appointment observations	9,446
Appointments	1,083
Hiring-fund clusters	257

Notes: Equation (13). The country is held by the hiring fund before appointment and has no continuing manager educated there. The outcome is the recruit's previous teams' adjusted return in that country over four full pre-hire quarters. The education-country link equals one if the recruit studied there. The linked foreign-minus-U.S. comparison is −274.00 (SE 100.61), the sum of the two focal slopes with joint covariance. This model conditions on prior overall NAV performance. Each appointment has equal total weight across countries with observed prior performance. The 152 linked observations cover 145 appointments at 85 funds. Hiring-fund-clustered standard errors appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table 14: Career outcomes and fund performance

Dependent variable	Departure	New assignment	Departure	New assignment	New assignment	New assignment
Units	Probability in pp					
	(1)	(2)	(3)	(4)	(5)	(6)
Specification	Baseline	Baseline	Local return control	Local return control	Common sample	Common sample
Return window	One quarter	One quarter	One quarter	One quarter	One quarter	Four-quarter mean
Foreign-educated × fund return (pp)	0.145***	−0.107**	0.138***	−0.108**	−0.163***	−0.278***
	(0.044)	(0.053)	(0.044)	(0.054)	(0.054)	(0.101)
Local return observed	—	—	−0.17	0.65*	—	—
			(0.35)	(0.39)		
Local return observed × local return (pp)	—	—	0.017	0.002	—	—
			(0.013)	(0.016)		
Log recorded fund spell length	2.70***	−2.43***	2.71***	−2.43***	−0.01	0.01
	(0.28)	(0.35)	(0.28)	(0.35)	(0.58)	(0.58)
Female × fund return (pp)	0.071	−0.029	0.071	−0.030	−0.089	−0.352*
	(0.056)	(0.076)	(0.056)	(0.076)	(0.089)	(0.210)
Unknown gender × fund return (pp)	0.006	0.468**	0.007	0.469**	0.395**	0.633**
	(0.139)	(0.210)	(0.139)	(0.210)	(0.176)	(0.286)
CFA × fund return (pp)	−0.050	−0.044	−0.050	−0.044	−0.004	−0.106
	(0.043)	(0.050)	(0.043)	(0.050)	(0.057)	(0.134)
Unknown biography × fund return (pp)	−0.107	−0.577	−0.109	−0.580	−0.496***	−0.795***
	(0.248)	(0.380)	(0.249)	(0.380)	(0.188)	(0.301)
Log(1 + experience) × fund return (pp)	0.045	−0.027	0.045	−0.027	−0.062	−0.074
	(0.029)	(0.036)	(0.029)	(0.036)	(0.044)	(0.097)
Log(1 + manager experience)	−3.15***	1.74***	−3.15***	1.73***	2.23***	2.19***
	(0.41)	(0.56)	(0.41)	(0.56)	(0.77)	(0.77)
Fund × quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Fund × manager FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	154,677	154,677	154,677	154,677	116,550	116,550
Fund clusters	1,334	1,334	1,334	1,334	1,016	1,016

Notes: All columns estimate Equation (14). Outcomes are departures or new assignments in the following quarter, in probability percentage points. Columns 3–4 add the local-return-observed indicator and its interaction with local performance. Columns 5–6 share a common sample with four consecutive quarterly return observations and managers present at each quarter-end from $q-4$ through q, and fixed-effect singletons are excluded. In column 6, fund return is the arithmetic mean of quarterly benchmark-relative returns from $q-3$ through q, and every interaction with fund return uses this mean. The other columns use the quarter-q return. Return measures are in percentage points. Coefficients are probability points per return point. The foreign interaction measures a difference in performance responses relative to U.S.-educated colleagues. Columns 1–4 weight each fund-quarter equally across its managers. Columns 5–6 keep those weights without renormalizing them in the smaller sample. With the four-quarter measure, the departure interaction is 0.056 (standard error 0.108, discussed in Section 4.4), and the assignment interaction in column 6 has $p = 0.012$ after a Holm adjustment for testing both career outcomes. Standard errors, clustered by economic fund, appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%, without adjustment for multiple outcomes.

A Descriptive Statistics

Tables A1–A4 describe fund mandates, holdings and recorded education links. All tests other than the comparison in Appendix D start from the same 1,663 funds. Estimation samples differ with data availability and the variation required for each comparison.

B Supplementary Results

The supplementary tests address three questions behind the main interpretation. First, does the NAV association survive changes in controls and weighting, and can mandate fit or staffing timing explain it? Second, how do portfolio construction, country weights and calendar time affect the local and outside-country findings? Third, what follows linked appointments, and can commercial benefits, narrower prior-performance comparisons or country coverage explain the hiring and career patterns? The appendix follows that order. Panel models use economic-fund clustering. Portfolio alphas use the time-series inference in Section 3.4.

B.1 Country relevance measured before appointment

This extension of Equation (3) asks whether a poor match between managers' education countries and the fund's mandate explains the shortfall. If foreign-educated managers often run mandates in which their education countries carry little weight, the association should be more favorable where those countries matter more to the benchmark. Let a_{ih} denote the share of benchmark capital in manager i's distinct non-U.S. education countries, averaged over the four quarter-ends strictly before appointment h. Education includes undergraduate and graduate degrees. The denominator is total benchmark capital, including U.S. equities, and multiple degrees in the same country count once. All four quarters must have at least 95% of benchmark weight mapped to countries. Otherwise relevance is unknown. The recorded benchmark is fixed over time, so a_{ih} uses the historical weights of that index.

The added regressor Relevance_{jq} is the sum of measured a_{ih} across current foreign-educated managers divided by the number of current managers with known education. Unmeasured contributions are set to zero and identified separately by $\text{UnknownRelevance}_{jq}$, the share of known-education managers who are foreign-educated and have unknown relevance. We add $\theta\text{Relevance}_{j,q-1}$ and $\eta\text{UnknownRelevance}_{j,q-1}$ as separate terms to Equation (3), with its existing $\rho\text{ForeignShare}_{j,q-1}$ retained. Unknown education remains a distinct control. Relevance is fixed within each appointment spell. Changes in the current team's membership and composition generate variation within funds. No fund-month is dropped for missing relevance, which affects 50.4% of foreign manager-quarter observations.

Table A5, column 2, also includes foreign education-country shares. A manager with K distinct foreign education countries contributes $1/K$ to each, aggregated over the same team denominator. Their sum is ForeignShare, so the U.K. share is omitted.

The coefficient of interest, θ, measures whether a given foreign-educated share performs better when its education is more relevant to the mandate. The additional controls separate where managers studied from relevance. For example, the team share educated in Japan is distinct from how important Japan was to the benchmark before those managers joined. The change from column 1 to column 2 therefore reflects conditioning on the mix of education countries, rather than a new sample or risk model. The positive foreign-share coefficients describe the reference point of zero measured relevance, and in column 2 also of U.K. education. They are not the average share comparison in Table 2. The foreign share with unknown relevance separates managers by whether their benchmark's country weights are available before appointment, so we do not interpret its large negative coefficient.

The raw units of θ are annual NAV basis points per unit of team relevance. A unit change would move a fully foreign-educated team from education countries with no benchmark weight to countries that make up the entire benchmark, so we report the interquartile contrast. For a fully foreign-educated team with known relevance, this contrast, 0.1050θ, is -48.6 in column 1 and -72.3 in column 2, with 95% intervals $[-109.4, 12.2]$ and $[-122.7, -21.9]$. For a partly foreign-educated team, the contrast scales with the foreign-educated share when each foreign-educated manager's relevance changes by the same amount.

The contrast compares foreign-educated teams with one another and does not add to the −70-basis-point gap in Table 2. Returns are weaker, not stronger, where managers' education countries matter more to the mandate.

B.2 Further net-return specifications

The NAV model controls for the preceding quarter's flow. Quarterly flow is $\text{Flow}_{jq} = 100[\text{TNA}_{jq} - \text{TNA}_{j,q-1}(1 + R^{\text{net}}_{jq})]/\text{TNA}_{j,q-1}$, where TNA is quarter-end assets and R^{net}_{jq} the compounded quarterly net return in decimals. Thus Flow is in percentage points of beginning-quarter assets. The NAV model uses the preceding quarter's $\text{Flow}_{j,q-1}$ for the months of quarter q, without capping and with an indicator for missing flows.

We also check whether the NAV association depends on measured portfolio composition or extreme returns. Table A6 retains Equation (3), its sample and fixed effects. Column 1 adds portfolio size and momentum tilts, the holdings-weighted average size and past-return terciles (coded 0 to 2) of the fund's foreign stocks within the country groups that define the stock benchmarks in Section 2.4, to the model without expenses, turnover and concentration. Column 2 instead adds top-ten concentration and turnover. Missing-value indicators keep observations whose tilts, concentration or turnover are unavailable. Column 3 uses the full controls and winsorizes the factor-adjusted outcome at the 1st and 99th percentiles within benchmark-month cells. All use equal fund-month weights and economic-fund clustering. The focal education-share coefficients are −66, −69 and −69 annual basis points.

B.3 Staffing timing and selection sensitivity

Two diagnostics examine staffing timing and selection on observed characteristics. Funds are more likely to hire managers educated in a country after local losses (Section 4.1). The first diagnostic checks whether such staffing responses could produce the fund-level coefficient. If weak current returns forecast a future shift toward foreign-educated managers, the current share could stand in for that later response because team composition is persistent. Future team composition would then help explain the current association. We add future foreign-educated share and future unknown-education share to Equation (3), measured four quarters after the current team, as $\rho_{+4}\text{ForeignShare}_{j,q+3}$ and $\xi_{+4}\text{UnknownShare}_{j,q+3}$, where q is again the quarter containing month m.

The controls X are those of Equation (3), including the current unknown-education share. The coefficient ρ_{+4} measures the additional predictive content of future foreign-educated share, conditional on the current team and controls. On the 193,986 fund-months with teams observed four quarters ahead, the current-share coefficient is −75 basis points (Table A7, column 1). Including future foreign-educated and unknown-education shares leaves it at −72. The future foreign-educated share has a coefficient of −3 with a standard error of 37 (column 2). The current-share coefficient is thus stable when future team composition is included. The future share shows no clear additional association with current returns.

The second diagnostic checks whether observed fund characteristics account for the negative coefficient. With only the unknown-education fraction controlled, the estimate is −59 basis points (Table A7, column 3). Adding the other controls makes it −70 (Table 2). Within the proportional-selection calibration of Oster (2019), the adjustment at equal observed and unobserved selection gives −73 basis points, while the selection parameter needed for a zero coefficient is −20.6.[32] Because the observed controls move the coefficient away from zero, unobserved factors would have to push it in the opposite direction, with about 20 times their influence, to bring it to zero. Both checks show a coefficient that is stable to the staffing and fund conditions measured in Equation (3).

B.4 Performance beyond the initial appointment period

To test whether the NAV association is confined to newly appointed managers, Table A8 modifies Equation (3) by replacing $\text{ForeignShare}_{j,q-1}$ with three shares, $\text{ForeignShare}^{\text{new}}_{j,q-1}$, $\text{ForeignShare}^{\text{est}}_{j,q-1}$ and $\text{USShare}^{\text{new}}_{j,q-1}$. Their coefficients are ρ^{new}_F, ρ^{est}_F and ρ^{new}_{US}, respectively.

[32] We set $R_{\max} = 1.3$ times the within-fund R^2 of the controlled regression.

The three shares count new foreign-educated managers, established foreign-educated managers and new U.S.-educated managers, respectively, divided by all managers with known education. Established U.S.-educated managers form the omitted category. New means within the first two years (eight quarters) at the fund, and established means later. As in the NAV equation, q contains the return month and the shares are measured at quarter-end $q-1$. The reduced controls X contain unknown education, team and asset scale with spline terms, and the five team traits. Fund and benchmark-month effects, the predetermined thirteen-factor outcome and fund clustering are retained. OLS and precision-weighted columns use the same model. The focal established-share coefficient ρ_F^{est} is -81 annual basis points in OLS and -82 in WLS, significant at 5% and 1%. The new foreign-minus-U.S. contrast, $\rho_F^{\text{new}} - \rho_{US}^{\text{new}}$, is -35 in OLS and imprecise. The NAV association is present among teams with established foreign-educated managers.

B.5 Alternative local-portfolio constructions

Table A9 varies how conviction is measured, which holdings and education links enter, and the risk model. Each column changes one element of the baseline and re-estimates the top-portfolio alpha with Equation (8).

The first two columns change the conviction measure. The baseline measure in Equation (7) subtracts each stock's capitalization share among the fund's holdings in the country from its portfolio weight within that country. Column 1, the benchmark reference, uses the assigned benchmark's weights instead, renormalized over the same held stocks:

$$I^B_{jkq} = \frac{\omega_{jkq}}{\sum_{k' \in \Omega_{jcq}} \omega_{jk'q}} - \frac{b_{jkq}}{\sum_{k' \in \Omega_{jcq}} b_{jk'q}},$$

where b_{jkq} is stock k's weight in fund j's assigned benchmark at quarter-end q. When the benchmark holds none of the fund's stocks in the country, the second term is dropped and positions are ranked by their weights within the country. Column 2, the two-year reference, keeps capitalization weights but widens the reference set from the stocks held at q to Ω^8_{jcq}, all stocks the fund held in the country at any quarter-end from $q-8$ through q:

$$I^U_{jkq} = \frac{\omega_{jkq}}{\sum_{k' \in \Omega^8_{jcq}} \omega_{jk'q}} - \frac{\text{MC}_{kq}}{\sum_{k' \in \Omega^8_{jcq}} \text{MC}_{k'q}}.$$

In both columns, the top quartile of the alternative measure within each fund-country-quarter defines the high-conviction positions.

The remaining columns keep the baseline measure. Column 3 removes U.K. equities from portfolio construction, because the United Kingdom is the most common foreign education country. Column 4 defines education countries by undergraduate degrees only. Columns 5 and 6 add developed and emerging momentum factors, retaining or omitting the common portfolio benchmark. Column 7 retains only that benchmark, and column 8 deducts estimated bid-ask trading costs. Each column reports one top-portfolio regression with all loadings. The table notes report the paired Wald tests of the top-minus-ordinary alpha. The baseline is shown only in Table 6.

Alphas range from 2.75% to 5.74%. The top-minus-ordinary spread remains significant at 5% in columns 3 through 8. It is marginal with the two-year reference ($p = 0.057$) and imprecise with the benchmark reference ($p = 0.251$).

B.6 Contribution of local positions and country weighting

This subsection measures what education-country positions add to the funds' returns and explains why that contribution is much smaller than the strategy alphas in Section 3.4 would suggest. We measure the contribution per dollar of pooled covered equity capital, which is the value of sample funds' disclosed stocks with usable returns, including U.S. stocks. Let s^L_m be the education-country share of covered equity capital in month m and R^L_m the return on those holdings at actual capital weights. Their monthly contribution to the covered equity return is

$$C^L_m = s^L_m R^L_m, \qquad R^{\text{all}}_m = C^L_m + C^O_m. \tag{15}$$

Here R^{all}_m is the return on the entire covered equity portfolio and C^O_m the contribution of the other covered equities. The annualized raw contribution is the average of C^L_m over the 300 months, times twelve, expressed

in basis points. Weighting each month's return by that month's capital share captures comovement between capital shares and returns, which multiplying the average share by the average return would miss. To measure the contribution beyond risk exposures, we use the excess-return contribution $C^{\text{ex}}_{pm} = s_{pm}(R_{pm} - R^f_m)$ as the dependent variable in Equation (8), where s_{pm} is the share of covered equity capital in group p, all local, top local or ordinary local positions, and R_{pm} is now the group's return at actual capital weights. The intercept is then the alpha these positions add per dollar of pooled covered equity, rather than per dollar invested in them (Table A10).

Education-country holdings represent 8.7% of covered capital. They contribute 57.5 basis points a year of raw equity return but no clearly positive alpha, as their alpha contribution is −10 basis points (standard error 11). The high-conviction positions occupy 3.6% of covered capital, and their alpha contribution is 0.3 basis points a year (standard error 6.1), well below the 14 basis points implied by applying their 4.0% strategy alpha to that share.

The difference comes from country weights. The 4.0% strategy alpha holds the selected stocks in benchmark country proportions. Table A11 re-estimates Equation (8) for six portfolios built from identical fund-country-month cells. Columns 1–3 use benchmark country shares for top, ordinary and combined local holdings. Columns 4–6 use actual capital weights. Risk factors, the common portfolio benchmark and time-series inference are those of Table 6. Because the stock baskets are the same, the comparison changes only the capital assigned to their countries. The combined portfolios are estimated directly because the capital shares of top and ordinary holdings change over time.

On 77,188 fund-country-month cells at 749 fund ISINs, the top local portfolio earns 407 basis points a year under benchmark country weights but 17 (standard error 159) at funds' actual country weights. Actual weights lower the combined local alpha by 234 basis points, significant at 1% ($p = 0.007$). The strategy alpha thus measures the value of the stock choices. The contribution measures how much of that value reaches the portfolio at the allocations funds choose. The comparison is per dollar invested in education-country holdings, not per dollar of total fund capital.

B.7 Changes in the country-return gap over time

This subsection tracks the foreign-versus-U.S. comparison in outside countries over the twenty-five-year sample. To make the local and outside comparisons consistent, we retain the full country panel in Equation (6) and allow both education associations to vary with calendar time:

$$\begin{aligned} R_{jcq} = \alpha_{jc} + \lambda_{cq} + \beta \mathbb{1}^{\text{Link}}_{jcq} + \gamma \text{ForeignShare}_{jq} \\ + \beta^T \mathbb{1}^{\text{Link}}_{jcq} \times t_{q+1} + \gamma^T \text{ForeignShare}_{jq} \times t_{q+1} + \Gamma' X_{jcq} + \varepsilon_{jcq}. \end{aligned} \tag{16}$$

Here t_q is the year of quarter q minus 2012, in single years. The superscript T marks the time-interaction coefficients. Holdings and education at quarter-end q predict returns in $q+1$, so the interactions use the payoff year t_{q+1}. The sample, controls, weights, fixed effects and clustering are those of the joint country model in Section 3.3.

The outside-country gap at date t is $g_{\text{out}}(t) = \gamma + \gamma^T t$. For a fully foreign-educated team in an education country, the comparison with a U.S.-educated team is $g_{\text{local}}(t) = \beta + \gamma + \beta^T t + \gamma^T t$. Standard errors for these sums use the joint covariance of their component estimates. The endpoint gaps are fitted values from the full-sample trend regression.

The outside-country gap becomes more negative over time. In Table A12, column 1, γ^T is −7.06 annualized basket basis points per calendar year, significant at 5%. The fitted gap reaches −124.5 by 2024, with a standard error of 48.3. Inverse-residual-variance weighting gives a similar deterioration and endpoint gap (column 2). Figure A1 plots the outside and local comparisons from these same regressions. The late-sample outside shortfall is about 1.25 percentage points per dollar of the country basket.

The education-country comparison remains imprecise in 2024 under both weighting schemes. The evidence for local expertise comes from the direct conviction-portfolio contrasts in Section 3.4.

B.8 Familiarity results with raw country returns

The adjusted country return in Equation (2) subtracts the return on a matched portfolio of local stocks with similar size and past returns. Table A13 re-estimates Equation (9) with the raw return on the fund's holdings in each country instead. The observations, controls, fixed effects, weights and clustering are those of Table 7. Country-quarter effects remain, so the comparison is still among funds holding the same country in the same quarter. Only the matching on stock size and past returns is dropped.

The familiarity pattern remains. The difference between the foreign- and U.S.-educated slopes, θ, is -346 basis points for a shared border (standard error 155, significant at 5%), -195 for a common official language (standard error 69, significant at 1%) and -81 basis points per log point of social connectedness (standard error 41, significant at 5%). The foreign-educated slopes, $\delta + \theta$, are -157 for borders (standard error 74, significant at 5%), -62 for official language (standard error 64) and -35 per log point of social connectedness (standard error 17, significant at 5%). The raw differences are 15 to 30 percent smaller than those in Table 7. For borders, the change comes mainly from U.S.-educated teams' holdings in Canada and Mexico, whose size and past-return characteristics earned lower returns. Without the stock-characteristic adjustment, the U.S.-educated border slope falls from +300 to +188 basis points, while the foreign-educated slope moves only from -184 to -157.

B.9 Performance following linked appointments

Section 4.2 assesses the payoff to seeking local expertise after disappointing performance. Starting with the fund-country-quarters of Equation (10), we retain appointments of managers educated in the held country (Table 8, column 1) through 2023Q4, allowing four subsequent payoff quarters through 2024Q4. Disappointing prior performance means a negative twelve-month country-market return or a negative own country-relative return. Each appointment is matched to up to three other economic funds sharing the investment country, assigned benchmark, appointment quarter and disappointing-performance status, with no appointment that quarter. We select the closest comparison funds using standardized differences in prior country-relative returns, log country capital, log stock count and incumbent education coverage. Matching uses only information available before appointment. Comparison funds may already employ foreign-educated managers and may hire later.

Equation (11) compares changes within each fund-case with changes at its matched funds in the same payoff quarter. We retain quarters -4 through -1 and $+1$ through $+4$, requiring at least three jointly observed quarters on each side and omitting quarter zero. The common panel contains 13,237 observations in 511 cases, 267 hiring funds and 610 economic-fund clusters. Requiring observations with all three outcomes does not change the matches. Comparison funds that lack these observations are not replaced. Each hiring fund-quarter has equal total weight, divided across its country cases, hiring and comparison sides, pre- and post-appointment phases, and observed quarters within each phase. The controls are those of the main NAV model. Its foreign-share regressor is replaced by the hiring-by-post interaction. The missing-concentration indicator is zero throughout this panel and drops out. Fund clustering accounts for repeated hiring and comparison roles.

The local outcome follows the characteristic-adjusted country-return construction in Section 2.4. The outside outcome pools the corresponding adjusted country returns at their monthly capital weights. For each case, the education-country set consists of the non-U.S. education countries of the hiring team's current managers and dated appointees in the appointment quarter. We hold this set fixed for all quarters and apply it to both sides of the comparison, so appointment does not mechanically reclassify holdings. Outside countries here are outside that common set, rather than outside each comparison fund's own education countries. NAV uses the factor-adjusted monthly return of Equation (1), averaged within the actual payoff quarter. All outcomes use the return construction of the main tests. Complete stock-return coverage is not required and missing country baskets are not assigned zero returns.

Table 9 reports one regression per outcome on exactly these observations. Section 4.2 discusses the estimates. Using the same design across outcomes shows whether improvement after a linked appointment extends beyond the target country. This comparison concerns changes around a hire, whereas the main NAV coefficient concerns team composition over the full sample.

To assess persistence, Table 10 extends the same matches through eight subsequent quarters. Appointments must occur by 2022Q4 to allow two full years within the sample period. We require at least three jointly observed local-return and NAV quarters in each of three phases, quarters -4 to -1, $+1$ to $+4$ and $+5$ to $+8$, omitting quarter zero. The common cohort contains 430 country-appointment cases at 231 hiring funds, 1,339 fund-case units, 15,862 observations and 514 economic-fund clusters. We keep the matches of the one-year comparison, without requiring that recruits stay or that comparison funds refrain from later hiring. The cohort is smaller than the first-year comparison because appointments must occur earlier and returns must be observed in all three periods.

We replace $\beta_H \mathbb{1}_{ja}^{\text{Hire}} \times \mathbb{1}_{at}^{\text{Post}}$ in Equation (11) with $\beta_{H,1} \mathbb{1}_{ja}^{\text{Hire}} \times \mathbb{1}_{at}^{\text{Year1}} + \beta_{H,2} \mathbb{1}_{ja}^{\text{Hire}} \times \mathbb{1}_{at}^{\text{Year2}}$. The year indicators select quarters $+1$ to $+4$ and $+5$ to $+8$, respectively. Both coefficients measure improvement relative to the pre-appointment year and to comparison funds. Year two is the average return during months 13–24, not the cumulative two-year return. The controls and fixed effects are unchanged, and equal hiring-fund-quarter weights are divided across cases, hiring/comparison sides, the three phases and observed quarters. The contrast $\beta_{H,2} - \beta_{H,1}$ uses their joint fund-clustered covariance. The estimated local decline is -205.34 basis points (standard error 96.72, $p = 0.034$). The NAV contrast is $+26.03$ (standard error 34.02), so the local fading pattern does not establish a decline in whole-fund performance.

B.10 Inherited holdings and the source of local recovery

The appointment comparison conditions on disappointing prior country performance. A subsequent rebound can therefore come from stocks already held, even if new stock choices add no value. Table 11 separates these sources using the same appointments, comparison funds, controls, fund-case effects, case-quarter effects and fund clustering as Table 10.

Let the appointment quarter be zero. For post-appointment returns, the passive reference portfolio is the country basket at quarter-end -1. Its stocks are held without rebalancing, with capital weights drifting by observed returns through each evaluated month. For the pre-appointment year, the reference is quarter-end -6, which gives the four pre-period formation quarters the same one-to-four-quarter distances from their reference as the first post-appointment year. The second post-appointment year continues to use the pre-hire reference, so it compares the evolving observed holdings with a progressively older passive portfolio. Monthly returns on both portfolios are adjusted by the same contemporaneous size-and-past-return benchmarks used in Equation (2), and are averaged and annualized in basis points. Missing evaluated returns are excluded and each portfolio is normalized over its covered capital. A missing intermediate return leaves that stock's weight unchanged during the carry-forward calculation.

For each observation, write the adjusted return on observed holdings as R, the passive return as R^P, and their difference as R^Δ. The difference can be further divided into returns on stocks added since the reference date, C^B, the foregone passive returns on stocks no longer held, C^S, and the change in contributions from retained stocks, C^W:

$$R = R^P + R^\Delta, \qquad R^\Delta = C^B - C^S + C^W. \tag{17}$$

Each term is measured per dollar of the covered country basket. Reweighting retained stocks includes the effect of normalizing the observed and passive portfolios over their respective covered capital. These terms describe differences between quarterly holdings.

We estimate the two-year version of Equation (11) separately for each outcome on the same observations. The 258 observations without both returns are excluded, retaining all 430 cases, 1,339 fund-case units and 514 fund clusters. Each unit's original phase weight is divided over its remaining observations. Because the regressions use the same design matrix and weights, their coefficients obey Equation (17) exactly. Tests of sums and differences use the joint fund-clustered covariance. Table A14 also reports a version with stricter data requirements.

The first-year improvement of 178.45 basis points comprises 174.37 for the passive portfolio and 4.08 for portfolio changes. Dividing by 178.45 gives point-estimate shares of 97.7% and 2.3%. A Fieller interval using the joint covariance is 7.3% to 364.3% for the passive share, and an accounting share can exceed 100% when the other component is negative. The informative comparison is that the estimated gain is already present in inherited stocks and the increment from portfolio changes is imprecise.

Table A14 checks whether the conclusion depends on return availability or the horizon. Columns 1–2 require at least three jointly observed quarters per phase for every fund-case unit and restore equal hiring-fund-quarter weights among the retained cases. This leaves 401 cases and 481 fund clusters. The first-year coefficients are 128.92 for observed holdings, 107.99 for the passive portfolio and 20.93 for portfolio changes, each imprecise. Columns 3–4 instead use the larger one-year appointment cohort and drop only unavailable paired observations. On 511 cases and 610 fund clusters, the corresponding estimates are 163.67, 250.07 and −86.40 basis points. Because these use only quarters in which both portfolio returns are observed, the observed-holdings estimate differs from the 228 in Table 9. In the table, the observed-holdings coefficient is the sum of the passive and portfolio-change columns. Both comparisons attribute most of the point estimate to the inherited portfolio, with no precisely estimated positive increment from changes in holdings.

Finally, we estimate separate coefficients for each event quarter on the common observations. A joint test that the pre-appointment coefficients are zero gives $p = 0.467$ for observed holdings and $p = 0.046$ for the passive portfolio. Most of the estimated first-year relative recovery can be earned by carrying forward stocks selected before the appointment, consistent with mean reversion in inherited holdings.

B.11 Monthly flows controlling for prior performance

A commercial payoff could make the background valuable even without stronger investment performance. Investor preferences can shape fund incentives (Gantchev et al., 2024). The monthly flow test separates a potential commercial value of foreign education from investors' responses to prior returns. Let TNA_{jm} be end-of-month assets and R^{net}_{jm} the monthly net return in decimal units. Flow in percentage points is $\text{Flow}_{jm} = 100[\text{TNA}_{jm} - \text{TNA}_{j,m-1}(1 + R^{\text{net}}_{jm})]/\text{TNA}_{j,m-1}$. Before estimation, this monthly flow measure is winsorized at the pooled 1st and 99th percentiles. The cutoffs appear in Table A15. This treatment applies to the monthly flow outcome, not the raw quarterly flow control in the NAV model. We estimate

$$\begin{aligned}\text{Flow}_{jm} = \alpha_j + \lambda_{bm} + \delta\text{ForeignShare}_{j,q-1} \\ + \Gamma'_S S_{j,m-1} + \Gamma'_X X_{j,m-1} + \Gamma'_Z Z_{j,q-1} + u_{jm}.\end{aligned} \tag{18}$$

$\text{Rank}_{j,m-1}$ is the zero-to-one percentile rank of the twelve-month compounded return ending in $m-1$ among same-benchmark funds. For brevity, we omit the rank's subscripts below. The three terms in $S_{j,m-1}$ are $\min(\text{Rank}, .2)$, $\min(\max(\text{Rank} - .2, 0), .6)$ and $\max(\text{Rank} - .8, 0)$. These terms allow flows to respond differently to poor, middle and strong prior performance. X contains log lagged assets, log fund age, the standard deviation of the preceding twelve monthly returns, lagged annual expenses and log(1+turnover). Z contains log team size, unknown-education share and the five team traits. As in Equation (3), b is the assigned benchmark and team measures use the quarter-end $q-1$ preceding the quarter q that contains month m. The observed-flow sample has 185,748 months at 1,494 funds.

The coefficient δ measures the additional flows associated with foreign-educated representation among funds with comparable prior performance and characteristics. Table A15 gives $\delta = 0.207$ monthly percentage points for the full-share difference, with a standard error of 0.153 and a 95% interval from −0.093 to +0.508. A positive commercial premium is possible but not precisely established. Column 2 omits expenses and turnover, which may themselves reflect staffing choices, on the same observations. The foreign-share coefficient remains 0.21 percentage points (standard error 0.15). The quarterly and appointment comparisons in Table A16 use their own outcome definitions and samples. With the prior-flow record in Section 4.3, these tests cover fundraising both before and after appointment.

B.12 Flows and fees at quarterly and appointment horizons

Table A16 looks for a commercial payoff at quarterly and appointment horizons. The outcomes $Y^{(v)}_{jq}$ pair the team observed at quarter-end q with subsequent flows and with the reported expense ratio, namely flow in quarter $q+1$ (columns 1–2), the sum of flows over quarters $q+1$ to $q+4$ (column 3), and the expense

ratio reported for the previous year (column 5). Flow is measured in percentage points of beginning-quarter assets and expenses in annual basis points. For each outcome v, we estimate

$$Y_{jq}^{(v)} = \alpha_j + \lambda_{bq} + \delta_v \text{ForeignShare}_{jq} + \Gamma_v' X_{jq} + u_{jq}^{(v)}. \tag{19}$$

Fund effects α_j and benchmark-quarter effects λ_{bq} follow Equation (3). The focal coefficient δ_v compares an entirely foreign-educated team with an entirely U.S.-educated team, conditional on the stated controls. A positive flow coefficient would indicate greater subsequent asset gathering. A positive expense coefficient would indicate a higher reported fee rate. The latter is a comparison of existing fees, not a subsequent fee change. The controls X contain UnknownShare, log team size and lagged assets with their spline terms, and the five team traits. Column 2 adds R_{jq}^{past} and the interaction $\text{ForeignShare}_{jq} \times R_{jq}^{\text{past}}$ as separate regressors, where R_{jq}^{past} is the prior-year relative NAV return, the same variable as the NAV-model control, in annual percentage points. It differs from the single-quarter career signal R^{rel}. Its interaction measures the foreign-minus-U.S. difference in flow-performance sensitivity.

Column 4 asks the more specific question of whether foreign-educated appointments are followed by stronger fundraising. Its outcome is the sum of flows over the four quarters after appointment. In Equation (19), it replaces current-team ForeignShare_{jq} with the appointee fraction s_{jq}^H and benchmark-quarter effects with appointment-year effects. We denote the appointee-share coefficient by δ_H.

The fraction s_{jq}^H is the foreign-educated share of known-education appointees, from zero to one. Thus δ_H compares subsequent four-quarter flows after all-foreign versus all-U.S. appointments, conditional on prior fund conditions. The controls X contain pre-appointment four-quarter flows, the prior-year relative NAV return, lagged assets, the size and traits of the hiring fund's team before the appointment, and unknown education among new appointees. All commercial models use equal observation weights and cluster by economic fund. The full-share differences are -0.29 flow percentage points next quarter, -0.26 over four quarters, and $+0.22$ over four quarters after appointment. The expense difference is -0.39 annual basis points. These estimates do not establish a commercial premium. The flow-performance interaction is -0.04 percentage points per annual NAV percentage point (standard error 0.06), also imprecise. It tests whether investors also respond less to performance when more managers are foreign-educated.

B.13 Robustness of the prior-performance comparisons

The main hiring comparison controls for the benchmarks of recruits' previous funds. The checks here ask whether the prior-performance gap persists after a finer comparison with other funds facing the same investment opportunities, or after comparing only recruits appointed together. Both use outside-country performance. The main comparison also includes education countries. For each previous fund's country-quarter return, we subtract the mean adjusted return of other economic funds sharing its assigned benchmark, investment country and quarter. All share classes of the previous fund are excluded. Peer share classes are combined before averaging economic funds. We then apply the same equal-country, equal-quarter and equal-economic-origin weights to form $\bar{R}_h^{\text{peer}}$, which replaces $\bar{R}_h^{\text{pre}}$ as the outcome in Equation (12).

The outside-country measure and its peer-adjusted counterpart use the same countries, 1,321 appointments and controls, with hiring-fund and year effects, equal appointment weights and fund clustering. Neither controls for prior NAV. The main all-country measure adds education-country holdings for the same appointments. Benchmarks are those recorded in the data.

The appointee controls in both columns of Table A17 are those of Equation (12). The school-rank indicators mark degrees from the top 100 and from ranks 101–500 of the 2003 Academic Ranking of World Universities, with other schools omitted. The peer-adjusted foreign-education coefficient is -71.8 annualized basket basis points, with a standard error of 41.0, significant at 10% (Table A17, column 1). The gap remains negative after comparing the previous teams with funds facing the same country, quarter and benchmark conditions. The corresponding outside-country estimate without peer subtraction is -100.8. The all-country main estimate is -86.4. Comparing the three estimates separates the effect of peer adjustment from that of including education-country holdings.

A separate comparison is limited to recruits accepted together. Table A17, column 2 uses the outside-country version of $\bar{R}_h^{\text{pre}}$ as the outcome in Equation (12), replaces separate fund and year effects with

hiring-fund-by-appointment-quarter effects α_{jq}, where q is the appointment quarter, and omits $\mathbf{B}_h$. We denote its foreign-education coefficient by δ_{round}.

Hiring-fund-by-appointment-quarter effects hold common recruitment conditions fixed. The hiring team's size and unknown-education share are absorbed. This model uses all available outside-country returns and no origin benchmark or prior NAV control. Equal appointment weights and hiring-fund clustering are retained. On 832 appointees, δ_{round} is -81 annualized basket basis points, with a standard error of 41 ($p = 0.053$). The negative gap remains among appointees accepted together, providing a separate check on recruitment conditions from the origin-mandate adjustments above.

The education-country comparison in Equation (13) holds the previous funds' overall performance fixed through their benchmark-relative NAV return. The gap does not depend on that choice. On the 995 appointments observed in at least two such countries (of the 1,083 in Table 13), the gap $\beta_G + \beta_L$ is -251 annualized basket basis points with the prior-NAV control, -268 without it and -274 when the previous funds' benchmark shares replace it, as in Equation (12). The standard errors are 84, 87 and 94, and all three estimates are significant at 1%.

B.14 Country coverage and career evaluation

A fund may retain a specialist because colleagues cannot replace the country knowledge. We ask whether the relative departure response is concentrated among managers who supply country knowledge that no colleague shares. Table A18 separates the career comparison by education coverage. A foreign-educated manager has unique coverage if the fund holds at least one of the manager's education countries in which no other current manager studied. Coverage is duplicated when the fund holds some of the manager's education countries but a colleague also studied in each of those held countries. The manager has no match when the fund holds none of them, and coverage is unclassified when prior holdings are missing. These categories are measured each quarter from the previous quarter's holdings and the current team. The equation replaces $\mathbb{1}_i^{\text{Foreign}} \times R_{jq}^{\text{rel}}$ in Equation (14) with category interactions:

$$100 \times \mathbb{1}_{ijq}^{\text{Exit}} = \mu_{jq} + \nu_{ij} + \sum_g \kappa_g \mathbb{1}_{ijq}^{g} \times R_{jq}^{\text{rel}} + \sum_{g \neq \text{NoMatch}} \eta_g \mathbb{1}_{ijq}^{g} + \Gamma' X_{ijq} + e_{ijq}. \tag{20}$$

Here g indexes unique, duplicated, no-match and unclassified coverage, and $\mathbb{1}_{ijq}^{g}$ is the corresponding foreign-educated-manager indicator. U.S.-educated colleagues are the comparison. Because the four indicators sum to $\mathbb{1}_i^{\text{Foreign}}$, which the fund-manager effects absorb, the no-match level is omitted, and all four performance interactions enter alongside the career controls, fund-quarter and fund-manager effects, the weights of Equation (14) and fund clustering.

The κ_g coefficients, relative exit responses per percentage point of better performance, are +0.180 for unique coverage, +0.006 for duplicated coverage and +0.122 for managers whose education countries the fund does not hold. A ten-point deterioration lowers the foreign-minus-U.S. departure gap by 1.80, 0.06 and 1.22 percentage points. The unique and no-match slopes are each significantly different from zero, but the differences between groups are imprecise, and a joint test that the duplicated and no-match slopes are both zero gives $p = 0.141$. Relative protection is thus not confined to managers who supply country knowledge that no colleague has. It also appears among managers whose education countries the fund does not hold.

Column 2 drops the interactions of performance with the other team traits, on the same sample. The unique-coverage slope remains +0.177 (standard error 0.056), significant at 1%. The no-match slope is +0.113 (standard error 0.061), significant at 10%. The similar point estimates preserve the pattern, while the no-match comparison is less precisely estimated.

C Variable Definitions and Timing

Table A19 defines the main variables. A basis point is one hundredth of a percentage point. Coefficients on ForeignShare compare foreign-educated shares of one and zero among managers with known education, holding the unknown-education fraction fixed. Fund returns, adjusted country returns and portfolio alphas

are all annualized but have different capital denominators and cannot be added. Missing education is coded as unknown, and an unavailable return is not treated as zero.

Fractions are coded from zero to one unless a percentage-point unit is stated. Return controls and signals use percentage points, and calendar interactions use years from the stated reference date. Social connectedness enters in log units without standardization. Economic changes such as ten percentage points or one standard deviation are calculated in the text. Log experience is $\log(1 + \text{years of experience})$. The team measure is its mean. The two nonlinear asset terms are restricted-cubic-spline bases in log lagged assets, with knots $(0.5921, 4.1586, 5.8772, 8.4417)$. The corresponding team-size knots are $\log(1, 2, 3, 5)$. The linear log term is reported separately.

The prior relative NAV control in the NAV and quarterly commercial models is the preceding year's annualized monthly mean in percentage points. The monthly flow model instead uses ranks of compounded twelve-month returns. Prior relative volatility is annualized in percentage points before applying $\log(1 + \cdot)$. The main and peer-adjusted prior-performance models contain the previous funds' benchmark shares and no prior NAV control. The prior-flow and education-country models instead control for the previous funds' mean quarterly benchmark-relative NAV return, in percentage points, over the quarters from five to two quarters before the appointment quarter. Prior flow enters directly in percentage points of beginning-of-quarter assets, with a missing-information indicator. Separate indicators identify observations with unavailable controls, allowing those observations to remain in the sample.

Table notes identify regressors absorbed by fixed effects. The factor exposures used to construct factor-adjusted NAV are estimated before each evaluated return and enter the outcome, so they are not reported as regression coefficients. The portfolio-alpha tables report every factor loading, in dimensionless units, alongside annualized alpha.

D Comparison with Jagannathan et al. (2022)

D.1 Samples, education links and measures

Jagannathan et al. (2022), hereafter JJK, study 1,090 U.S. actively managed international mutual funds over 1991–2014 and identify a manager's home country by the country of undergraduate education. Managers whose home country is linked to a fund's geographic mandate are home-linked, and their funds are home-linked funds. Their "Global" group contains all international funds, including the Regional and Country groups, and Regional funds include Country funds, so the three groups are nested. We keep these definitions in every direct reconstruction. Our main analysis differs in three ways. It covers 1,663 funds over 2000–2024, uses all recorded post-secondary education, and separates characteristic-adjusted stock selection from country allocation.

We rebuild JJK's tests in four samples, using their fund list over 1991–2014, the same list over 2000–2014, our funds over 2000–2014 and our funds over 2000–2024. The second and third samples differ only in the fund universe, and the fourth adds later years and later entrants. The 1,663 funds are all distinct sample funds observed over 2000–2024. Each window contains only the funds active and usable in that window, and Table A20 reports those counts. Of the 1,090 identifiers on JJK's list, 1,066 match share classes in our Morningstar files, 1,021 have stock holdings, and 884 also have a current manager with a recorded undergraduate country and lagged assets at some date. Holdings and managers are linked by economic fund, so share classes are not counted twice. Education comes from the records behind the main text, which rest on a manager-by-manager review of the biographies. Because JJK's coding is not available to us, we also report an earlier, less thoroughly reviewed coding of the same biographies, which shows how sensitive the reconstruction is to coding choices.

JJK limit their sample to funds with Morningstar information on the country of managers' bachelor's degrees. Applying the same rule to their list gives 24,878 fund-quarters and 464,242 held fund-country-quarters over 1991–2014, against their 24,422 and 464,578, and 401,906 observations with fund controls, against their 401,785. Without the rule the panel has 587,995 observations. Similar sample sizes do not establish that the same funds and positions enter, and the home-linked measures are somewhat higher in our data. Home-linked funds are 34% of fund-quarters against their 28%, and the mean home-linked manager share is 0.0107 against their 0.0092. The earlier coding gives 30% and 0.0087.

Four elements of JJK's design remain approximate. First, their mandate rule depends on each fund's geographic objective. Our baseline treats a country as within the mandate if the fund holds it at formation, and Table A21 reports the alternative that uses category mandates. Second, Morningstar categories are a static snapshot, not a dated history. Third, JJK assign firms by domicile, while we assign them by primary business activity. The ISIN-prefix check in Table A21 uses an alternative country definition, which is not identical to domicile. Fourth, the Global ex-U.S. factor series we hold, from the June 2019 version of Ken French's library, ends in that month. For the 2000–2024 column we therefore use the Developed ex-U.S. series. Over the 234 months through June 2019 the all-international long–short alpha is −14.1 basis points ($t = -1.13$) with the original factors and −13.8 ($t = -1.10$) with the extended factors on the same months, so the change of factor file has little effect.

D.2 Home-country stock selection: JJK Table 4

JJK compare home-country stocks held by home-linked funds with stocks from the same countries held by unlinked funds in the same Morningstar category. The calendar-time portfolios use positions at the previous quarter-end and are held for three months. Within a fund, stocks are weighted by dollar holdings. Across funds, baskets are weighted by total net assets at the end of the previous quarter. We follow this two-stage weighting using monthly fund assets. We treat a fund as unlinked if none of its managers is home-linked. JJK report a four-factor alpha of 31 basis points per month for the all-international long–short portfolio ($t = 1.69$, $p \approx 0.09$), which their table marks at the 10% level and their text describes as insignificantly positive, and 65 and 59 basis points in the Regional and Country groups ($t = 2.21$ and $t = 2.29$).

Table A20 reports the reconstruction. On JJK's list over 1991–2014, the long-only alpha is +10.6 basis points ($t = 0.56$) and the all-international long–short alpha is −24.0 ($t = -1.22$, $p = 0.22$). JJK's 31 lies outside this estimate's 95% interval of [−62.8, +14.8]. The Regional and Country long–short alphas are +23.4 ($t = 0.81$) and −28.5 ($t = -0.81$). The estimates in the later columns are −14.6 ($t = -1.02$) on JJK's list over 2000–2014, −21.3 ($t = -1.33$) on our funds over the same years, and −7.9 ($t = -0.73$) on our funds over 2000–2024. All four all-international long–short estimates are negative and none is statistically distinguishable from zero. Funds eligible by 2014 and followed through 2015–2024 earn −2.4 ($t = -0.24$), which is a surviving subset rather than an unbiased later universe.

The outside-country comparison is closer to JJK's. On their list the outside-country long–short alphas are −3.8, +22.8 and +30.2 basis points in the three groups ($t = -0.39$, 1.54 and 2.13), against their 0, 29 and 49. The all-international result over 2000–2024 is +9.7 ($t = 1.95$).

The construction of the short leg matters, and Table A21 reports the alternatives for the all-international portfolio. JJK describe the unlinked portfolio as stocks in home-linked managers' home countries held by unlinked managers in the same category and quarter. Two choices remain open. One is whether a fund linked to a different country counts as unlinked, and the other is whether the short leg carries the long leg's country weights. Admitting funds linked to other countries gives −29.9 ($t = -1.51$), and matching the country weights gives −12.4 ($t = -0.79$). Across the eleven constructions in the upper block of the table, the 1991–2014 estimate ranges from −29.9 to +19.8 basis points. The only positive estimate combines the earlier education coding with the country-matched short leg, at +19.8 ($t = 0.79$). JJK's education data are not available to us, so we cannot say which coding is closer to theirs. Requiring the link to lie inside the category mandate raises the Regional long-only alpha to +57.6 ($t = 1.85$), against JJK's 72, but the Regional long–short alpha falls to −75.7 ($t = -1.99$).

D.3 Stock-months without a total-return index

Our monthly stock returns start from the Datastream total-return index. When a stock-month has no index, we use its price return plus the average dividend return of other stocks in that country and month, and we leave the month missing if price and market-capitalization returns differ by more than 10 basis points. JJK state that stock returns come from Datastream and CRSP and do not describe a comparable step. These filled stock-months are 60% of long-leg value with returns on JJK's list, and 58% of that filled value is in U.K. stocks, led by GlaxoSmithKline, Vodafone, Shell, Tesco and BP.

Dropping the filled stock-months brings our long-only alphas closer to JJK's in magnitude. Table A22 reports +49.0, +51.2 and +59.6 basis points in the three groups, against their 54, 72 and 47, and the earlier education coding gives +52.7, +59.7 and +57.4. JJK's values lie inside each 95% interval, but the estimates are less precise than theirs, with t-statistics of 1.77, 1.28 and 1.27 against their 2.67, 2.33 and 1.89, and only 34% of long-leg formation value remains. The restriction does not reproduce their positive long–short alphas, which are −6.8, −26.6 and −37.9, and it does not move the outside-country alphas closer to JJK's. Those alphas are instead closest on the filled stock-months, at +2.9, +32.2 and +53.3 against their 0, 29 and 49, so the two sets of JJK estimates point to opposite subsets of stock-months and no single restriction matches both.

The higher long-only alpha primarily reflects which stock-months remain in the portfolio, rather than the dividend estimates used to fill missing returns. The imputation uses an equal-weighted country mean with the largest values trimmed. Replacing it with size-matched dividends on the same stock-months raises the all-international long-only alpha only to +16.3 ($t = 0.86$) and leaves the long–short alpha at −23.9 ($t = -1.21$). Restoring the months that the 10-basis-point rule drops gives long-only and long–short alphas of −0.7 and −20.4. On the 276 months with both portfolios, the long-only alpha on index stock-months exceeds that on filled stock-months by +37.1 basis points ($t = 1.62$), which is not statistically distinguishable from zero.

The closer match in long-only alphas also depends on the early years. The all-international long-only alpha on index stock-months is +90.9 ($t = 1.13$) over 1991–1999, when the index covers 15% of long-leg formation value, and +20.9 ($t = 1.25$) over 2000–2014. Their sample would match this restriction only if their Datastream returns lacked the same large stocks, which we cannot verify. Return coverage is therefore a candidate explanation for the long-only gap, not an established one, and it does not account for the long–short gap.

D.4 Home-country allocation: JJK Table 2

JJK regress a fund's raw weight in a country, measured as a fraction of total equity and missing when the country is not held, on the share of its managers home-linked to that country. Our main allocation tests instead use benchmark-relative weights in percentage points, setting the fund's own weight to zero when it does not hold a benchmark country, so the magnitudes are not comparable with those in the main text. Table A23 applies JJK's design, including their fund controls and their clustering by country and year.

The allocation tilt has the same sign as JJK's in every specification, and on JJK's list every magnitude is smaller. On JJK's list over 1991–2014, the four all-international coefficients are 0.135, 0.105, 0.097 and 0.065, against their 0.174, 0.139, 0.138 and 0.121, and the earlier education coding gives 0.156, 0.120, 0.109 and 0.088. The country-by-quarter estimate on JJK's list is significant at the 10% level, and our two-way standard errors are larger than those implied by JJK's t-statistics. The coefficients without fixed effects are 0.152 for Regional and 0.401 for Country funds, against JJK's 0.338 and 0.499. Requiring the country to lie in the category mandate gives 0.294 and 0.496. Panel D is an extension rather than a replication. With category-by-quarter and country effects together the coefficient is 0.029, and with fund and country-by-quarter effects together it is 0.020. The tilt remains positive, at 0.036 and 0.028 on our funds over 2000–2024, which is consistent with the education-country overweight in the main text.

D.5 Comparison of performance measures on common samples

JJK's portfolio alpha and our stock-selection measure answer related but different questions. JJK's alpha compares linked and unlinked funds' portfolios against factor returns, in basis points per month. Our measure compares a fund's country basket with similar local stocks, in annualized basis points per dollar of the basket, with fund-by-country and country-by-quarter effects. In the joint specification of Table 5, the comparison of a fully foreign-educated team in its education country with a fully U.S.-educated team in the same country is $\beta + \gamma$, not β alone. The two estimates, +46.52 and −46.31, sum to +0.21 annualized basis points (standard error 56.61), using their joint covariance.

Table A24 estimates both measures on common funds and quarters. The country panel exists only for funds in our universe, so the first column uses the funds on JJK's list that are also among our 1,663. The

alpha row uses the fund-quarters of each panel. The all-degree contrast is −105.1 (standard error 81.3) on the JJK-list sample, −83.5 (70.7) over 2000–2014 on our funds, and +0.2 (56.6) over 2000–2024. With undergraduate links and shares it is −28.7 (85.5), −30.3 (75.6) and +34.6 (62.6). The alpha on the same fund-quarters is −15.2, −19.6 and −6.1 basis points per month. None of these estimates differs from zero at conventional levels, and their standard errors allow sizable effects in either direction.

D.6 Interpretation and unresolved differences

Four conclusions follow. First, the allocation tilt appears in every sample. On JJK's fund list and on our broader universe, funds hold more of their managers' education countries, though the magnitude falls as fixed effects are added. Second, we do not obtain a positive baseline home-country long–short alpha for the all-international group in any of the four samples, including their list and period. The only positive baseline subgroup estimate is the Regional +23.4 on that list ($t = 0.81$). Third, our long-only alphas move closer to theirs when stock-months without a total-return index are dropped, mainly through the 1990s. Fourth, the outside-country comparison, particularly for Country funds on JJK's list, is the closest match to their long–short results.

We cannot identify the source of the difference in home-country alphas. JJK's holdings, education and fund-asset data are not available to us, so we cannot verify that the same funds, positions and education links enter each cell. No single choice we examined delivers both JJK's long-only and long–short alphas. We do not examine their purchases test in Table 4, Panel B. The comparisons here describe what our data show under JJK's design and do not bear on the results from their data. Our focus is different. We ask how far the value of local expertise extends across an international mandate and how funds recruit and evaluate managers who have it.

Table A1: Fund assets by assigned benchmark

Morningstar Index	Mean assets (USD billions)	Mean active funds
Morningstar Gbl NR USD	4.672	51.55
Morningstar Gbl xUS Val TME NR USD	2.853	86.63
Morningstar Gbl Growth TME NR USD	2.790	51.36
Morningstar Gbl xUS Growth TME NR USD	1.787	86.45
Morningstar Gbl xUS NR USD	1.260	164.27
Morningstar EM Americas NR USD	1.091	2.57
Morningstar DM APAC xJpn NR USD	0.800	14.87
Morningstar EM NR USD	0.757	125.59
Morningstar Gbl Val TME NR USD	0.606	38.64
Morningstar Gbl xUS SMID NR USD	0.550	56.65
Morningstar DM Eur NR USD	0.528	10.16
Morningstar Gbl SMID NR USD	0.460	18.04
Morningstar Jpn NR USD	0.367	6.99
Morningstar DM APAC NR USD	0.257	3.15
Morningstar China NR USD	0.180	13.05
Morningstar India NR USD	0.121	2.34
Morningstar Gbl Lrg NR USD	0.093	28.97

Notes: This table reports the mean fund-level total net assets and the mean number of funds in each Morningstar benchmark category, computed at quarterly frequency over the 1,663-fund sample. Because the counts are per-quarter averages, the "Mean active funds" column sums to about 760, the average number of funds active in a quarter. The total is below the 1,663 distinct funds observed over the full sample because funds enter and exit over time. In the Morningstar index names, Gbl is global, xUS excludes the United States, DM and EM are developed and emerging markets, APAC is Asia-Pacific, xJpn excludes Japan, Lrg is large-cap, SMID is small- and mid-cap, Val is value, TME denotes Morningstar's target-market-exposure indexes, and NR USD denotes net returns in U.S. dollars.

Table A2: Portfolio holdings

	Pooled over all observations			
	Mean	Median	Std. dev.	N
Sample Funds				
# of countries held	18.99	19	8.44	67,380
Country-level portfolio share (%)	4.83	2.17	9.47	1,279,345
# of stocks held	210.26	81	589.15	71,742
Stock-level portfolio share (%)	0.45	0.09	0.88	15,084,718
Morningstar Gbl xUS NR USD				
# of countries held	49.47	51	3.46	88
Country-level portfolio share (%)	1.97	0.58	3.35	4,353
# of stocks held	5,136.20	5,104	982.78	88
Stock-level portfolio share (%)	0.02	0.00	0.06	451,986
All Benchmarks (equal weights)				
# of countries held	27.55	26	20.28	1,947
Country-level portfolio share (%)	3.43	0.57	10.15	53,639
# of stocks held	1,887.58	1,344	1,841.74	1,947
Stock-level portfolio share (%)	0.05	0.01	0.23	3,675,116

Notes: This table reports country-level and stock-level holding counts and portfolio shares for the sample funds, the Morningstar Global ex-US benchmark, and all Morningstar benchmarks assigned to sample funds. The sample-fund rows are computed on the same 1,663-fund sample and 2000–2024 window as Table 1. Each statistic pools all observations counted in N. The rows counting countries or stocks held pool fund-quarters (benchmark-quarters in the lower two panels), and the "portfolio share" rows pool individual positive-weight country or stock positions. The last panel gives each benchmark-quarter equal weight. Country and stock statistics use their respective available observations.

Table A3: Education countries of managers

Country	# Mgrs	Share	Country	# Mgrs	Share
USA	2188	0.7313	ESP	6	0.0020
GBR	580	0.1939	NZL	6	0.0020
CAN	77	0.0257	AUT	5	0.0017
IND	70	0.0234	CHL	5	0.0017
CHN	63	0.0211	POL	4	0.0013
FRA	51	0.0170	PAK	4	0.0013
AUS	44	0.0147	VEN	4	0.0013
DEU	35	0.0117	TUR	3	0.0010
JPN	32	0.0107	ARG	3	0.0010
IRL	26	0.0087	NGA	3	0.0010
HKG	18	0.0060	NOR	3	0.0010
ZAF	16	0.0053	EGY	2	0.0007
ITA	14	0.0047	CZE	2	0.0007
SGP	12	0.0040	GRC	2	0.0007
KOR	11	0.0037	PRT	2	0.0007
CHE	11	0.0037	BGR	1	0.0003
NLD	10	0.0033	ROU	1	0.0003
DNK	10	0.0033	GTM	1	0.0003
SWE	9	0.0030	UKR	1	0.0003
BRA	8	0.0027	THA	1	0.0003
TWN	7	0.0023	EST	1	0.0003
RUS	7	0.0023	LBN	1	0.0003
MEX	7	0.0023	COL	1	0.0003
BEL	7	0.0023	IDN	1	0.0003
ISR	7	0.0023			
Total # of Managers = 2,992					

Notes: Countries are identified by ISO 3166-1 alpha-3 codes (for example, GBR is the United Kingdom, IND India and HKG Hong Kong). Each Share column divides the number of managers educated in the listed country by the 2,992 managers with recorded education countries (49 countries after hand review). The shares sum to more than one because a manager can have education ties to multiple countries.

Table A4: Education-country links across funds

Country	# Funds	Share	Country	# Funds	Share
USA	1492	0.898	AUT	15	0.009
GBR	639	0.384	MEX	14	0.008
IND	188	0.113	POL	14	0.008
CAN	173	0.104	BRA	12	0.007
CHN	133	0.080	VEN	8	0.005
FRA	129	0.078	NGA	8	0.005
AUS	88	0.053	CZE	8	0.005
JPN	67	0.040	PRT	8	0.005
ITA	51	0.031	CHL	7	0.004
IRL	51	0.031	EGY	7	0.004
HKG	46	0.028	ARG	6	0.004
DEU	44	0.026	GRC	6	0.004
SGP	41	0.025	PAK	6	0.004
ZAF	39	0.023	TUR	6	0.004
SWE	38	0.023	ROU	6	0.004
KOR	32	0.019	NOR	6	0.004
ISR	32	0.019	IDN	3	0.002
NLD	32	0.019	THA	2	0.001
CHE	29	0.017	LBN	2	0.001
NZL	28	0.017	GTM	2	0.001
DNK	28	0.017	UKR	2	0.001
ESP	23	0.014	EST	1	0.001
RUS	20	0.012	COL	1	0.001
TWN	19	0.011	BGR	1	0.001
BEL	16	0.010			
Total # of Funds = 1,662					

Notes: Countries are identified by ISO 3166-1 alpha-3 codes, as in Table A3. Each Share column divides the number of funds with at least one manager educated in the listed country by the 1,662 funds with at least one manager whose education country is recorded. This descriptive denominator includes funds with a recorded education link and does not restrict any regression sample. The shares sum to more than one because a fund can have education ties to multiple countries through its managers.

Table A5: Pre-appointment country relevance and net fund returns

Dependent variable	Factor-adjusted net NAV return	
Units	Annualized bps	
	(1)	(2)
Specification	Baseline controls	Add country shares
Pre-appointment country relevance	−463.18	−688.30***
	(295.12)	(244.59)
Foreign-educated share	32.55	11.82
	(48.28)	(63.37)
Foreign share with unknown relevance	−147.74**	−224.94***
	(58.44)	(61.37)
Log team size	−87.43	−78.48
	(59.12)	(59.75)
Unknown-education share	−106.00	−84.24
	(68.38)	(68.57)
Log lagged assets (USD millions)	−21.75**	−22.49**
	(9.83)	(9.80)
Assets spline basis 1	−101.28***	−105.86***
	(25.89)	(25.73)
Assets spline basis 2	571.88***	602.09***
	(157.85)	(155.38)
Team-size spline basis 1	−80.16	−86.49
	(137.49)	(139.47)
Team-size spline basis 2	688.02	708.37
	(591.04)	(601.64)
Female share (known gender)	−56.00	−58.83
	(38.09)	(38.36)
Unknown-gender share	−72.03	−208.71**
	(91.37)	(96.61)
CFA share (observed biographies)	6.87	4.11
	(31.42)	(31.69)
Unknown-biography share	209.45*	350.28***
	(107.00)	(108.49)
Mean log(1 + experience)	−29.37*	−32.39*
	(16.59)	(16.89)
Prior-year relative NAV return (pp)	−4.63***	−4.83***
	(1.45)	(1.45)
Log(1 + prior relative volatility)	62.83**	60.31**
	(30.99)	(30.65)
Prior-quarter flow (pp)	1.55×10^{-6}	1.57×10^{-6}
	(2.25×10^{-6})	(2.27×10^{-6})
Missing prior return history	99.70	93.63
	(62.75)	(62.37)
Missing prior-quarter flow	−41.63	−41.04
	(55.61)	(55.58)
Lagged expense ratio (bps)	−0.41	−0.45
	(0.33)	(0.33)
Lagged log turnover	85.11**	88.08**
	(35.28)	(34.70)
Top-ten equity concentration	65.36	125.57
	(137.46)	(136.30)
Missing lagged expense ratio	−111.35**	−117.46**
	(53.86)	(53.49)
Missing lagged turnover	73.17**	74.54**
	(36.67)	(36.03)
Missing equity concentration	23.62	36.41
	(42.02)	(41.86)
Education-country composition controls	No	Yes
Fund FE	Yes	Yes
Benchmark × month FE	Yes	Yes
Observations	206,847	206,847
Fund clusters	1,418	1,418

Notes: Both columns add pre-appointment country relevance and the foreign share with unknown relevance to Table 2, column 1. Column 2 also controls for education-country shares. All degree levels enter. Definitions appear in Appendix B.1. The relevance coefficient is conditional on foreign-educated share. The foreign-share slopes describe zero measured relevance. Outcomes, sample, baseline controls and equal fund-month weights follow Table 2, column 1. Fund-clustered standard errors appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table A6: Portfolio controls and return tails

Dependent variable	Factor-adjusted net NAV return		
Units	Annualized bps		
	(1)	(2)	(3)
Specification	Style tilts	Top ten, turnover	Winsorized
Foreign-educated share	-65.55^{*}	-68.72^{*}	-68.66^{**}
	(36.51)	(35.44)	(33.08)
Log team size	−89.65	−87.40	-89.27^{*}
	(59.49)	(59.07)	(54.09)
Unknown-education share	−110.43	−106.10	−97.95
	(71.04)	(68.58)	(67.21)
Log lagged assets (USD millions)	-20.96^{**}	-21.48^{**}	-21.04^{**}
	(9.87)	(9.87)	(9.39)
Assets spline basis 1	-100.14^{***}	-97.97^{***}	-92.84^{***}
	(25.91)	(26.13)	(24.74)
Assets spline basis 2	555.51^{***}	547.88^{***}	506.05^{***}
	(158.18)	(158.47)	(149.87)
Team-size spline basis 1	−79.80	−79.90	−68.71
	(138.05)	(137.39)	(129.38)
Team-size spline basis 2	694.90	698.49	647.42
	(592.19)	(589.83)	(560.66)
Female share (known gender)	−49.50	−50.55	−43.03
	(38.37)	(38.53)	(37.14)
Unknown-gender share	−94.23	−88.81	−84.47
	(92.32)	(92.95)	(88.79)
CFA share (observed biographies)	7.61	7.26	3.59
	(31.38)	(31.45)	(30.19)
Unknown-biography share	228.55^{**}	226.85^{**}	213.12^{**}
	(107.30)	(107.68)	(102.99)
Mean log(1 + experience)	-35.24^{**}	-33.29^{**}	-34.78^{**}
	(16.95)	(16.76)	(15.21)
Prior-year relative NAV return (pp)	-4.45^{***}	-4.63^{***}	-4.20^{***}
	(1.54)	(1.47)	(1.37)
Log(1 + prior relative volatility)	62.66^{**}	60.26^{*}	54.03^{*}
	(31.36)	(31.10)	(27.87)
Prior-quarter flow (pp)	1.57×10^{-6}	1.49×10^{-6}	1.65×10^{-6}
	(2.24×10^{-6})	(2.25×10^{-6})	(2.24×10^{-6})
Missing prior return history	73.20	71.98	79.14
	(56.93)	(59.88)	(57.92)
Missing prior-quarter flow	−44.21	−46.10	−37.84
	(55.44)	(55.55)	(53.42)
Holdings size tilt	−15.74	—	—
	(65.56)		
Holdings momentum tilt	−23.54	—	—
	(38.03)		
Missing size tilt	249.20	—	—
	(371.87)		
Missing momentum tilt	−292.73	—	—
	(346.07)		
Top-ten equity concentration	—	76.34	116.12
		(138.37)	(124.50)
Lagged log turnover	—	82.80^{**}	70.81^{**}
		(35.43)	(33.73)
Missing equity concentration	—	26.52	36.72
		(42.11)	(39.17)
Missing lagged turnover	—	40.67	57.81^{*}
		(31.75)	(33.67)
Lagged expense ratio (bps)	—	—	−0.39
			(0.32)
Missing lagged expense ratio	—	—	-95.70^{*}
			(51.41)
Fund FE	Yes	Yes	Yes
Benchmark × month FE	Yes	Yes	Yes
Observations	206,847	206,847	206,847
Fund clusters	1,418	1,418	1,418

Notes: All columns use thirteen-factor-adjusted NAV. Columns 1–2 add style or concentration and turnover. Column 3 adds expenses and 1st/99th-percentile benchmark-month winsorization. Table 2 uses uncapped NAV. All columns retain missing indicators, raw prior flow and equal weights. Fund-clustered standard errors appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table A7: Timing and coefficient stability of the fund-return gap

Dependent variable	Factor-adjusted net NAV return		
Units	Annualized bps		
Specification	(1) Lead sample	(2) With future team	(3) Unknown share only
Foreign-educated share	-74.53^{**}	-72.28^{*}	-58.61^{*}
	(37.81)	(43.87)	(35.54)
Future foreign-educated share	—	−2.60	—
		(37.28)	
Future unknown-education share	—	−25.23	—
		(61.67)	
Log team size	−77.10	−77.42	—
	(60.76)	(60.79)	
Unknown-education share	-134.59^{*}	−116.77	−69.42
	(75.46)	(80.49)	(48.05)
Log lagged assets (USD millions)	-21.89^{**}	-21.86^{**}	—
	(9.95)	(9.94)	
Assets spline basis 1	-105.21^{***}	-105.21^{***}	—
	(25.92)	(25.91)	
Assets spline basis 2	587.47^{***}	587.16^{***}	—
	(157.98)	(158.01)	
Team-size spline basis 1	−111.24	−110.75	—
	(142.09)	(142.14)	
Team-size spline basis 2	824.35	822.52	—
	(612.93)	(613.13)	
Female share (known gender)	−56.32	−56.17	—
	(39.90)	(39.93)	
Unknown-gender share	−65.51	−65.84	—
	(101.26)	(101.33)	
CFA share (observed biographies)	1.45	1.44	—
	(33.24)	(33.24)	
Unknown-biography share	226.39^{*}	225.87^{*}	—
	(123.04)	(123.07)	
Mean log(1 + experience)	-39.42^{**}	-39.55^{**}	—
	(17.74)	(17.74)	
Prior-year relative NAV return (pp)	-4.78^{***}	-4.78^{***}	—
	(1.55)	(1.55)	
Log(1 + prior relative volatility)	62.86^{**}	62.82^{**}	—
	(31.62)	(31.61)	
Prior-quarter flow (pp)	1.19×10^{-6}	1.19×10^{-6}	—
	(2.27×10^{-6})	(2.27×10^{-6})	
Missing prior return history	105.34	105.20	—
	(64.36)	(64.37)	
Missing prior-quarter flow	−40.02	−40.01	—
	(56.29)	(56.30)	
Lagged expense ratio (bps)	−0.38	−0.38	—
	(0.43)	(0.43)	
Lagged log turnover	91.39^{**}	91.57^{**}	—
	(35.99)	(35.97)	
Top-ten equity concentration	48.94	49.82	—
	(143.21)	(143.24)	
Missing lagged expense ratio	-110.58^{*}	-110.62^{*}	—
	(62.49)	(62.46)	
Missing lagged turnover	60.13	60.08	—
	(37.94)	(37.93)	
Missing equity concentration	16.94	17.33	—
	(43.68)	(43.71)	
Fund FE	Yes	Yes	Yes
Benchmark × month FE	Yes	Yes	Yes
Observations	193,986	193,986	206,847
Fund clusters	1,418	1,418	1,418

Notes: Factor-adjusted net NAV is in annualized bps. Columns 1–2 require an observed team four quarters ahead. Column 2 adds the future team's foreign-educated and unknown-education shares to Equation (3). Column 3 uses only those two current shares on the full sample. Against Table 2, column 1, Oster's δ for a zero coefficient is −20.6 and the bias-adjusted coefficient at Oster's $\delta = 1$ is −72.87 (Oster, 2019), with $R_{\max} = 1.3R^2$. Standard errors clustered by economic fund are in parentheses. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table A8: Net performance beyond the initial appointment period

Dependent variable	Factor-adjusted net NAV return	
Units	Annualized bps	
	(1)	(2)
Specification	OLS	WLS
Foreign-educated share: established	−80.91**	−81.82***
	(38.05)	(24.28)
Foreign-educated share: first two years	−37.91	−50.15*
	(40.87)	(27.14)
U.S.-educated share: first two years	−2.56	−12.13
	(21.51)	(14.54)
Log team size	−89.98	−70.02*
	(57.87)	(37.58)
Unknown-education share	−91.04	−123.01***
	(68.25)	(43.69)
Log lagged assets (USD millions)	−16.14*	−12.65*
	(9.44)	(7.33)
Assets spline basis 1	−101.79***	−63.72***
	(26.09)	(18.25)
Assets spline basis 2	548.67***	321.81***
	(158.23)	(106.71)
Team-size spline basis 1	−71.06	−25.95
	(134.60)	(87.51)
Team-size spline basis 2	648.80	388.29
	(578.47)	(371.78)
Female share (known gender)	−51.66	−38.00
	(38.38)	(27.27)
Unknown-gender share	−95.04	−1.06
	(91.55)	(52.86)
CFA share (observed biographies)	9.43	32.91
	(31.25)	(20.91)
Unknown-biography share	230.43**	81.34
	(106.28)	(66.35)
Mean log(1 + experience)	−25.58	−33.77***
	(17.42)	(11.53)
Fund FE	Yes	Yes
Benchmark × month FE	Yes	Yes
Observations	206,847	206,847
Fund clusters	1,418	1,418

Notes: The net NAV model separates managers in their first eight quarters at the fund from established managers. Established U.S.-educated managers are the omitted category. Outcomes use the predetermined factor adjustment and reduced controls, in annualized basis points. OLS gives fund-months equal weight. WLS weights by inverse fund residual variance. The first-two-years foreign-minus-U.S. contrasts, the new foreign-educated coefficient minus the new U.S.-educated coefficient, are −35.35 (SE 42.14) and −38.02 (27.52), using the joint covariance. Standard errors, clustered by economic fund, appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table A9: Conviction portfolios: alternative constructions

Dependent variable	Portfolio excess return							
Units	Annual alpha in bps, factor loadings dimensionless							
Specification	(1) Benchmark reference	(2) Two-year reference	(3) Ex-U.K.	(4) Undergrad.	(5) Momentum	(6) Momentum, no benchmark	(7) Benchmark only	(8) Net costs
Alpha (annualized bps)	275.4*	318.3**	574.1***	492.8***	380.5***	369.8***	379.9***	341.4***
	(147.5)	(132.1)	(148.0)	(134.3)	(136.8)	(136.8)	(124.8)	(124.8)
Global ex-U.S. benchmark excess return	0.101	0.158	0.095	0.182*	0.146	—	1.004***	0.184*
	(0.117)	(0.105)	(0.117)	(0.106)	(0.107)		(0.021)	(0.099)
Developed ex-U.S. market	0.790***	0.775***	0.781***	0.752***	0.780***	0.902***	—	0.749***
	(0.107)	(0.096)	(0.107)	(0.097)	(0.098)	(0.043)		(0.090)
Developed ex-U.S. size	−0.079	−0.065	0.020	−0.001	−0.015	−0.015	—	0.007
	(0.066)	(0.059)	(0.066)	(0.060)	(0.062)	(0.062)		(0.055)
Developed ex-U.S. value	−0.051	−0.028	−0.100	−0.092	−0.001	0.012	—	−0.015
	(0.076)	(0.068)	(0.076)	(0.069)	(0.072)	(0.071)		(0.064)
Developed ex-U.S. investment	−0.104	0.070	0.021	0.087	0.020	0.017	—	−0.073
	(0.103)	(0.092)	(0.104)	(0.094)	(0.096)	(0.097)		(0.087)
Developed ex-U.S. profitability	−0.125	0.052	−0.156	−0.092	0.011	0.025	—	−0.024
	(0.102)	(0.092)	(0.103)	(0.093)	(0.093)	(0.093)		(0.087)
Emerging market	0.176***	0.101**	0.162***	0.101**	0.104**	0.128***	—	0.108***
	(0.048)	(0.043)	(0.048)	(0.044)	(0.044)	(0.040)		(0.041)
Emerging size	0.044	0.009	−0.056	0.014	−0.004	−0.014	—	0.015
	(0.071)	(0.064)	(0.071)	(0.065)	(0.065)	(0.064)		(0.060)
Emerging value	−0.113	−0.029	0.002	−0.034	−0.012	−0.013	—	−0.069
	(0.073)	(0.065)	(0.073)	(0.066)	(0.066)	(0.066)		(0.062)
Emerging investment	0.203**	−0.027	−0.080	−0.015	−0.046	−0.048	—	0.082
	(0.085)	(0.076)	(0.086)	(0.078)	(0.078)	(0.078)		(0.072)
Emerging profitability	0.125	−0.081	−0.100	0.040	−0.076	−0.080	—	0.022
	(0.094)	(0.084)	(0.094)	(0.085)	(0.087)	(0.087)		(0.079)
Developed ex-U.S. momentum	—	—	—	—	0.001	0.007	—	—
					(0.043)	(0.043)		
Emerging momentum	—	—	—	—	0.025	0.021	—	—
					(0.050)	(0.050)		
Observations	300	300	300	300	300	300	300	300

Notes: Each column reports one top-portfolio factor regression. Annual alpha is $12 \times 10^4 \alpha_p$ basis points. Loadings are dimensionless. Column 1 replaces the capitalization-share reference in Equation (7) with the assigned benchmark's weights, renormalized over the stocks the fund holds in the country. Column 2 widens the reference set to all stocks the fund held in the country at any quarter-end from $q-8$ through q. Columns 3 and 4 exclude U.K. equities and use undergraduate links only. Columns 5 and 6 add developed and emerging momentum factors, with and without the common Global ex-US benchmark. Columns 7 and 8 use only that benchmark or deduct estimated bid-ask trading costs. All use January 2000–December 2024 and the baseline portfolio-weighting procedure. Portfolio standard errors are conventional OLS. Direct Wald tests use the robust stacked covariance. Standard errors appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%. Top-versus-ordinary Wald p-values for the eight columns are 0.251, 0.057, 0.007, < 0.001, 0.025, 0.025, 0.024 and 0.003, using the robust stacked covariance.

Table A10: Actual equity-capital contributions of education-country holdings

Dependent variable	Excess-return contribution		
Units	Annual alpha in bps, factor loadings dimensionless		
	(1)	(2)	(3)
Specification	All local	Top local	Ordinary local
Alpha (annualized bps)	−9.8	0.3	−10.1*
	(10.8)	(6.1)	(5.8)
Global ex-U.S. benchmark excess return	0.015*	0.008	0.007
	(0.009)	(0.005)	(0.005)
Developed ex-U.S. market	0.078***	0.035***	0.043***
	(0.008)	(0.004)	(0.004)
Developed ex-U.S. size	−0.018***	−0.005*	−0.013***
	(0.005)	(0.003)	(0.003)
Developed ex-U.S. value	0.021***	0.008***	0.013***
	(0.006)	(0.003)	(0.003)
Developed ex-U.S. investment	−0.013*	−0.003	−0.010**
	(0.008)	(0.004)	(0.004)
Developed ex-U.S. profitability	0.029***	0.013***	0.015***
	(0.007)	(0.004)	(0.004)
Emerging market	−0.001	−0.004**	0.003*
	(0.004)	(0.002)	(0.002)
Emerging size	0.014***	0.006**	0.008***
	(0.005)	(0.003)	(0.003)
Emerging value	0.008	0.000	0.007**
	(0.005)	(0.003)	(0.003)
Emerging investment	−0.007	0.000	−0.007**
	(0.006)	(0.004)	(0.003)
Emerging profitability	−0.006	−0.003	−0.003
	(0.007)	(0.004)	(0.004)
Observations	300	300	300

Notes: Columns 1–3 report capital-weighted excess contributions per dollar of covered disclosed equity, including U.S. equities. Local capital shares average 8.71% overall, 3.62% top and 5.10% ordinary. Raw annual contributions are 57.5, 27.3 and 30.1 bps, respectively. All use 300 months, January 2000–December 2024. Loadings are dimensionless. Alphas are annualized bps with conventional OLS standard errors. Actual country weights and changing monthly capital shares differ from the benchmark country mix used for the normalized strategies in Table 6. These holdings outcomes are distinct from NAV. Standard errors appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table A11: Local alpha on a common sample under benchmark and actual country weights

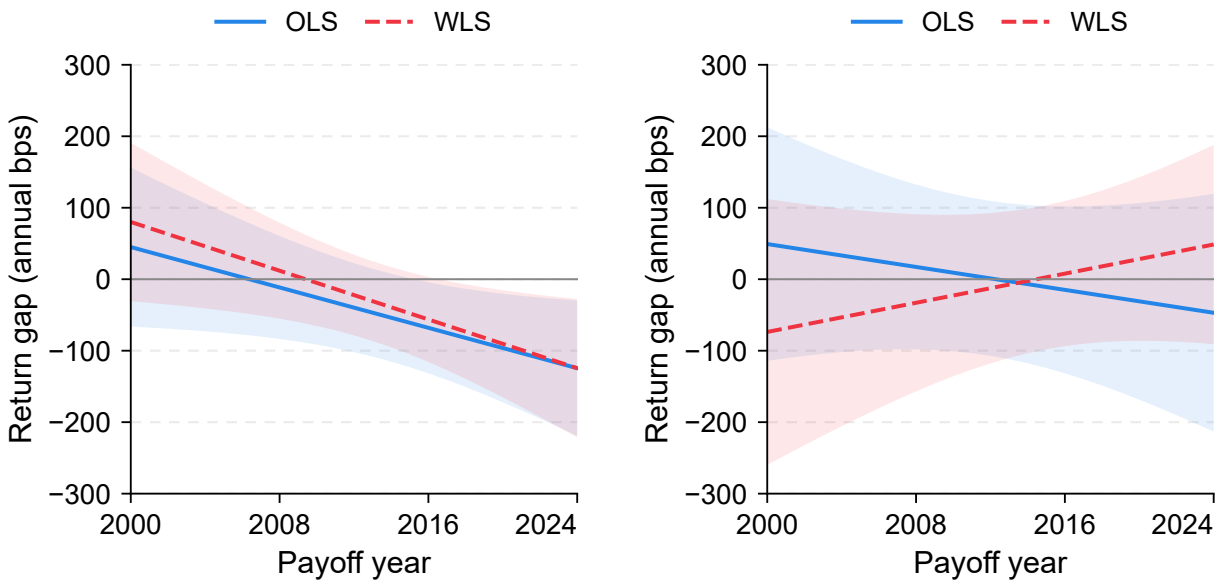
Dependent variable	Portfolio excess return					
Units	Annual alpha in bps, factor loadings dimensionless					
	(1)	(2)	(3)	(4)	(5)	(6)
Specification	Benchmark mix: top	Benchmark mix: ordinary	Benchmark mix: combined	Actual weights: top	Actual weights: ordinary	Actual weights: combined
Alpha (annualized bps)	406.7***	3.3	130.8	16.6	−177.5	−103.2
	(134.1)	(92.2)	(94.9)	(159.3)	(109.1)	(119.2)
Global ex-U.S. benchmark excess return	0.155	0.116	0.137*	0.249**	0.117	0.180*
	(0.106)	(0.073)	(0.075)	(0.126)	(0.086)	(0.094)
Developed ex-U.S. market	0.777***	0.731***	0.749***	0.895***	0.858***	0.872***
	(0.097)	(0.067)	(0.069)	(0.115)	(0.079)	(0.086)
Developed ex-U.S. size	−0.007	−0.138***	−0.085**	−0.155**	−0.326***	−0.248***
	(0.060)	(0.041)	(0.042)	(0.071)	(0.048)	(0.053)
Developed ex-U.S. value	−0.008	0.050	0.036	0.170**	0.136**	0.158**
	(0.069)	(0.047)	(0.049)	(0.082)	(0.056)	(0.061)
Developed ex-U.S. investment	0.008	−0.004	0.009	0.018	−0.026	−0.008
	(0.094)	(0.065)	(0.066)	(0.111)	(0.076)	(0.083)
Developed ex-U.S. profitability	0.007	0.033	0.050	0.267**	0.167**	0.220***
	(0.093)	(0.064)	(0.066)	(0.111)	(0.076)	(0.083)
Emerging market	0.101**	0.176***	0.138***	−0.113**	0.040	−0.031
	(0.044)	(0.030)	(0.031)	(0.052)	(0.036)	(0.039)
Emerging size	0.000	−0.037	−0.024	0.158**	0.121**	0.139**
	(0.065)	(0.045)	(0.046)	(0.077)	(0.053)	(0.058)
Emerging value	−0.020	−0.020	−0.016	−0.009	0.071	0.037
	(0.066)	(0.046)	(0.047)	(0.079)	(0.054)	(0.059)
Emerging investment	−0.018	−0.075	−0.056	0.057	−0.039	0.008
	(0.078)	(0.053)	(0.055)	(0.092)	(0.063)	(0.069)
Emerging profitability	−0.028	−0.006	0.001	0.012	−0.040	−0.014
	(0.085)	(0.059)	(0.060)	(0.101)	(0.069)	(0.076)
Observations	300	300	300	300	300	300

Notes: All portfolios use the same 77,188 fund-country-month cells at 749 fund ISINs and 300 payoff months. Columns 1–3 (benchmark mix) weight each country's holdings by its benchmark country share, as in Table 6. Columns 4–6 (actual weights) use the capital funds actually hold in each country. Alpha is annualized. Factor loadings are dimensionless monthly slopes with conventional OLS standard errors. Different top/ordinary country mixtures and time-varying capital shares prevent combining their alphas using a single average weight. Standard errors appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%. The combined actual-weights-minus-benchmark-mix alpha, column 6 minus column 3, is −233.98 bps (SE 85.45, $p = 0.007$), using the joint covariance.

Table A12: Changes in the foreign-education return gap over time

Dependent variable	Adjusted country return	
Units	Annualized basket bps	
	(1)	(2)
Specification	OLS	WLS
Foreign-educated share × year (γ^T)	-7.06^{**}	-8.52^{**}
	(3.50)	(3.72)
Foreign-educated share (γ)	−39.76	−22.20
	(31.72)	(28.75)
Education-country link × year (β^T)	3.05	13.62^{**}
	(4.98)	(5.40)
Education-country link (β)	40.89	9.61
	(48.78)	(49.80)
Log country stock count	-57.88^{***}	-53.49^{***}
	(11.41)	(11.19)
Log team size	-70.17^{***}	-44.30^{**}
	(21.50)	(18.92)
Unknown-education share	33.08	15.90
	(84.36)	(77.82)
Female share (known gender)	-90.37^{**}	-109.49^{***}
	(44.90)	(37.13)
Unknown-gender share	94.05	134.65
	(87.05)	(82.20)
CFA share (observed biographies)	54.13^{*}	49.53^{*}
	(30.10)	(28.45)
Unknown-biography share	−47.78	−68.59
	(123.27)	(111.68)
Mean log(1 + experience)	-53.72^{***}	-47.69^{***}
	(15.50)	(14.13)
Fund × country FE	Yes	Yes
Country × quarter FE	Yes	Yes
Observations	974,924	974,924
Fund clusters	1,428	1,428

Notes: Equation (16) adds both education-link and foreign-share interactions with payoff calendar year minus 2012 to the common panel in Table 5. The first coefficient is the annual change in the outside-country foreign-minus-U.S. return gap. The share coefficient gives that gap in 2012. The 2024 gaps use the entire fitted time profile and the joint coefficient covariance. Shares are in 0–1 units. Time interactions are per single year. Column 1 gives countries equal weight within each fund-quarter. Column 2 additionally uses inverse fund residual variance. Both columns use fund-country and country-quarter effects, eight displayed controls and economic-fund clustering. The payoff period is 2000–2024. Standard errors appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%. Fitted 2024 outside gaps are −124.50 (SE 48.28) and −124.50 (49.38). Linked-country gaps are −46.96 (84.80) and 48.59 (71.06), in columns 1 and 2. These sums use joint covariance and are plotted in Figure A1.



(a) Outside education countries

(b) Education countries

Figure A1: Education and outside-country selection gaps over time

Notes: Both panels use the joint country regressions in Table A12. Panel (a) plots $\gamma + \gamma^T t$. Panel (b) plots $\beta + \gamma + \beta^T t + \gamma^T t$, where t is years from 2012. Each compares an all-foreign-educated team with an all-U.S.-educated team. Panel (b) additionally requires a current manager's education link to the country. Solid lines are OLS, dashed lines WLS. Shaded bands are 95% intervals based on the complete economic-fund-clustered covariance matrix.

Table A13: Relatedness and raw country returns outside education countries

Dependent variable	Raw country return		
Units	Annualized basket bps		
	(1)	(2)	(3)
Specification	Shared border	Official language	Social connectedness
Foreign education × tie (θ)	-345.60^{**}	-194.62^{***}	-81.09^{**}
	(154.53)	(68.89)	(40.85)
Tie (δ)	188.34	132.31^{*}	46.21
	(126.83)	(74.09)	(42.15)
Foreign-educated share (γ)	−27.33	1.41	613.09^{*}
	(33.83)	(34.68)	(318.73)
Missing-tie share (MissingShare)	5.70	34.80	-302.82^{*}
	(47.49)	(48.03)	(168.92)
Log country stock count	-53.07^{***}	-53.25^{***}	-58.83^{***}
	(12.25)	(12.25)	(12.14)
Log team size	-85.29^{***}	-85.81^{***}	-80.88^{***}
	(21.64)	(21.65)	(22.41)
Unknown-education share	−10.45	−9.43	−21.72
	(93.67)	(94.10)	(93.54)
Female share (known gender)	−76.51	−77.01	-90.77^{*}
	(47.08)	(47.02)	(48.93)
Unknown-gender share	37.16	31.78	53.70
	(76.06)	(76.15)	(79.30)
CFA share (observed biographies)	70.73^{**}	70.11^{**}	74.14^{**}
	(31.65)	(31.69)	(31.92)
Unknown-biography share	79.16	85.82	69.45
	(123.40)	(123.29)	(121.99)
Mean log(1 + experience)	-57.50^{***}	-57.83^{***}	-52.22^{***}
	(16.73)	(16.76)	(17.07)
Fund × country FE	Yes	Yes	Yes
Country × quarter FE	Yes	Yes	Yes
Observations	936,076	936,076	876,532
Fund clusters	1,423	1,423	1,421

Notes: The specifications, samples, fixed effects, weights and clustering are those of Table 7. Only the dependent variable differs. The raw country return is the return on the fund's holdings in the country over the three months after quarter-end, annualized in basis points, without subtracting the size and past-return matched benchmark of Equation (2). Country-quarter effects still remove the national market return common to all funds holding the country in that quarter. Column 1 is a shared border. Column 2 is a common official language. Column 3 is log social connectedness from the October 2021 snapshot. The first row is θ, the foreign-minus-U.S. difference in slopes. The tie coefficient δ gives the U.S.-educated slope. Standard errors in parentheses are clustered by economic fund. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table A14: Passive recovery and portfolio changes: alternative samples

	Stricter coverage		One-year cohort	
Dependent variable	Passive holdings	Portfolio changes	Passive holdings	Portfolio changes
Units	Annualized basket bps			
	(1)	(2)	(3)	(4)
Linked appointment × Year 1	107.99	20.93	250.07***	−86.40
	(93.72)	(63.09)	(89.07)	(69.05)
Linked appointment × Year 2	73.92	−101.38	—	—
	(92.85)	(76.00)		
Log team size	−388.00	106.22	−563.39	159.61
	(396.43)	(294.34)	(526.06)	(321.16)
Unknown-education share	1162.57**	55.89	2550.68***	−1320.56***
	(527.71)	(456.94)	(757.10)	(485.16)
Log lagged assets (USD millions)	3.96	−94.14	−43.87	26.75
	(91.65)	(94.06)	(132.26)	(142.19)
Assets spline basis 1	−191.02	376.19*	−186.67	59.49
	(220.69)	(191.74)	(312.13)	(278.38)
Assets spline basis 2	411.11	−2795.11**	−349.13	58.69
	(1445.77)	(1125.26)	(1975.79)	(1520.88)
Team-size spline basis 1	−486.56	72.95	−120.71	−18.38
	(1047.34)	(802.33)	(1189.61)	(832.23)
Team-size spline basis 2	3548.15	−1509.19	2366.76	−1050.23
	(4286.95)	(3274.90)	(4671.67)	(3455.35)
Female share (known gender)	546.55**	−570.26***	37.89	−247.53
	(261.44)	(202.96)	(325.85)	(215.75)
Unknown-gender share	−881.00	−12.08	−1341.06*	255.91
	(817.14)	(540.97)	(761.80)	(711.19)
CFA share (observed biographies)	570.76**	−258.59	448.16	−99.93
	(233.51)	(185.24)	(285.85)	(255.94)
Unknown-biography share	602.24	−266.73	231.67	−196.44
	(802.84)	(651.28)	(1013.82)	(769.73)
Mean log(1 + experience)	−53.79	−295.03***	−206.72	−208.85*
	(108.21)	(103.94)	(134.33)	(111.91)
Prior-year relative NAV return (pp)	−10.52	−12.49**	−21.82**	−8.66
	(9.10)	(5.87)	(10.72)	(6.31)
Log(1 + prior relative volatility)	173.46	−3.79	152.90	142.88
	(163.45)	(122.10)	(206.58)	(139.86)
Prior-quarter flow (pp)	-1.79×10^{-4}	-2.26×10^{-5}	-2.31×10^{-4}	6.81×10^{-5}
	(2.25×10^{-4})	(1.32×10^{-4})	(2.23×10^{-4})	(9.11×10^{-5})
Missing prior return history	1312.71	−316.13	528.13	239.66
	(806.82)	(293.09)	(591.16)	(282.39)
Missing prior-quarter flow	2107.21	−1595.42	997.45	−987.26
	(2451.79)	(1073.85)	(2140.82)	(971.91)
Lagged expense ratio (bps)	0.38	−2.23	2.25	−1.03
	(2.85)	(2.98)	(3.49)	(3.08)
Lagged log turnover	112.67	−118.14	182.26	−28.21
	(253.19)	(230.97)	(251.42)	(200.53)
Top-ten equity concentration	−839.73	−682.11	−792.40	−552.60
	(1046.53)	(911.03)	(1209.95)	(810.49)
Missing lagged expense ratio	−106.99	−235.11	−175.51	67.33
	(454.52)	(472.03)	(521.36)	(459.22)
Missing lagged turnover	11.42	−116.80	314.20	−94.76
	(230.10)	(230.31)	(282.23)	(202.56)
Fund × matched-case FE	Yes	Yes	Yes	Yes
Matched-case × quarter FE	Yes	Yes	Yes	Yes
Observations	14,534	14,534	12,886	12,886
Country-appointment cases	401	401	511	511
Hiring funds	216	216	267	267
Fund clusters	481	481	610	610

Notes: Columns 1–2 require three paired quarters per phase and recompute equal hiring-fund-quarter weights. Columns 3–4 use the one-year cohort and available paired observations, preserving unit phase weights. Their Year 1 row is the hiring-by-post coefficient. Within each pair, columns sum to the observed-holdings coefficient. Outcomes follow Appendix B.10. In columns 1–2, Year 2 minus Year 1 is −34.07 (SE 102.74) and −122.30 (SE 68.49, significant at 10%), using the joint covariance. Concentration is never missing in this sample. Standard errors below coefficients are clustered by economic fund. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table A15: Foreign education and performance-conditioned monthly flows

Dependent variable	Monthly flow	Monthly flow
Units	Percentage points	
	(1)	(2)
Specification	Full controls	No operating controls
Foreign-educated share	0.207	0.207
	(0.153)	(0.153)
Unknown-education share	−0.458	−0.459
	(0.340)	(0.340)
Log lagged assets	−0.426***	−0.428***
	(0.039)	(0.039)
Log team size	−0.076	−0.078
	(0.087)	(0.087)
Log fund age	−1.991***	−1.986***
	(0.153)	(0.153)
Prior-return rank: bottom segment	5.586***	5.576***
	(0.605)	(0.605)
Prior-return rank: middle segment	1.642***	1.642***
	(0.124)	(0.124)
Prior-return rank: top segment	8.855***	8.863***
	(0.731)	(0.730)
Prior monthly-return volatility (pp)	−0.077*	−0.076*
	(0.043)	(0.043)
Lagged annual expense ratio (bps)	0.0004	—
	(0.0017)	
Log(1 + turnover)	0.104	—
	(0.188)	
Female share (known gender)	−0.361*	−0.363*
	(0.188)	(0.188)
Unknown-gender share	−0.220	−0.227
	(0.381)	(0.382)
CFA share (observed biographies)	−0.051	−0.052
	(0.138)	(0.138)
Unknown-biography share	0.467	0.471
	(0.494)	(0.494)
Mean log(1 + experience)	0.021	0.016
	(0.080)	(0.080)
Fund FE	Yes	Yes
Benchmark × month FE	Yes	Yes
Observations	185,748	185,748
Fund clusters	1,494	1,494

Notes: Equation (18) conditions monthly asset growth net of returns on prior performance ranks, characteristics and team traits. Prior-return rank is on a 0–1 scale, with knots at 0.2 and 0.8. The three spline terms enter the same regression. Column 2 omits expense ratio and turnover on the identical observations. All other regressors and fixed effects are unchanged. The pooled 1st/99th percentile flow caps are −21.435 and 47.589 pp. Fund age is measured from inception. Fund-months have equal weight. Standard errors, clustered by economic fund, appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table A16: Flows and fees

Dependent variable	Next-quarter flow	Next-quarter flow	Four-quarter flow	Flow after hire	Expense ratio
Units		Flows in pp, expense ratio in annual bps			
	(1)	(2)	(3)	(4)	(5)
Specification	Next quarter	Return sensitivity	Four quarters	After appointment	Expense ratio
Foreign-educated share	−0.29	0.38	−0.26	—	−0.39
	(0.81)	(0.64)	(2.76)		(1.49)
Foreign-educated share × prior-year relative NAV return (pp)	—	−0.04	—	—	—
		(0.06)			
Foreign-educated share of appointees	—	—	—	0.22	—
				(1.97)	
Unknown-education share	−4.36**	−3.14**	−14.41**	—	−3.27
	(1.85)	(1.51)	(6.61)		(3.43)
Log team size	−0.40	−0.26	−2.11	—	−3.73
	(1.40)	(1.18)	(4.65)		(2.35)
Log lagged assets (USD millions)	−8.08***	−4.64***	−22.84***	−21.25***	−0.14
	(0.53)	(0.48)	(1.42)	(2.37)	(0.43)
Assets spline basis 1	3.93***	0.80	0.61	—	−5.04***
	(0.97)	(0.85)	(2.79)		(1.19)
Assets spline basis 2	−8.98*	2.09	28.74*	—	22.27***
	(5.03)	(4.27)	(15.14)		(6.92)
Team-size spline basis 1	−1.86	−1.21	−8.79	—	1.86
	(3.20)	(2.68)	(10.48)		(5.48)
Team-size spline basis 2	10.81	7.75	55.47	—	9.70
	(13.43)	(11.30)	(43.82)		(22.90)
Female share (known gender)	−0.72	−1.38*	−5.29	1.65	−1.14
	(1.07)	(0.83)	(3.55)	(4.28)	(1.61)
Unknown-gender share	−1.11	−1.04	−2.38	−6.94	−1.41
	(1.93)	(1.46)	(6.77)	(12.95)	(2.52)
CFA share (observed biographies)	−0.88	−0.71	−3.65	−3.33	−1.33
	(0.73)	(0.60)	(2.44)	(4.38)	(1.26)
Unknown-biography share	3.12	1.97	8.18	15.67	4.87
	(2.51)	(2.08)	(9.20)	(18.28)	(4.02)
Mean log(1 + experience)	−1.59***	−0.76**	−2.70**	0.24	−1.24**
	(0.40)	(0.33)	(1.27)	(2.41)	(0.61)
Prior-year relative NAV return (pp)	—	0.38***	—	0.68***	—
		(0.03)		(0.17)	
Pre-appointment four-quarter flow (pp)	—	—	—	0.07*	—
				(0.04)	
Log pre-hire destination team size	—	—	—	0.55	—
				(2.62)	
Unknown education among appointees	—	—	—	−1.05	—
				(4.80)	
Fund FE	Yes	Yes	Yes	Yes	Yes
Benchmark × quarter FE	Yes	Yes	Yes	No	Yes
Year FE	No	No	No	Yes	No
Observations	67,343	63,869	62,657	2,973	61,639
Fund clusters	1,418	1,418	1,418	897	1,418

Notes: Flows are net flows as percentages of assets. The final column uses annual expense-ratio basis points. Columns 1–2 explain next-quarter flows, column 3 the next four quarters, and column 4 the four quarters following an appointment conditional on prior flows. Column 2 adds the interaction of foreign-educated share with prior relative performance, in annual NAV percentage points. Team information precedes subsequent flows. All five regressions use equal observation weights and economic-fund clustering. Standard errors appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table A17: Prior-performance comparisons within origin peers and hiring rounds

Dependent variable	Peer-adjusted return	Prior adjusted return
Units	Annualized basket bps	
	(1)	(2)
Specification	Origin peers	Same hiring round
Foreign-educated appointee	−71.77*	−80.52*
	(40.96)	(41.29)
Log pre-hire destination team size	−13.85	—
	(66.08)	
Pre-hire destination unknown education	422.18**	—
	(206.91)	
Log origin team size	−43.59	69.06
	(55.09)	(91.83)
Origin-team unknown education	161.62	−546.09**
	(330.54)	(260.01)
Log origin country stock count	−68.35	−126.69
	(77.72)	(138.88)
Appointee: top-100 school	54.35	−17.06
	(34.48)	(27.92)
Appointee: school ranked 101–500	134.55**	23.94
	(56.31)	(27.75)
Appointee: female	−18.41	−17.97
	(38.26)	(30.38)
Appointee: unknown gender	43.13	270.17**
	(70.74)	(105.25)
Appointee: CFA	−45.16	−14.23
	(35.06)	(25.84)
Log(1 + manager experience)	22.20	78.88***
	(31.79)	(29.10)
Prior benchmark: China	70.61	—
	(305.31)	
Prior benchmark: Developed Asia-Pacific	−1029.96	—
	(953.70)	
Prior benchmark: Developed Asia-Pacific ex Japan	32.28	—
	(179.00)	
Prior benchmark: Developed Europe	492.95**	—
	(234.47)	
Prior benchmark: Emerging Americas	707.99**	—
	(338.04)	
Prior benchmark: Emerging markets	31.50	—
	(128.52)	
Prior benchmark: Global growth	191.20	—
	(237.58)	
Prior benchmark: Global large cap	127.49	—
	(212.64)	
Prior benchmark: Global	96.97	—
	(164.44)	
Prior benchmark: Global small/mid cap	102.20	—
	(270.14)	
Prior benchmark: Global value	20.29	—
	(220.97)	
Prior benchmark: Global ex-U.S. growth	−66.31	—
	(118.35)	
Prior benchmark: Global ex-U.S. small/mid cap	134.34	—
	(145.04)	
Prior benchmark: Global ex-U.S. value	−23.98	—
	(101.81)	
Prior benchmark: Japan	908.97	—
	(2120.85)	
Hiring-fund FE	Yes	No
Appointment-year FE	Yes	No
Hiring-fund × appointment-quarter FE	No	Yes
Observations	1,321	832
Hiring-fund clusters	300	189

Notes: Both outcomes average four pre-appointment quarters of outside-country adjusted returns. Column 1 subtracts benchmark-country-quarter peers and controls for origin benchmarks (Global ex-U.S. omitted). Column 2 compares 832 appointments in 271 common hiring rounds, 126 with both education groups, without peer subtraction or origin-benchmark controls. In column 2, destination team size and unknown education are absorbed. Neither model controls for prior NAV. Appointments have equal weight. Hiring-fund-clustered standard errors appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table A18: Country coverage and relative departure responses

Dependent variable	Departure	Departure
Units	Probability in pp	
	(1)	(2)
Specification	Other trait slopes	No other trait slopes
Unique coverage × fund return (pp)	0.180***	0.177***
	(0.056)	(0.056)
Duplicated coverage × fund return (pp)	0.006	0.000
	(0.118)	(0.116)
No match × fund return (pp)	0.122**	0.113*
	(0.062)	(0.061)
Unclassified coverage × fund return (pp)	0.051	0.048
	(0.157)	(0.156)
Unique held-country coverage	−0.12	−0.12
	(0.41)	(0.41)
Duplicated held-country coverage	−0.02	−0.01
	(0.78)	(0.78)
Unclassified held-country coverage	−0.83	−0.82
	(0.81)	(0.81)
Log recorded fund spell length	2.70***	2.70***
	(0.28)	(0.28)
Female × fund return (pp)	0.067	—
	(0.056)	
Unknown gender × fund return (pp)	0.012	—
	(0.138)	
CFA × fund return (pp)	−0.051	—
	(0.044)	
Unknown biography × fund return (pp)	−0.111	—
	(0.249)	
Log(1 + experience) × fund return (pp)	0.045	—
	(0.029)	
Log(1 + manager experience)	−3.16***	−3.15***
	(0.41)	(0.41)
Fund × quarter FE	Yes	Yes
Fund × manager FE	Yes	Yes
Observations	154,677	154,677
Fund clusters	1,334	1,334

Notes: Both columns use the same observations and fixed effects. Column 2 omits the five other-trait interactions with performance, retaining spell length and experience. Coverage, measured from prior holdings and incumbent colleagues, is unique, duplicated, no-match or unclassified. Return interactions and their direct differences are percentage points per percentage point of better prior-quarter performance. These are relative foreign-minus-U.S. exit responses. Each fund-quarter receives equal weight, divided among its managers, and standard errors, clustered by economic fund, appear in parentheses below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%. In column 1, duplicated minus unique and no-match minus unique return slopes are −0.175 (SE 0.119) and −0.058 (0.076). Jointly testing that the duplicated and no-match slopes equal zero gives $F = 1.96$, $p = 0.141$.

Table A19: Principal variables, units and timing

Variable	Unit	Definition and observation window
Foreign-educated manager share ForeignShare_{jq}	Fraction of known-education managers	Any non-U.S. post-secondary education, including dual U.S./foreign degrees. Quarter-end team used for the following quarter's returns. Measures education location, not nationality or diversity.
Unknown-education share UnknownShare_{jq}	Fraction of all current managers	Managers without identifiable education divided by the recorded team count. Entered separately from ForeignShare.
Education-country link	Country indicator or team share	Current managers' recorded education ties to the country. An outside country has no such tie.
Factor-adjusted fund return, R^a_{jm}	Annualized fund NAV basis points	Monthly net NAV return less the risk-free rate and thirteen predetermined factor-exposure returns. Uses preceding returns and the stated peer fallback and covers the entire fund.
Adjusted country return, R_{jcq}	Annualized basket basis points	As in Equation (2), twelve times the mean of the three monthly size/past-return-adjusted returns after quarter-end. Positions drift within the quarter, and stocks without returns are dropped. Teams and holdings are updated at the next quarter-end.
Appointee's prior performance, $\bar{R}^{\text{pre}}_h$	Annualized basket basis points	Previous teams' adjusted stock returns over four full quarters before hiring. Equal country weights, then averages over quarters and distinct economic origin funds. The main hiring comparison includes all non-U.S. countries and controls for origin benchmarks. The $\bar{R}^{\text{peer}}_h$ check uses outside countries only and subtracts benchmark-country-quarter peers. A measure of the previous teams' performance, not an individual manager's return.
Career performance signal, R^{rel}_{jq}	Quarter-q return in percentage points	Compounded NAV in q relative to other funds sharing the benchmark. It predicts careers between the ends of q and $q+1$. Distinct from prior-year controls in the NAV and flow models.
Career outcomes, $\mathbb{1}^{\text{Exit}}_{ijq}$ and $\mathbb{1}^{\text{NewFund}}_{ijq}$	Probability in percentage points	Departure from q to $q+1$, or new recorded appearance at another economic fund in $q+1$. Assignments need not be promotions.
Education-country coverage	Prior-quarter holdings and team education	The country-specialty test uses held countries without a continuing manager educated there. The career test classifies a current manager's held-country links as unique, duplicated by colleagues or no-match. Missing holdings remain unclassified.
Conviction, I_{jkq}	Within-country active weight	Stock weight less capitalization weight in the same held-country basket. Quartiles precede future returns, and portfolio alphas use monthly country-reweighted, factor-adjusted returns.

Notes: The sources, fixed effects, controls and samples of each comparison are described with the corresponding analysis. Panel regressions cluster standard errors by economic fund. Calendar-time conviction portfolios use the time-series covariance specified in Table 6. Education comes from manager biographies assigned to dated fund spells.

Table A20: Home-country stock-selection performance: JJK and reconstructed samples

Dependent variable	Calendar-time portfolio return				
Units	Monthly four-factor alpha, bps				
	(1)	(2)	(3)	(4)	(5)
Sample	Published	JJK list	JJK list	Our funds	Our funds
Payoff period	1991–2014	1991–2014	2000–2014	2000–2014	2000–2024
Panel A: All international funds (JJK "Global")					
Home-country long-only	54***	10.6	4.2	0.3	1.5
	(2.67)	(0.56)	(0.31)	(0.02)	(0.15)
Home linked–unlinked	31*	−24.0	−14.6	−21.3	−7.9
	(1.69)	(−1.22)	(−1.02)	(−1.33)	(−0.73)
Home linked–unlinked, matched short	—	−12.4	−6.1	−8.5	1.9
		(−0.79)	(−0.60)	(−0.72)	(0.23)
Outside linked–unlinked	0	−3.8	5.4	8.0	9.7*
	(0.05)	(−0.39)	(0.89)	(1.11)	(1.95)
Months, home long–short	282	279	180	180	300
Linked funds per month	—	86	127	176	239
Unlinked funds per month	—	161	231	290	356
Funds with holdings in the window	—	1,019	1,018	1,255	1,629
Funds with degree country and assets	1,090	859	849	1,171	1,559
Panel B: Regional funds (nested subgroup)					
Home-country long-only	72**	50.7	45.5	39.4	14.0
	(2.33)	(1.48)	(1.23)	(1.28)	(0.66)
Home linked–unlinked	65**	23.4	−16.6	−16.7	−10.5
	(2.21)	(0.81)	(−0.52)	(−0.58)	(−0.55)
Home linked–unlinked, matched short	—	10.8	2.9	−12.1	−9.5
		(0.46)	(0.10)	(−0.44)	(−0.55)
Outside linked–unlinked	29	22.8	14.8	13.7	10.7
	(1.62)	(1.54)	(1.55)	(1.06)	(1.30)
Months, home long–short	237	249	180	180	300
Linked funds per month	—	24	34	45	66
Unlinked funds per month	—	35	46	60	83
Funds with holdings in the window	—	315	315	358	471
Funds with degree country and assets	—	247	244	332	451
Panel C: Country funds (nested subgroup)					
Home-country long-only	47*	3.3	−9.7	−8.7	5.0
	(1.89)	(0.10)	(−0.24)	(−0.21)	(0.19)
Home linked–unlinked	59**	−28.5	−19.9	−21.4	−6.9
	(2.29)	(−0.81)	(−0.51)	(−0.57)	(−0.27)
Home linked–unlinked, matched short	—	−31.7*	−28.0	−27.3	−5.0
		(−1.75)	(−1.43)	(−1.40)	(−0.39)
Outside linked–unlinked	49**	30.2**	36.6**	22.2	0.8
	(2.46)	(2.13)	(2.24)	(0.70)	(0.03)
Months, home long–short	207	219	180	180	300
Linked funds per month	—	7	9	10	12
Unlinked funds per month	—	10	11	13	14
Funds with holdings in the window	—	90	90	85	99
Funds with degree country and assets	—	67	66	80	96

Notes: Column 1 is copied from JJK Table 4, Panel A. Columns 2–5 are new estimates using undergraduate links from the education records behind the main text. Funds with holdings have a position at a formation date in the window, and the next row adds a current manager with a recorded undergraduate country and lagged assets. Long legs hold home-linked funds' stocks in their home countries, and the long-only row is net of the Treasury-bill rate. The short leg holds the same category-quarter's home countries in funds with no home-linked manager. The matched-short row instead values each linked fund's short leg at that fund's own country weights, using same-category unlinked funds' country returns. Stocks are weighted by dollar holdings within funds and funds by total net assets at the previous quarter-end. Home-country portfolios use the Fama–French Global ex-U.S. market, size, value and momentum factors in columns 1–4. Column 5 uses the Developed ex-U.S. series, an extended model, because the Global ex-U.S. file we hold ends in June 2019. Outside-country portfolios use the Developed (formerly Global) factors. t-statistics in parentheses are heteroskedasticity-robust, as in JJK. With Newey–West standard errors (three lags), two of the 48 alphas in columns 2–5 change significance level. Stock returns are observed for 85% of long-leg and 90% of short-leg formation value on JJK's list over 1991–2014 and for 89% and 93% on our funds over 2000–2024. Missing returns are excluded, not set to zero. The largest fund averages 30% of the all-international long leg in column 2 and 16% in column 5. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table A21: Sensitivity of the JJK home-country portfolio to its construction

Dependent variable	Home-country portfolio return			
Units	Monthly four-factor alpha, bps			
	(1)	(2)	(3)	(4)
Sample	JJK list	JJK list	Our funds	Our funds
Payoff period	1991–2014	2000–2014	2000–2014	2000–2024
All international funds, home linked–unlinked				
Baseline	−24.0	−14.6	−21.3	−7.9
	(−1.22)	(−1.02)	(−1.33)	(−0.73)
Country-relative short	−29.9	−20.3	−35.5*	−17.5
	(−1.51)	(−1.29)	(−1.82)	(−1.37)
Country-matched short	−12.4	−6.1	−8.5	1.9
	(−0.79)	(−0.60)	(−0.72)	(0.23)
Pooled stock dollars across funds	−12.4	−3.5	−10.9	−1.8
	(−0.70)	(−0.23)	(−0.71)	(−0.17)
Equal weights across funds	−5.4	−6.2	−7.9	−5.1
	(−0.37)	(−0.56)	(−0.73)	(−0.78)
Category mandate required for the link	−24.3	−19.8	−26.3	−8.4
	(−1.33)	(−1.39)	(−1.61)	(−0.73)
Earlier education coding	−19.1	−25.3	−23.7	−8.7
	(−0.86)	(−1.52)	(−1.50)	(−0.75)
Earlier coding, country-matched short	19.8	−7.3	−6.1	3.9
	(0.79)	(−0.58)	(−0.48)	(0.42)
Stock-months with a return index only	−6.8	−13.4	−19.2	−0.1
	(−0.28)	(−0.81)	(−0.94)	(−0.01)
Country from the ISIN prefix	−23.1	−21.5	−24.8	−8.0
	(−1.11)	(−1.25)	(−1.39)	(−0.68)
Months with three funds on each side	−16.0	−14.6	−21.3	−7.9
	(−0.93)	(−1.02)	(−1.33)	(−0.73)
Regional and Country funds, category mandate required for the link				
Regional, home long-only	57.6*	72.2*	90.5*	52.5
	(1.85)	(1.85)	(1.78)	(1.60)
Regional, home linked–unlinked	−75.7**	−23.5	−2.9	8.4
	(−1.99)	(−0.62)	(−0.06)	(0.29)
Country, home long-only	20.3	17.3	18.0	20.2
	(0.79)	(0.63)	(0.66)	(1.10)
Country, home linked–unlinked	−7.8	6.6	4.8	7.9
	(−0.36)	(0.27)	(0.21)	(0.47)

Notes: Each row changes one feature of the baseline in Table A20, except the row that combines the earlier coding with the country-matched short leg. The country-relative short leg admits any same-category fund with no manager linked to the country. The pooled-dollars specification weights positions by their own dollar value across funds rather than by fund assets. The category-mandate rows keep a link only if the country lies in the fund's Morningstar category mandate. Broad categories cover every non-U.S. country. The return-index row keeps returns only for stock-months with a total-return index. The last row of the upper block keeps months with at least three funds on each side. Factors and inference are as in Table A20. Column 4 uses the extended factor model. t-statistics are in parentheses. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table A22: JJK home-country portfolios by return coverage, JJK list, 1991–2014

Dependent variable	Calendar-time portfolio return					
Units	Monthly four-factor alpha, bps					
	(1)	(2)	(3)	(4)	(5)	(6)
Stock-months	Published	All	Index only	Index only	Filled only	All
Education coding		Main	Main	Earlier	Main	Main
Dividends on filled returns		Country			Country	Matched
Panel A: All international funds (JJK "Global")						
Home-country long-only	54***	10.6	49.0*	52.7*	1.1	16.3
	(2.67)	(0.56)	(1.77)	(1.81)	(0.06)	(0.86)
Home linked–unlinked	31*	−24.0	−6.8	−10.5	−17.5	−23.9
	(1.69)	(−1.22)	(−0.28)	(−0.40)	(−0.88)	(−1.21)
Outside linked–unlinked	0	−3.8	4.9	5.8	2.9	−3.1
	(0.05)	(−0.39)	(0.32)	(0.38)	(0.28)	(−0.31)
Months, long-only	282	285	276	276	285	285
Panel B: Regional funds (nested subgroup)						
Home-country long-only	72**	50.7	51.2	59.7	60.7	54.9
	(2.33)	(1.48)	(1.28)	(1.45)	(1.60)	(1.60)
Home linked–unlinked	65**	23.4	−26.6	−23.7	55.4*	24.8
	(2.21)	(0.81)	(−0.57)	(−0.53)	(1.66)	(0.86)
Outside linked–unlinked	29	22.8	16.3	17.5	32.2*	24.3
	(1.62)	(1.54)	(0.82)	(0.89)	(1.91)	(1.64)
Months, long-only	237	285	273	273	285	285
Panel C: Country funds (nested subgroup)						
Home-country long-only	47*	3.3	59.6	57.4	4.2	8.4
	(1.89)	(0.10)	(1.27)	(1.21)	(0.14)	(0.25)
Home linked–unlinked	59**	−28.5	−37.9	−31.5	2.8	−29.4
	(2.29)	(−0.81)	(−0.71)	(−0.57)	(0.09)	(−0.84)
Outside linked–unlinked	49**	30.2**	17.3	22.3	53.3***	29.0**
	(2.46)	(2.13)	(1.00)	(1.26)	(3.02)	(2.05)
Months, long-only	207	270	258	258	270	270

Notes: Column 1 is copied from JJK Table 4, Panel A. Columns 2–6 are new estimates on JJK's fund list. Column 2 repeats the baseline in Table A20. Column 3 keeps returns only for stock-months with a Datastream total-return index, and column 5 keeps only the others, so the two partition column 2's stock-months. Column 4 repeats column 3 with the earlier education coding. Columns 2 and 5 add a country-month mean dividend to filled returns. Column 6 keeps column 2's stock-months and instead uses a size-matched dividend, a value-weighted country-year mean, untrimmed, computed separately for the 20 largest stocks in a country-month and for other stocks. Portfolio construction, factors and inference are as in Table A20. t-statistics are in parentheses. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table A23: Education-linked raw country allocation: JJK Table 2 comparison

Dependent variable	Fund's raw weight in a held non-U.S. country				
Units	Fraction of total equity, regressor is the home-linked manager share				
	(1)	(2)	(3)	(4)	(5)
Sample	Published	JJK list	JJK list	Our funds	Our funds
Period	1991–2014	1991–2014	2000–2014	2000–2014	2000–2024
Panel A: All international funds (JJK "Global")					
No fixed effects	0.174***	0.135***	0.138***	0.143***	0.143***
	[16.10]	(0.018)	(0.021)	(0.019)	(0.027)
Category × quarter FE	0.139***	0.105***	0.106***	0.117***	0.107***
	[18.10]	(0.018)	(0.020)	(0.015)	(0.008)
Fund FE	0.138***	0.097***	0.096***	0.103***	0.099***
	[17.30]	(0.024)†	(0.025)†	(0.023)†	(0.012)†
Country × quarter FE	0.121***	0.065*	0.067*	0.066	0.080*
	[9.46]	(0.034)	(0.038)	(0.039)	(0.045)
Observations, no fixed effects	464,578	464,242	426,025	560,638	1,160,297
Observations, with controls	401,785	401,906	369,922	482,897	1,041,088
Panel B: Regional funds (nested subgroup)					
No fixed effects	0.338***	0.152**	0.156*	0.150*	0.193*
	[12.90]	(0.067)	(0.073)	(0.082)	(0.102)
Category × quarter FE	0.221***	0.081***	0.080**	0.079**	0.091**
	[12.90]	(0.028)	(0.031)	(0.036)	(0.040)
Fund FE	0.221***	0.053***	0.048***	0.042***	0.073**
	[12.40]	(0.015)†	(0.016)†	(0.015)†	(0.035)†
Country × quarter FE	0.203***	0.117***	0.119**	0.125**	0.165**
	[11.30]	(0.041)	(0.044)	(0.056)	(0.073)
Observations, no fixed effects	91,384	90,475	82,461	109,667	256,090
Observations, with controls	79,002	78,838	72,213	95,173	231,484
Panel C: Country funds (nested subgroup)					
No fixed effects	0.499***	0.401**	0.438**	0.472**	0.541***
	[10.37]	(0.149)	(0.154)	(0.164)	(0.142)
Category × quarter FE	0.307***	0.244***	0.251***	0.283***	0.251***
	[11.23]	(0.054)	(0.062)	(0.070)	(0.041)
Fund FE	0.323***	0.181***	0.178***	0.175**	0.218***
	[10.73]	(0.055)	(0.066)†	(0.067)	(0.051)
Country × quarter FE	0.096***	0.087***	0.093***	0.113**	0.150***
	[5.40]	(0.031)	(0.031)	(0.038)	(0.053)
Observations, no fixed effects	14,755	11,686	10,282	11,914	18,291
Observations, with controls	12,664	9,822	8,673	9,976	15,750
Panel D: Extensions on all international funds, not replications					
Category × quarter + country FE	—	0.029**	0.029*	0.032*	0.036**
		(0.013)	(0.015)	(0.016)	(0.017)
Fund + country × quarter FE	—	0.020***	0.018**	0.016**	0.028**
		(0.007)†	(0.008)†	(0.007)†	(0.012)

Notes: Column 1 is copied from JJK Table 2, with t-statistics in brackets. Columns 2–5 are new estimates with standard errors in parentheses, on the fractions scale of JJK's table. Undergraduate links come from the education records behind the main text, and the sample requires a current manager with a recorded undergraduate country. Rows with fixed effects include JJK's controls, lagged one quarter, which are log fund size, expense ratio, log turnover, log fund age and log number of managers. Each panel row is a separate regression. Standard errors are clustered by country and year, as in JJK, except cells marked †, where the two-way variance is not positive semi-definite and we report country clustering instead. Fund clustering gives smaller standard errors for most specifications. Panel D uses two sets of fixed effects at once and is not part of JJK's specification menu. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table A24: Home-country selection: comparison of empirical measures

Sample Payoff period	(1) JJK-list funds 2000–2014	(2) Our funds 2000–2014	(3) Our funds 2000–2024
JJK linked–unlinked alpha (bps per month)	−15.2	−19.6	−6.1
	(15.0)	(16.6)	(10.9)
Our characteristic-adjusted $\beta + \gamma$ (annualized bps)	−105.1	−83.5	0.2
	(81.3)	(70.7)	(56.6)
Our $\beta + \gamma$, undergraduate-only (annualized bps)	−28.7	−30.3	34.6
	(85.5)	(75.6)	(62.6)
Fund-country-quarters	350,475	457,477	974,924
Fund-quarters, panel	19,512	25,753	55,242
Funds, panel	758	1,041	1,428
Months, alpha	180	180	300

Notes: The first row is the monthly four-factor alpha of the home-country linked–unlinked portfolio in Table A20, restricted to the fund-quarters of the panel in that column, with robust standard errors in parentheses. Column 3 uses the extended factor model. The next two rows are the characteristic-adjusted contrast $\beta + \gamma$ from Equation (6), in annualized basket basis points, with fund-by-country and country-by-quarter effects, the controls of Table 5, equal country weights within fund-quarter, standard errors clustered by economic fund and the joint covariance of β and γ. Singleton cells are removed within each sample. The undergraduate row replaces both the link and the foreign-educated share with their undergraduate-only versions on the same rows. Rows differ in measure and units.